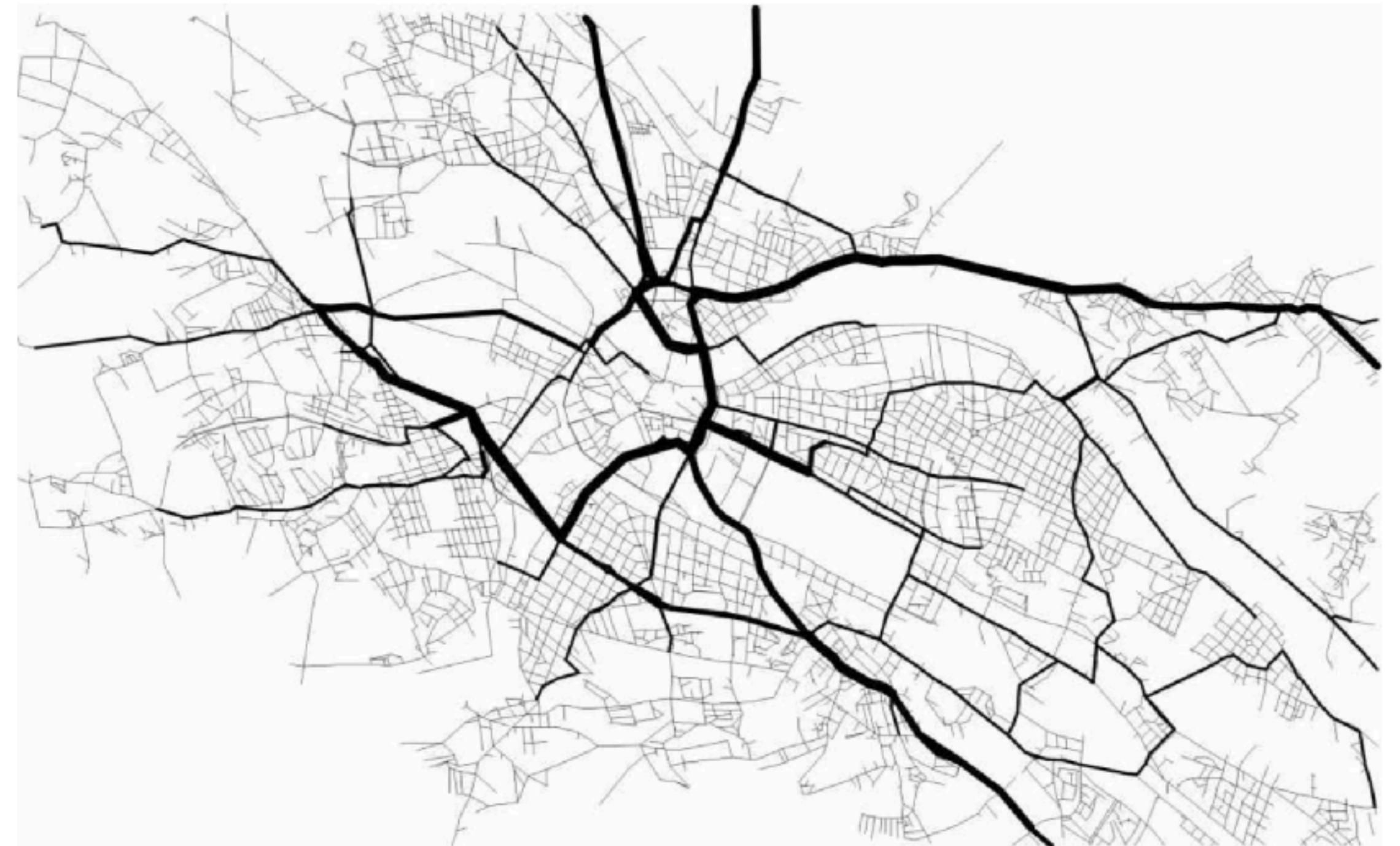


# Lecture 10: Spatial networks

Instructor: Michael Szell

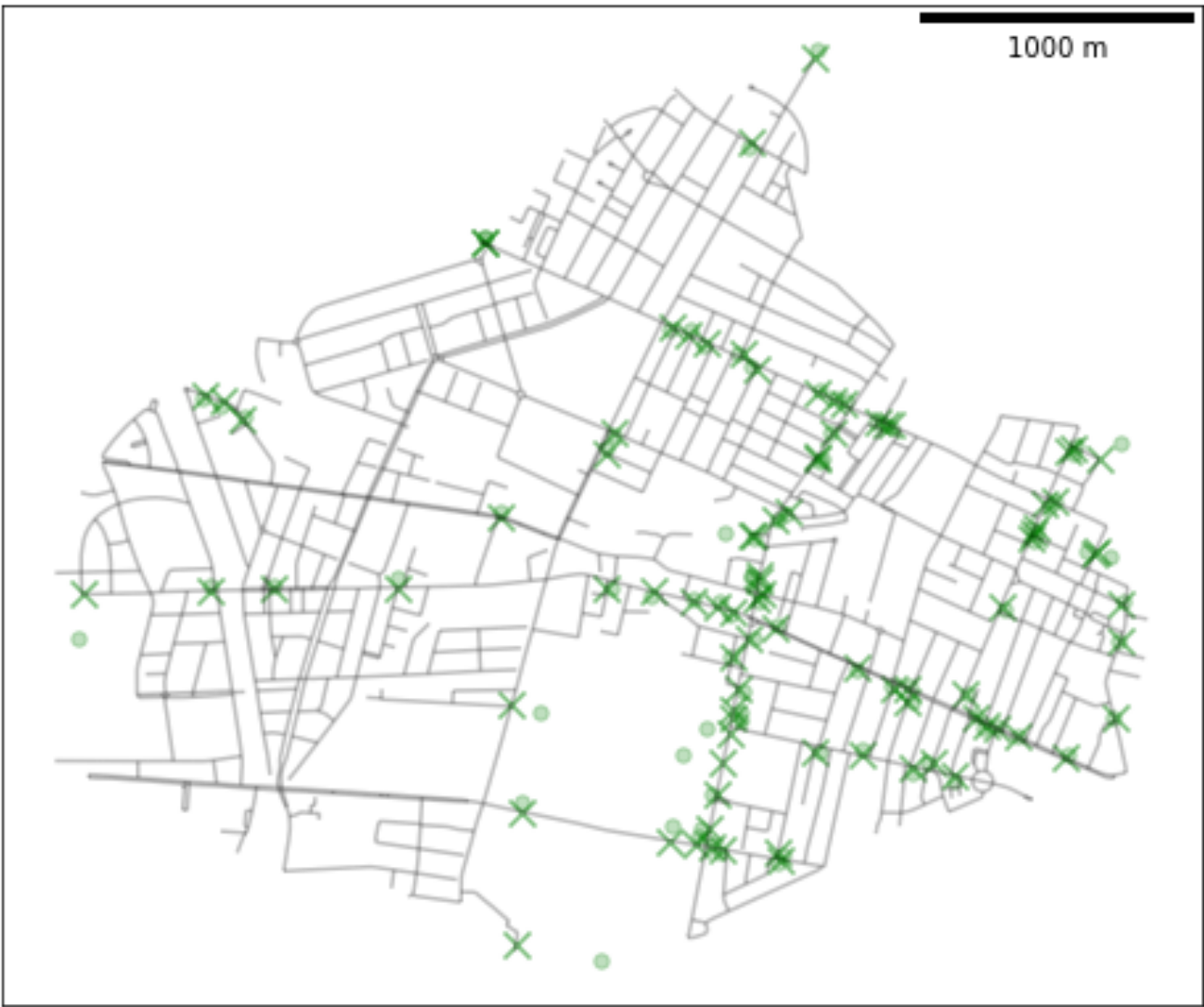


# Today you will learn about spatial networks

## Difference to non-spatial networks



spaghetti



## Optimal + transport networks and metrics

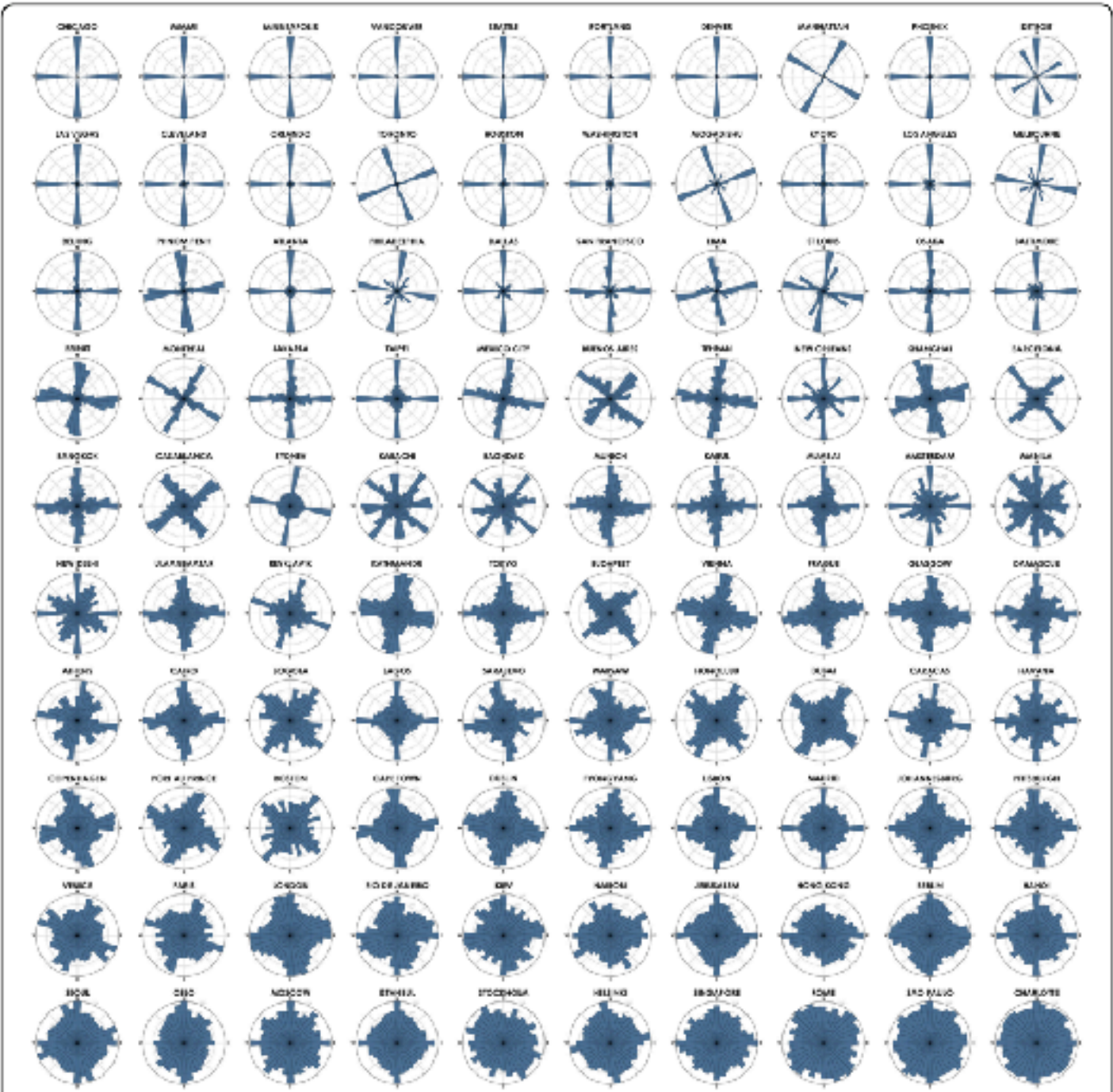
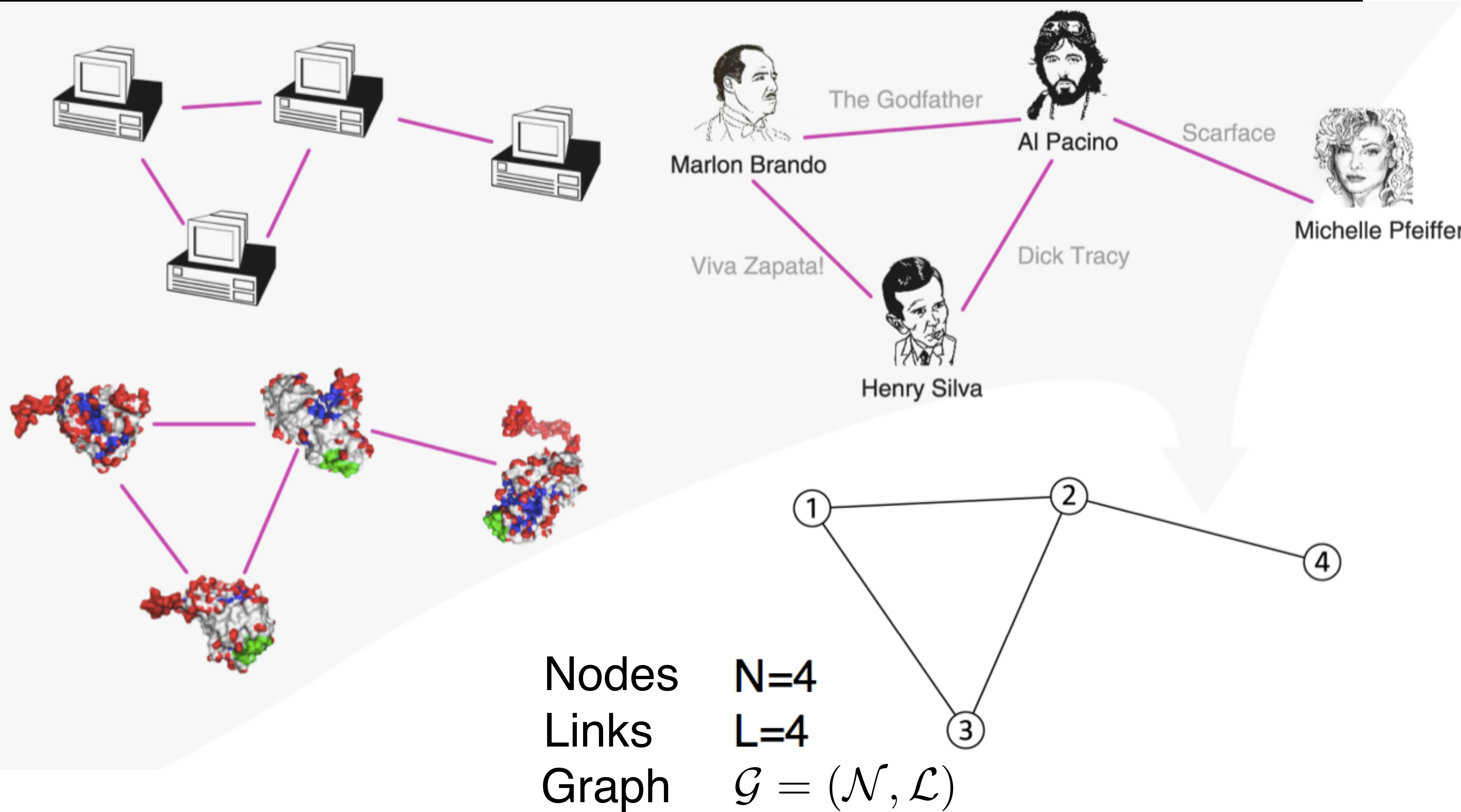


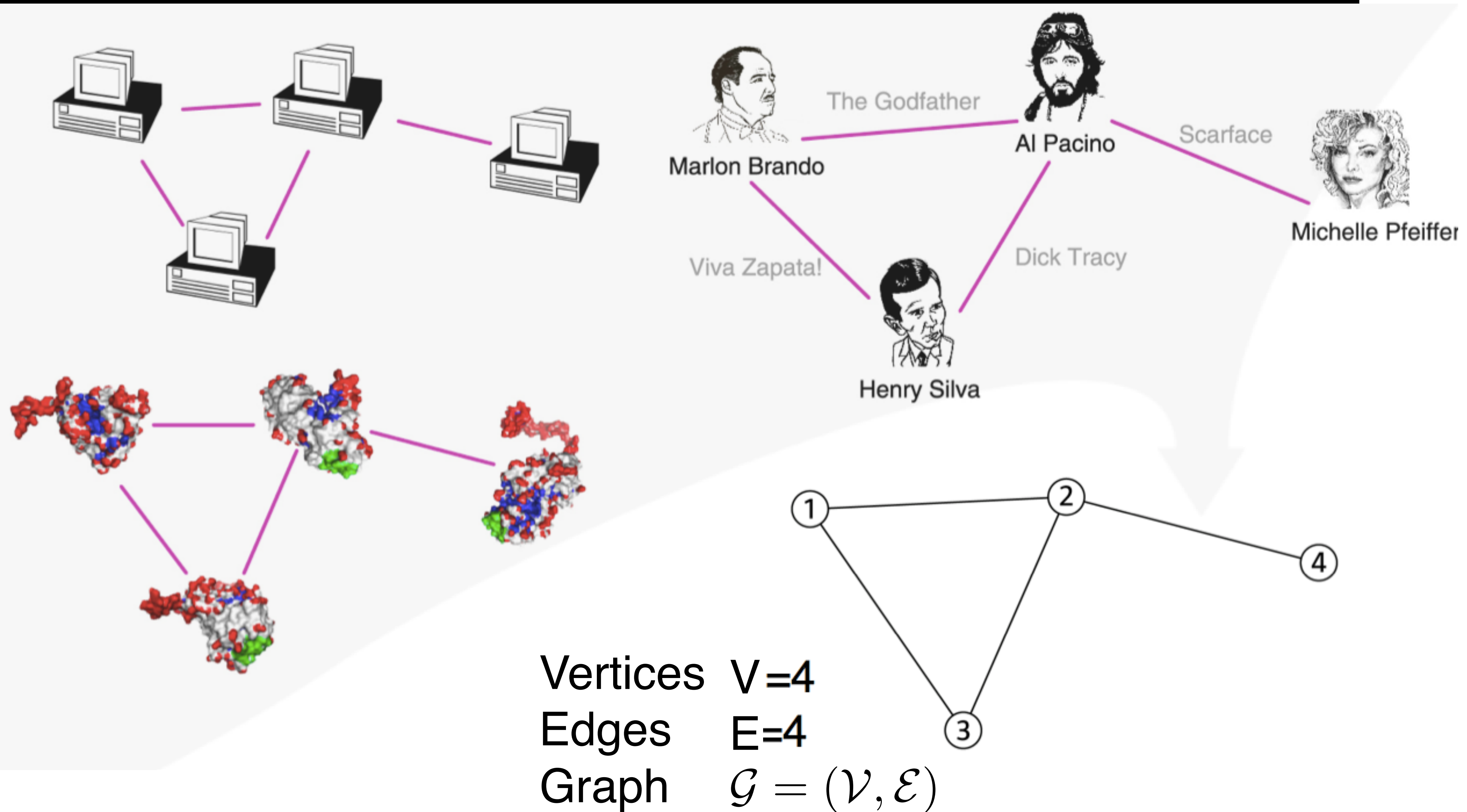
Fig. 5 Polar histograms from Fig. 4, resorted by descending  $\phi$  from most to least grid-like (equivalent to least to greatest entropy)



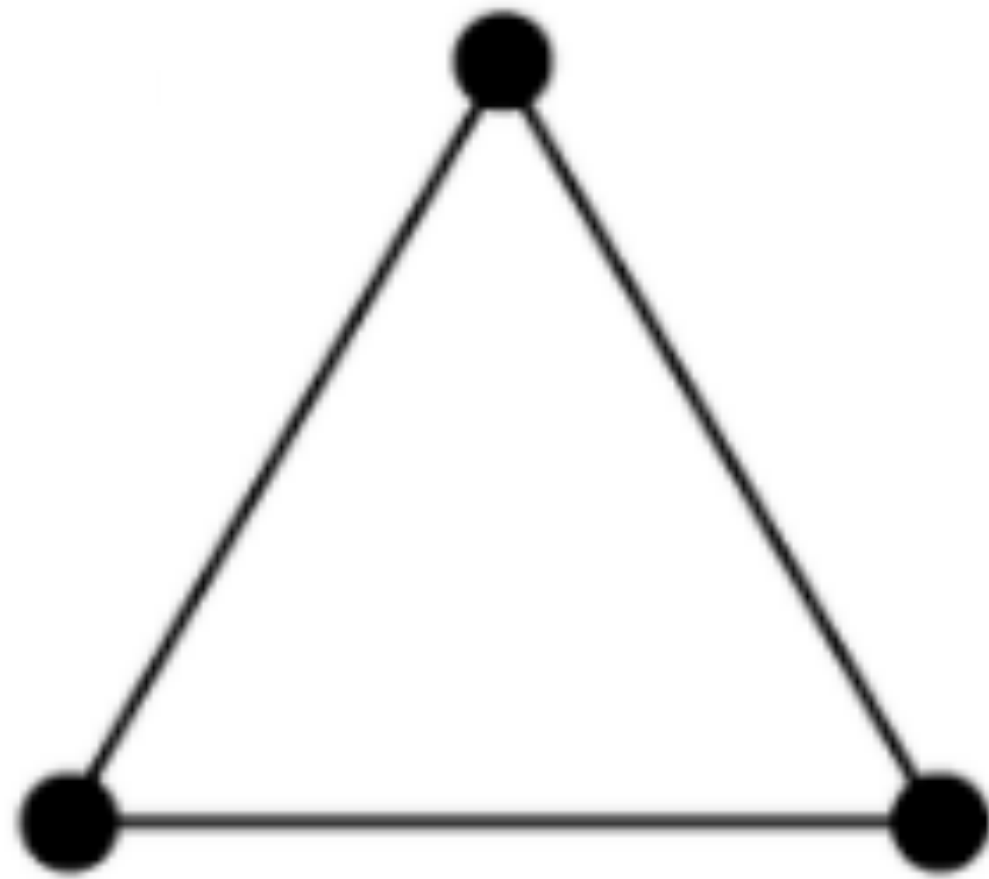
# Networks are a common language for different applications



# Network = Graph + real-world meaning



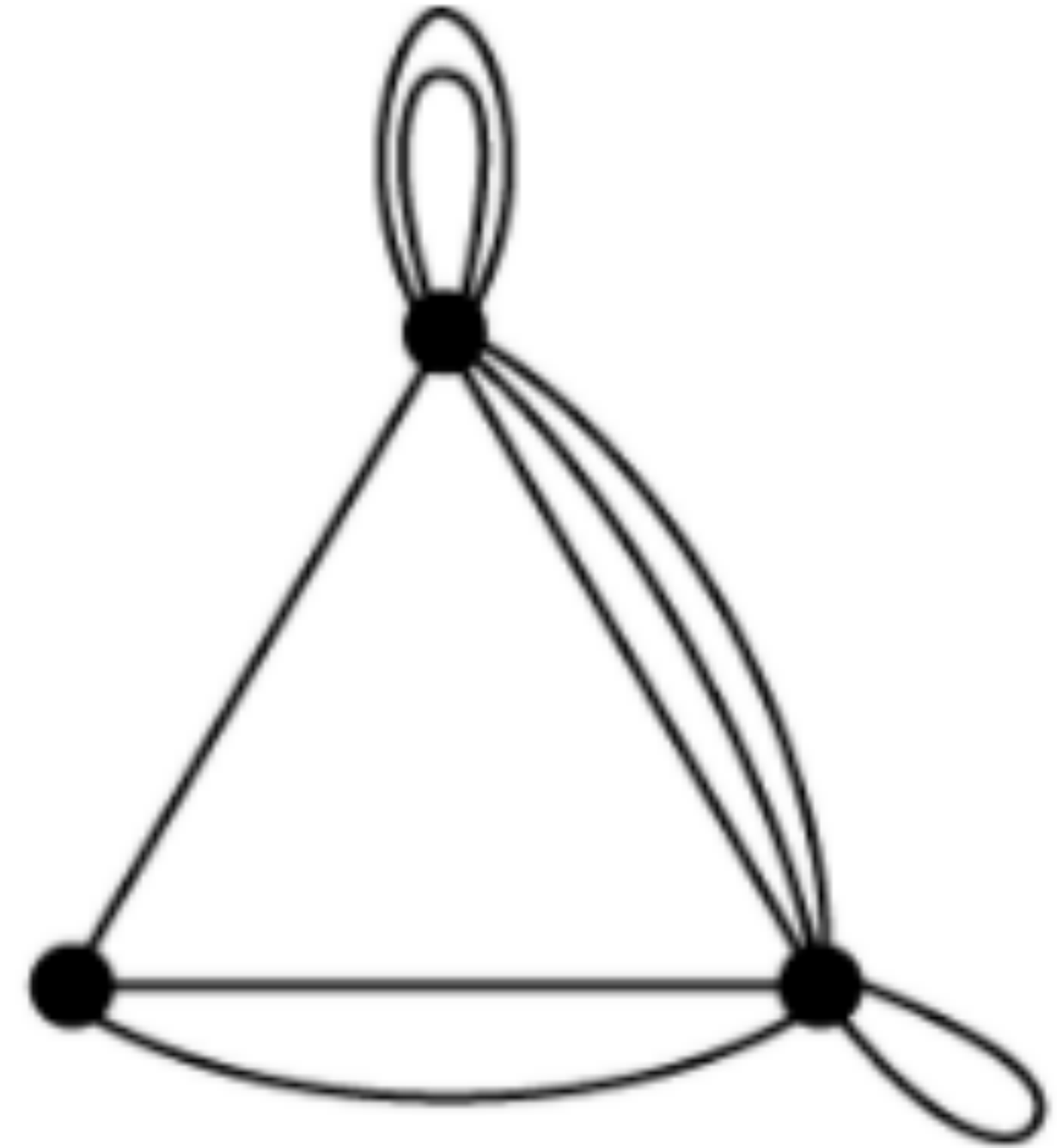
# Real spatial networks are usually pseudographs



Simple graph



+ parallel links  
= Multigraph



+ self-loops  
= Pseudograph

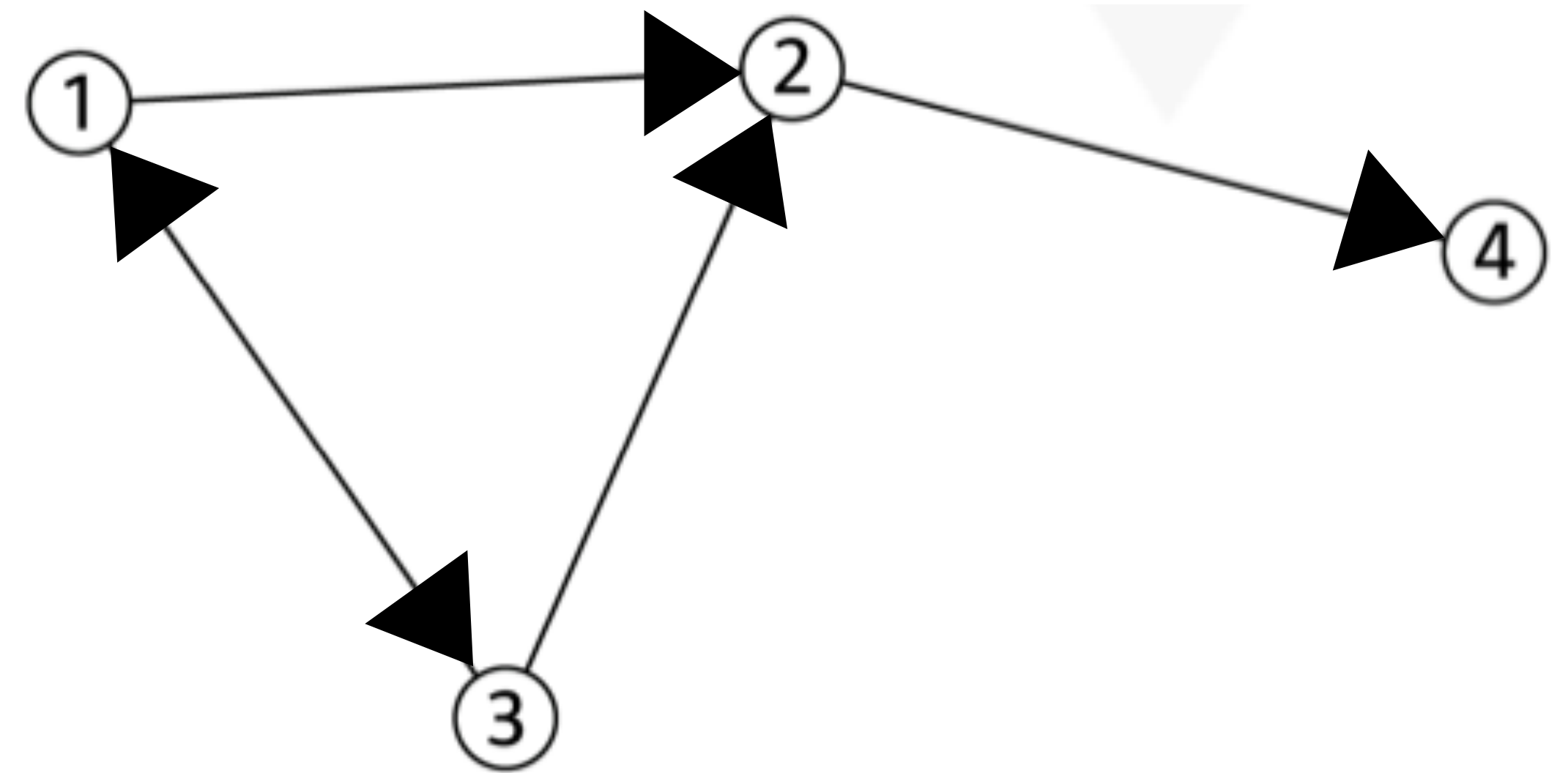


OSMnx calls this  
MultiGraph



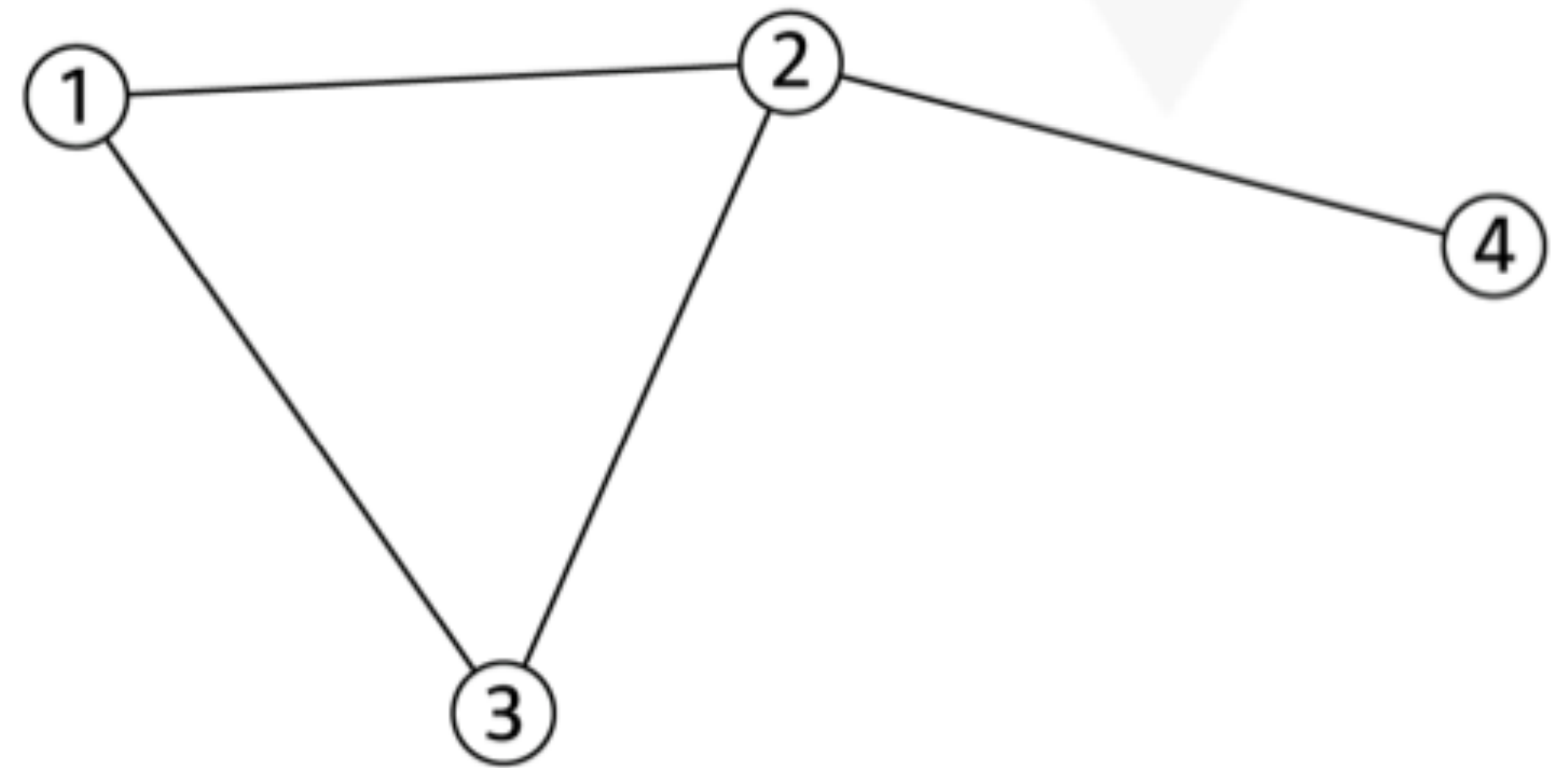
A directed graph (**digraph**) has links with a direction

Nodes             $N=4$   
Links (Arcs)    $L=5$



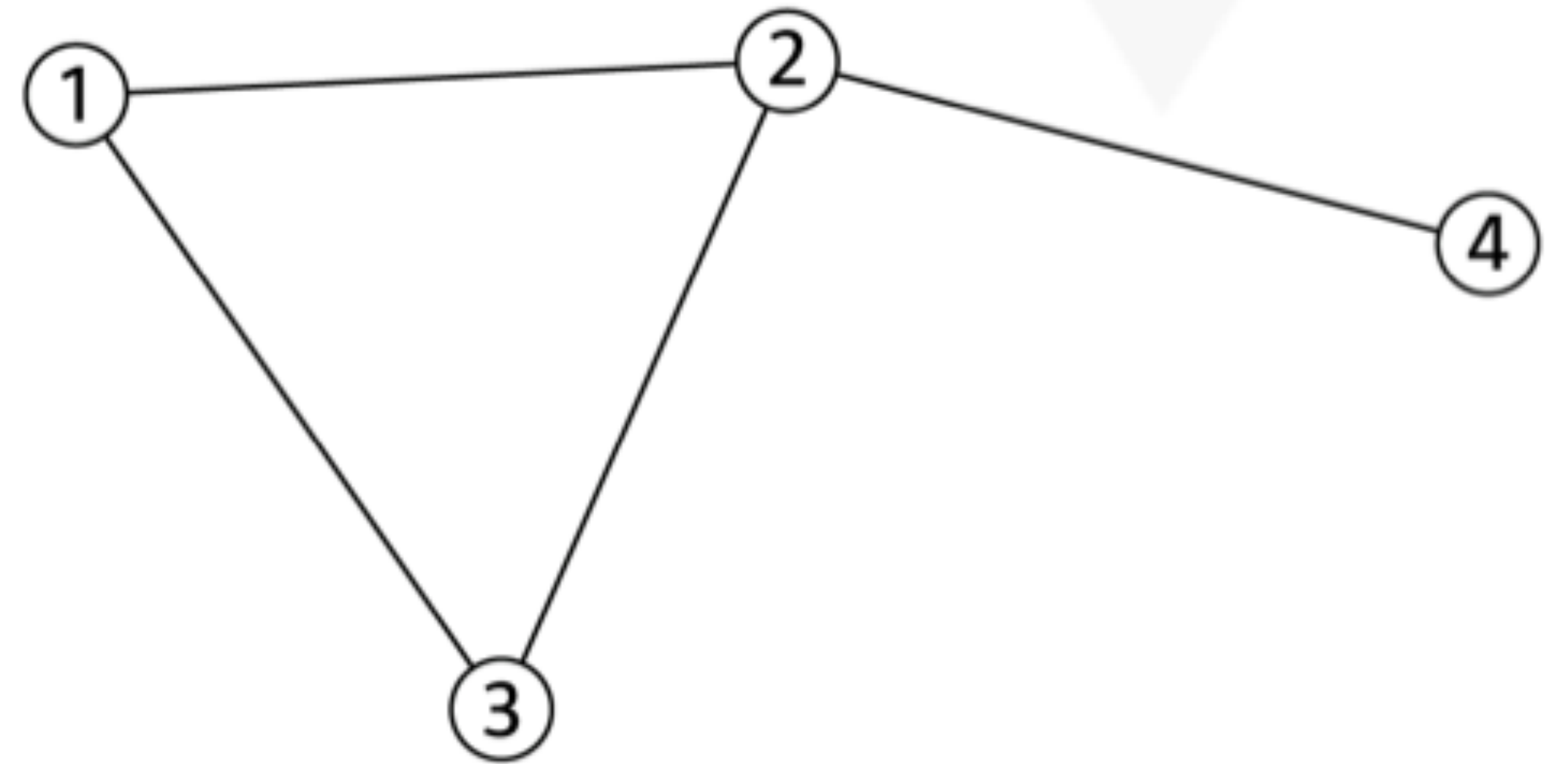
The **degree**  $k_i$  of a node  $i$  is the number of incident links

$$\begin{array}{ll} k_1 = 2 & k_3 = 2 \\ k_2 = 3 & k_4 = 1 \end{array}$$



Every network has an **average degree**  $\langle k \rangle$

$$\begin{array}{ll} k_1 = 2 & k_3 = 2 \\ k_2 = 3 & k_4 = 1 \end{array}$$

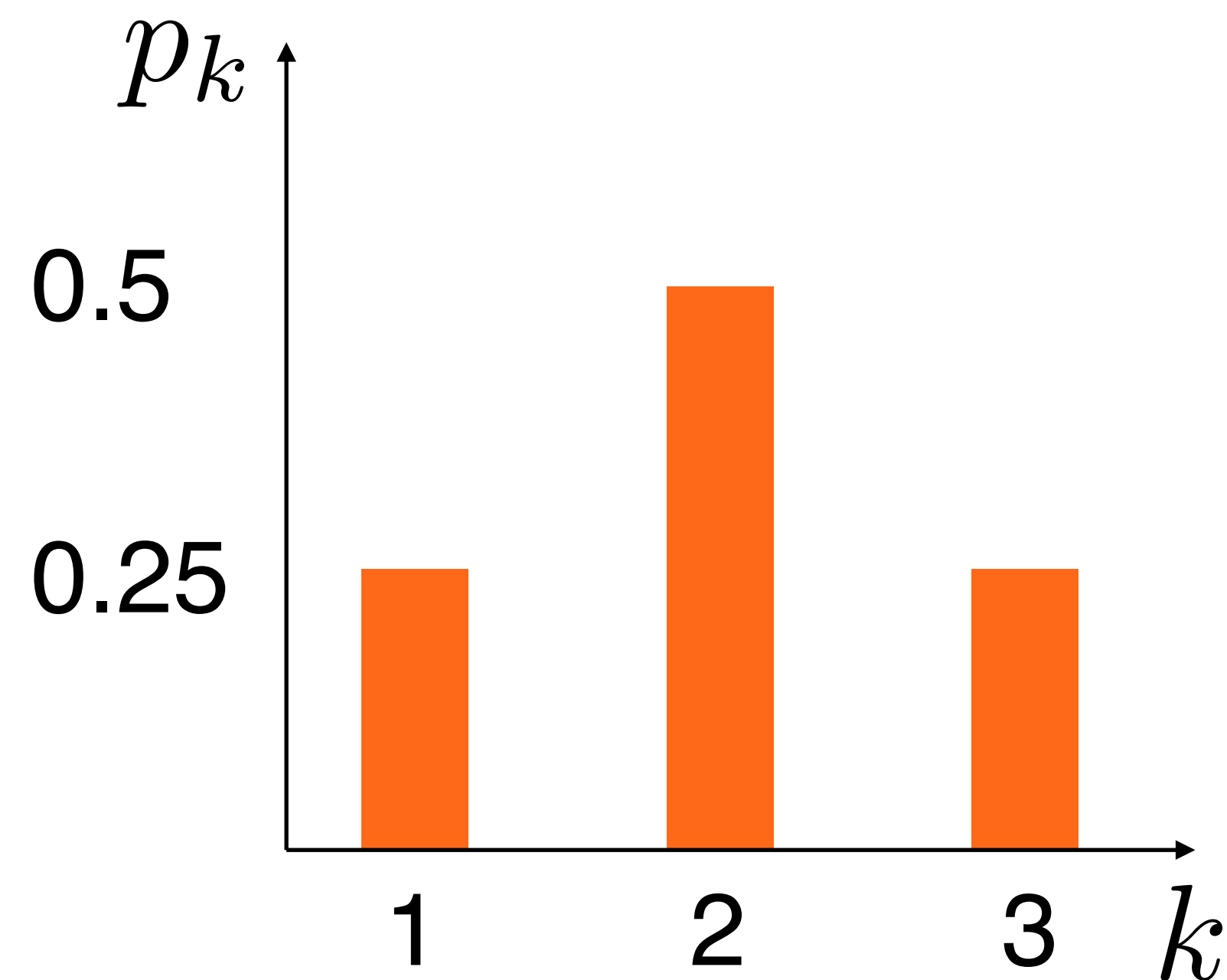
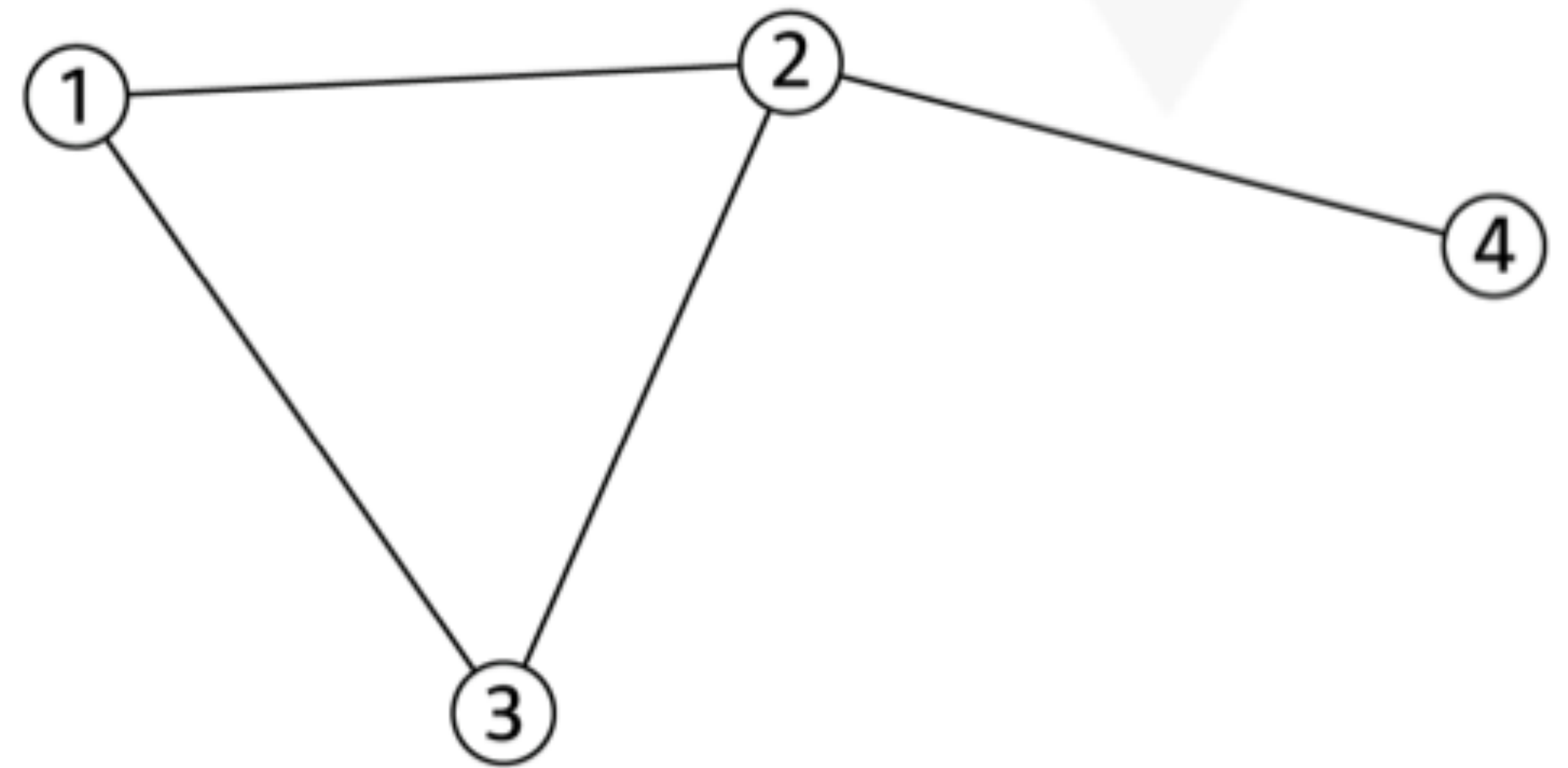


$$\langle k \rangle = \frac{1}{N} \sum_{i=1}^N k_i \quad \langle k \rangle = \frac{2 + 3 + 2 + 1}{4} = 2$$



The **degree distribution**  $p_k = N_k/N$  captures the probabilities that a node has a certain degree

$$\begin{array}{ll} k_1 = 2 & k_3 = 2 \\ k_2 = 3 & k_4 = 1 \end{array}$$



Usually:

A network's degree distribution tells us something fundamental about how individuals in the system connect

Usually:

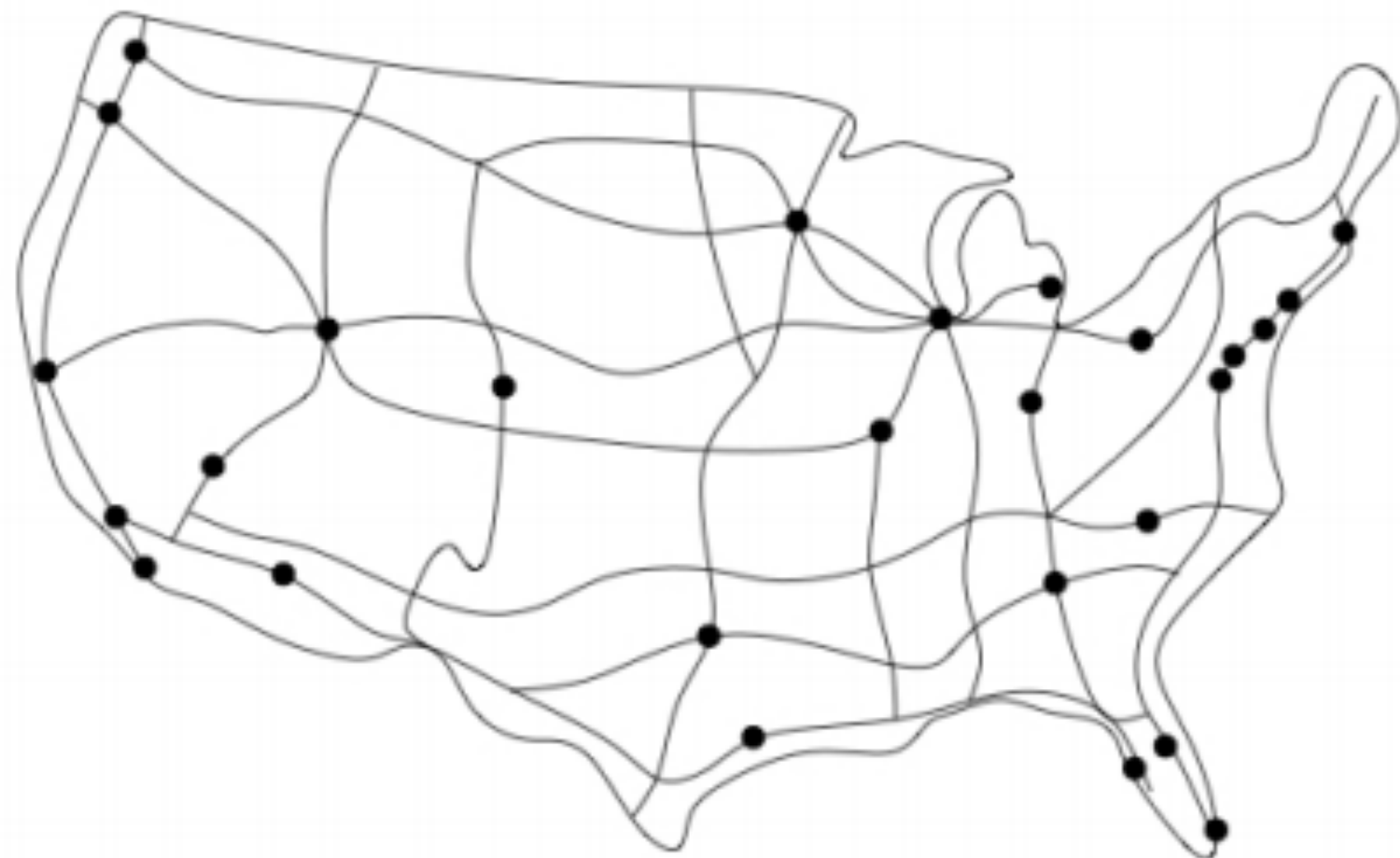
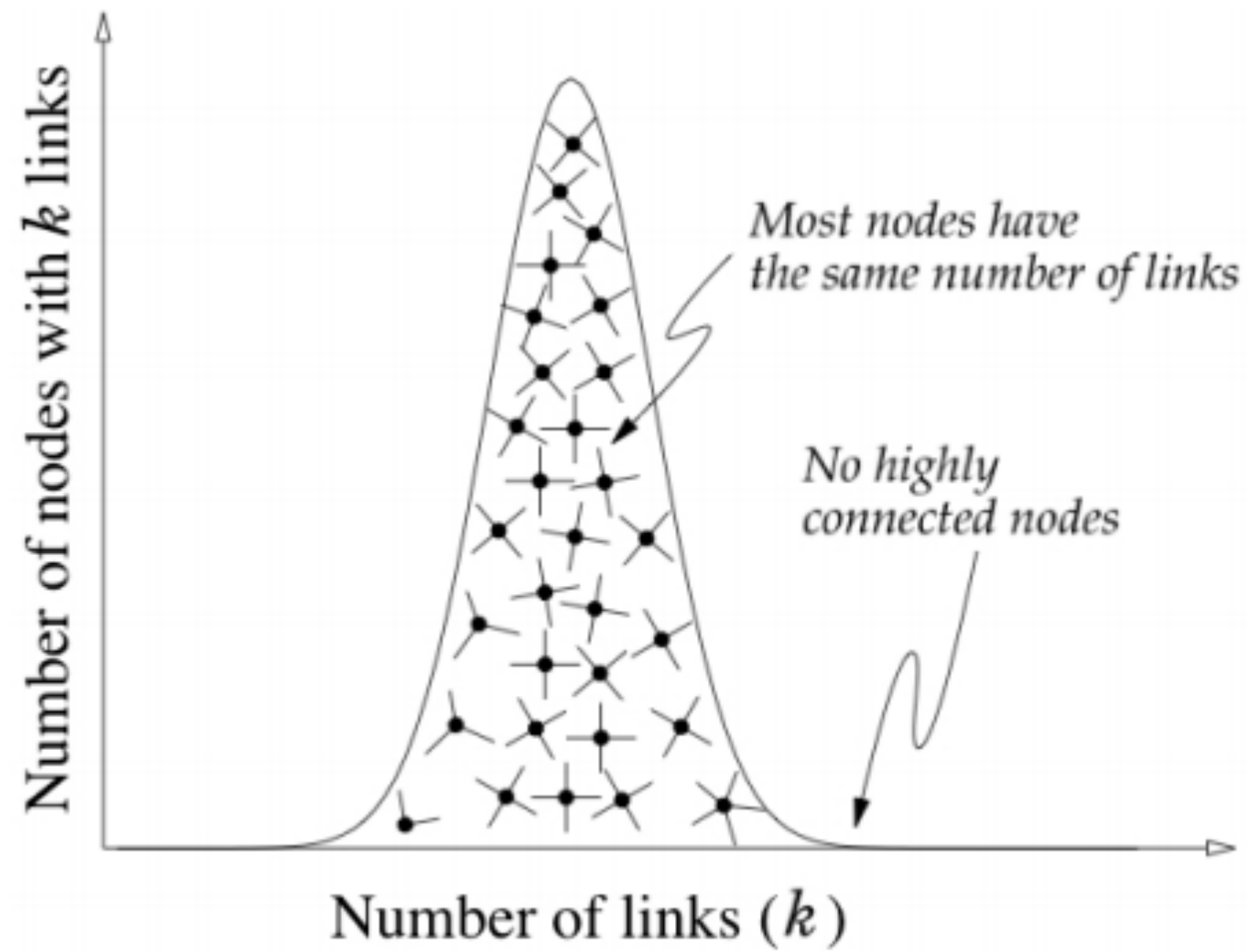
A network's degree distribution tells us something fundamental about how individuals in the system connect

However:

In many spatial networks the degree distribution is far less interesting

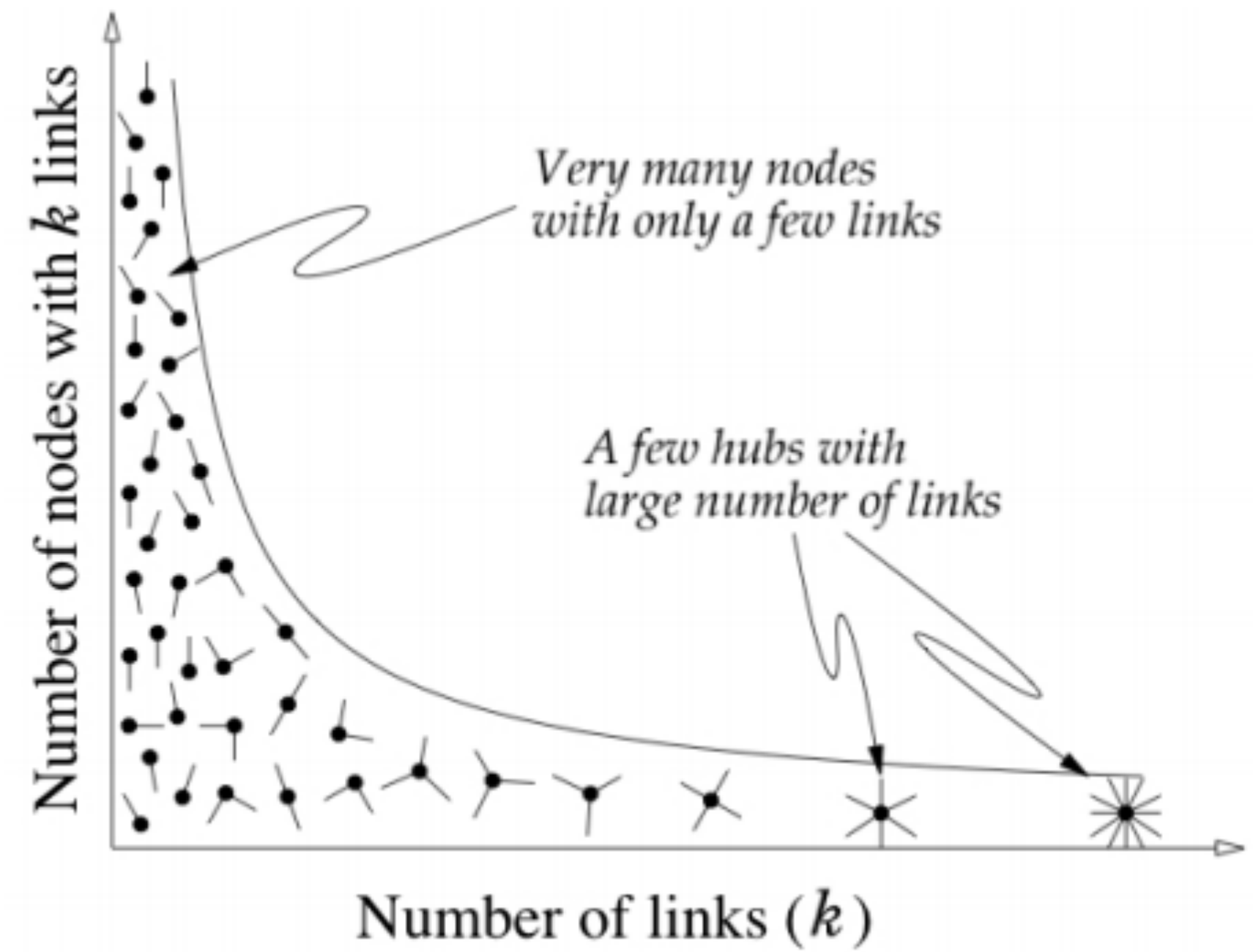


# Thin-tailed network



Streets

# Heavy-tailed network

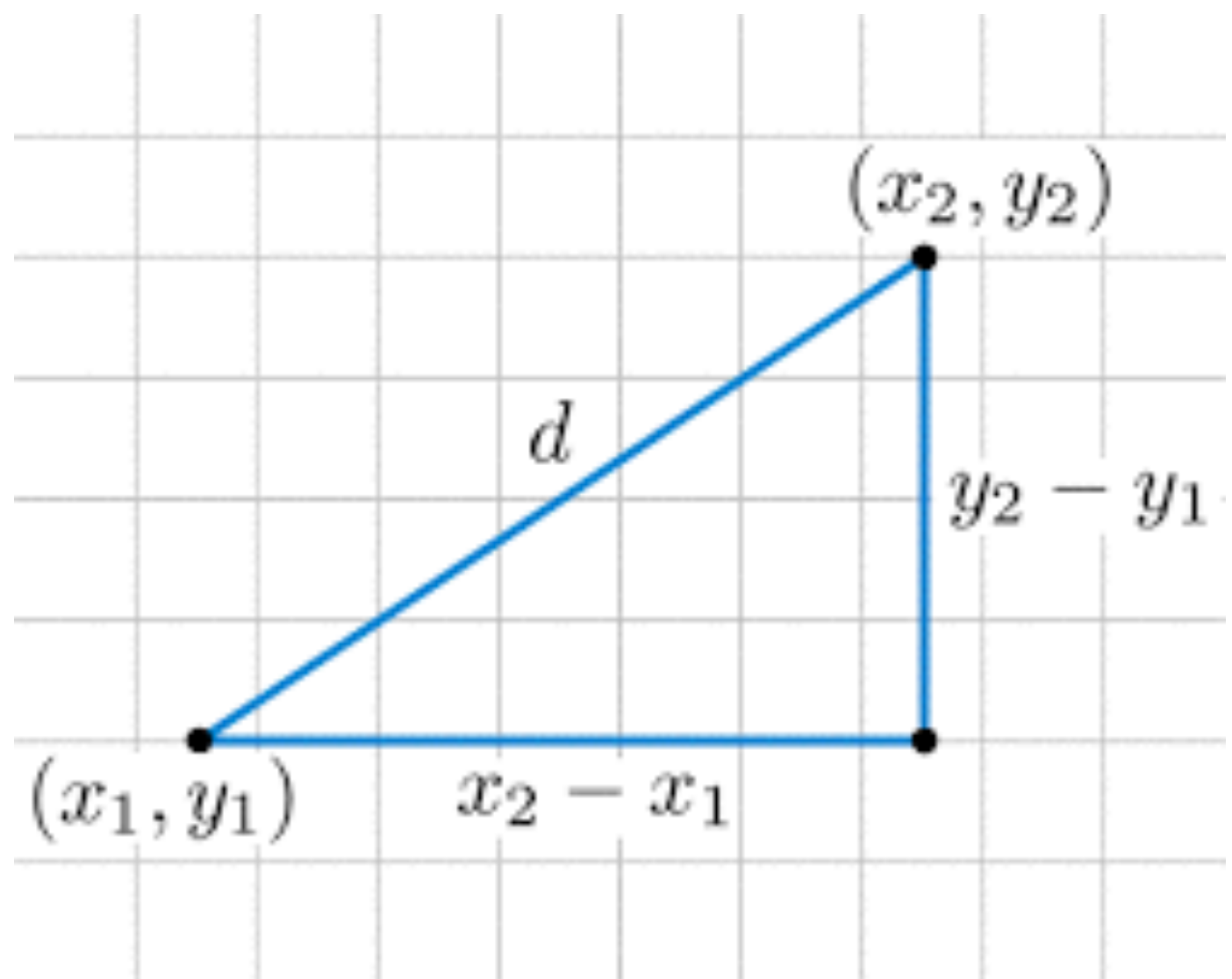


Airlines

A **spatial network** has nodes in a metric space

You can measure a distance between nodes

In practice: 2D-space with **Euclidian distance**

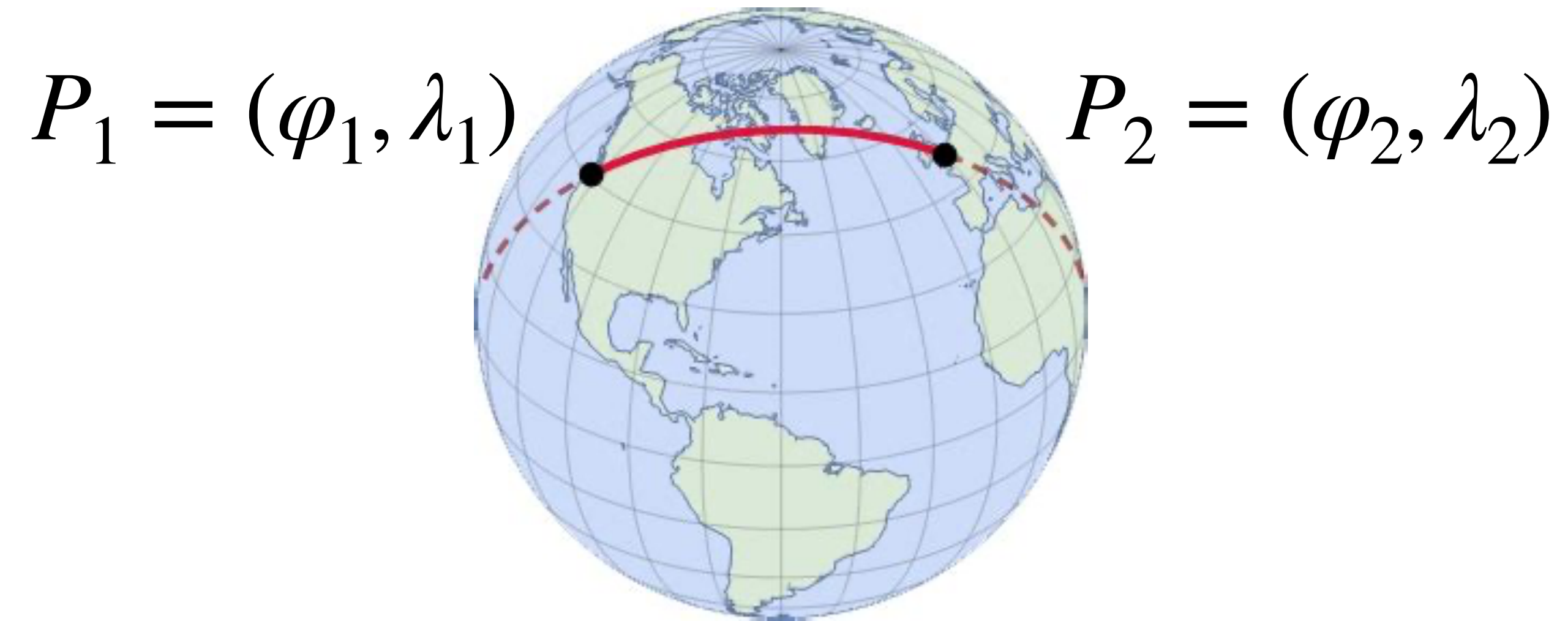


$$d(P_1, P_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

A **spatial network** has nodes in a metric space

You can measure a distance between nodes

For larger distances on a sphere use the **Haversine distance**

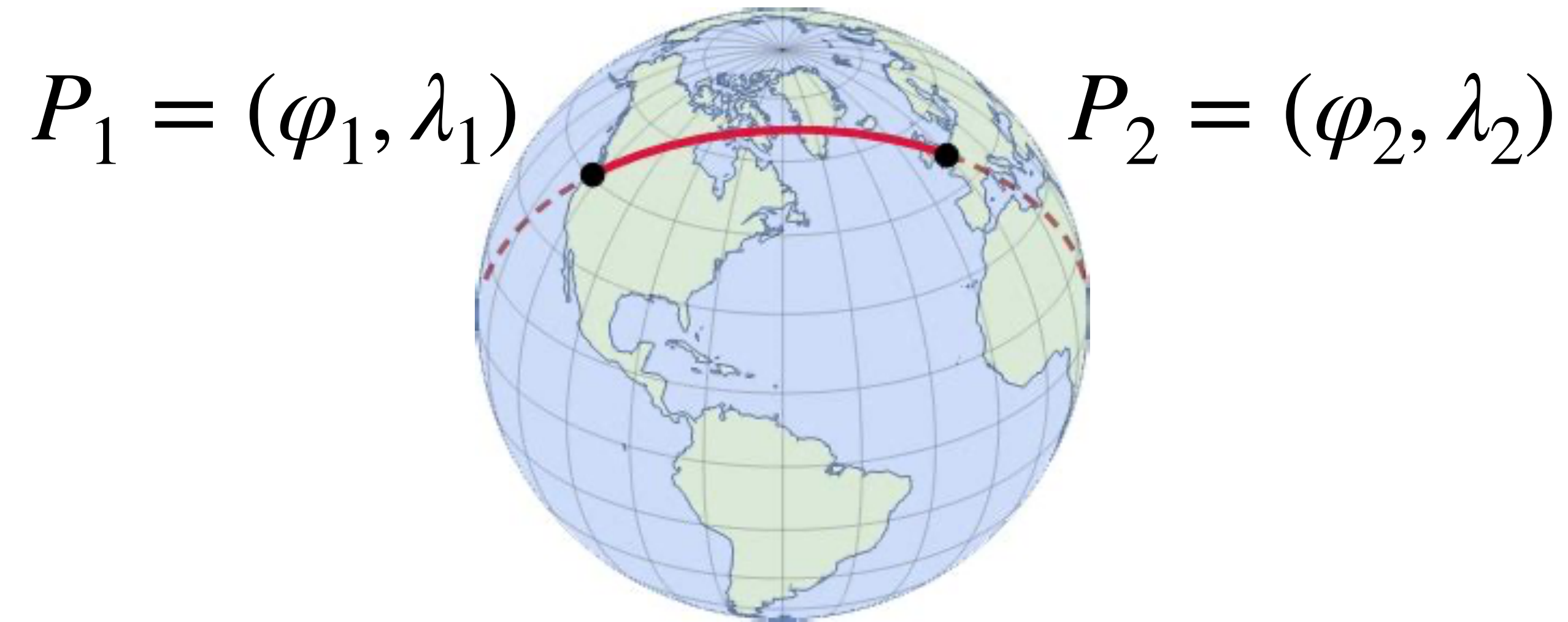




A **spatial network** has nodes in a metric space

You can measure a distance between nodes

For larger distances on a sphere use the **Haversine distance**



$$d = 2r \arcsin \left( \sqrt{\sin^2 \left( \frac{\varphi_2 - \varphi_1}{2} \right) + \cos \varphi_1 \cdot \cos \varphi_2 \cdot \sin^2 \left( \frac{\lambda_2 - \lambda_1}{2} \right)} \right)$$



# Planarity

**Three utilities problem:** Can you link each house to each utility?



**Three utilities problem:** Can you link each house to each utility?

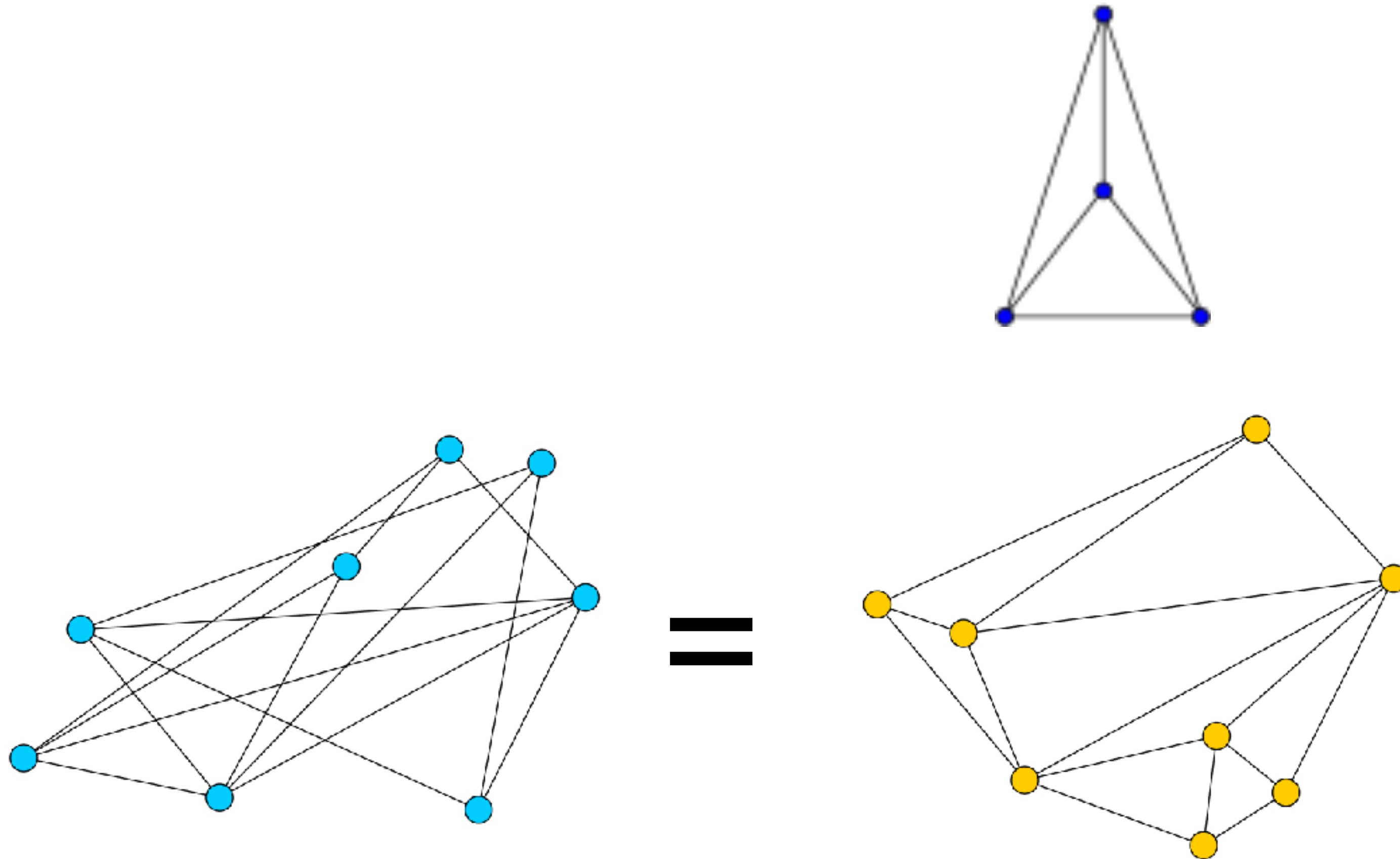


100+ years old, was solved in 1930 by Kuratowski: **No**



A graph is **planar** if it can be embedded in the plane

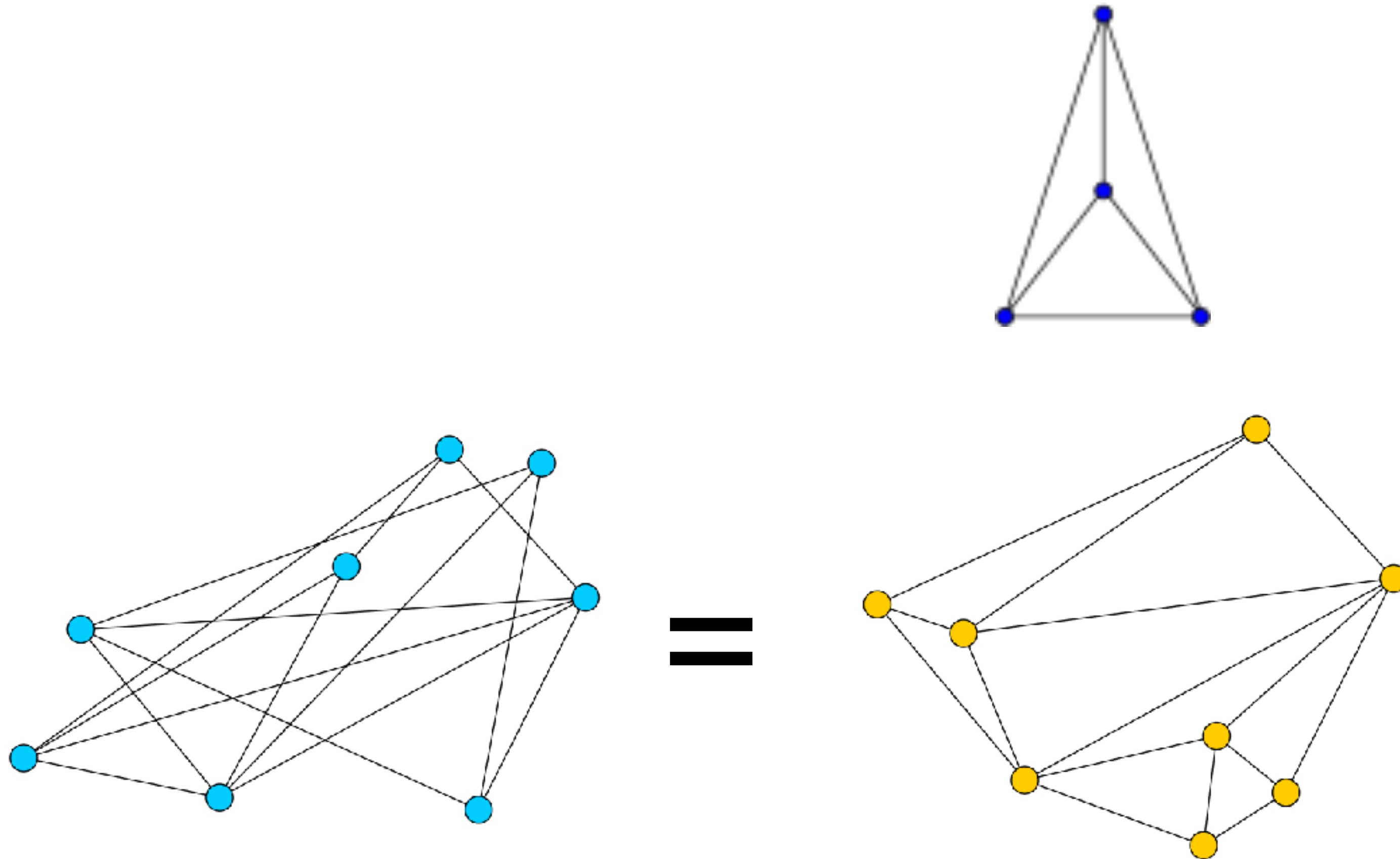
Planar: You can move the nodes around until no links cross





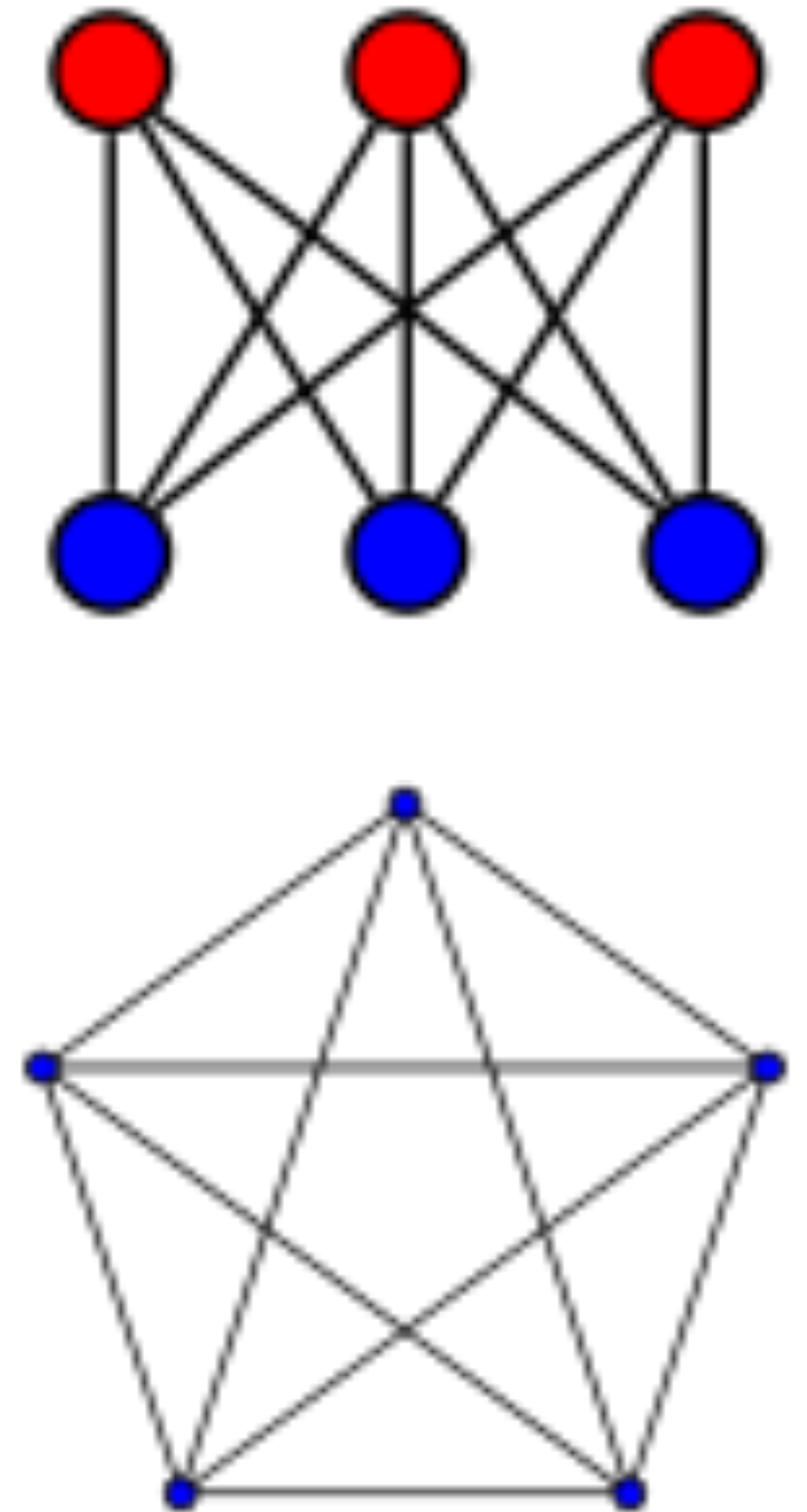
A graph is **planar** if it can be embedded in the plane

Planar: You can move the nodes around until no links cross



Not planar: you can't

Utility graph

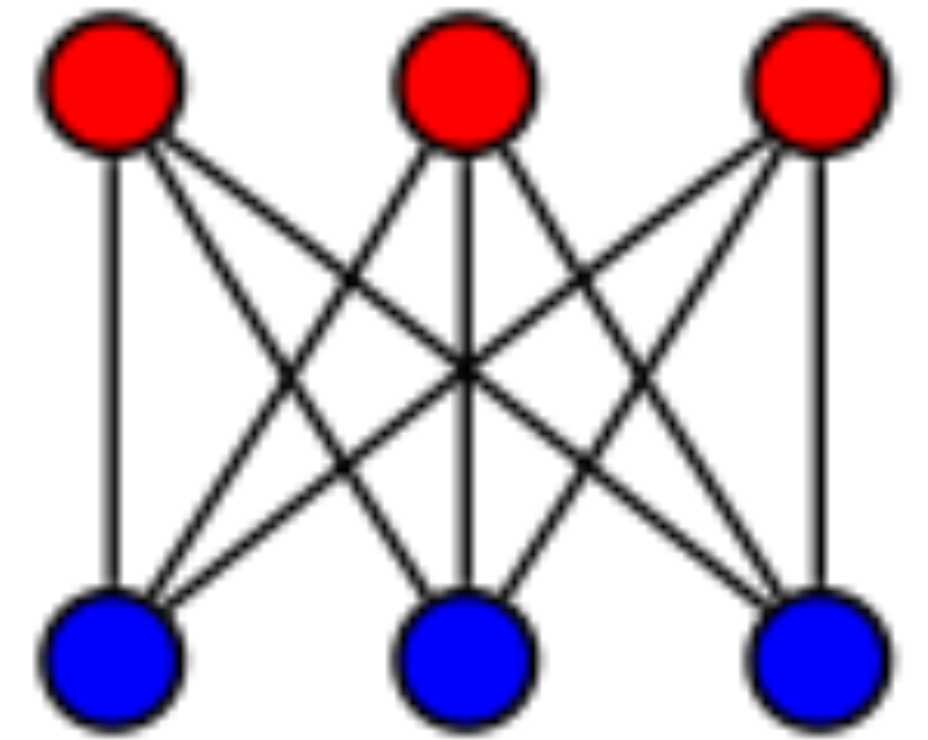


# Kuratowski's theorem gives an easy way to check planarity

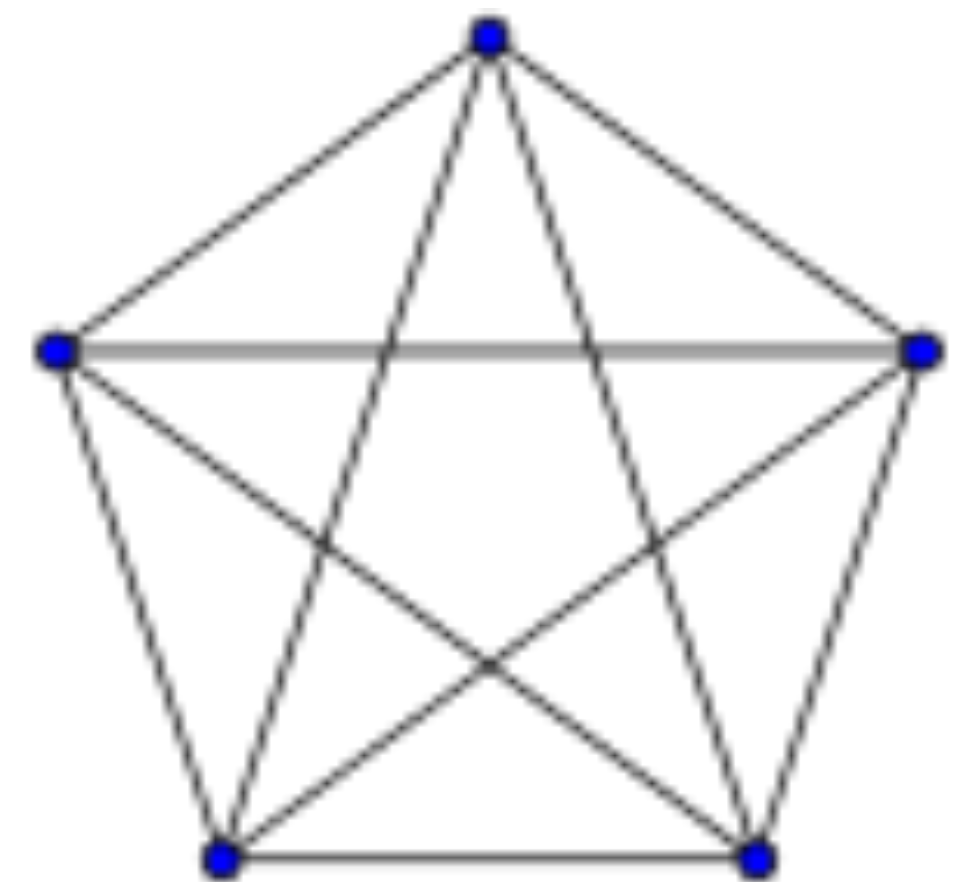
## Kuratowski's theorem

A graph is planar if and only if it does not contain (a subgraph that is a subdivision of)  $K_5$  or  $K_{3,3}$

$K_{3,3}$



$K_5$



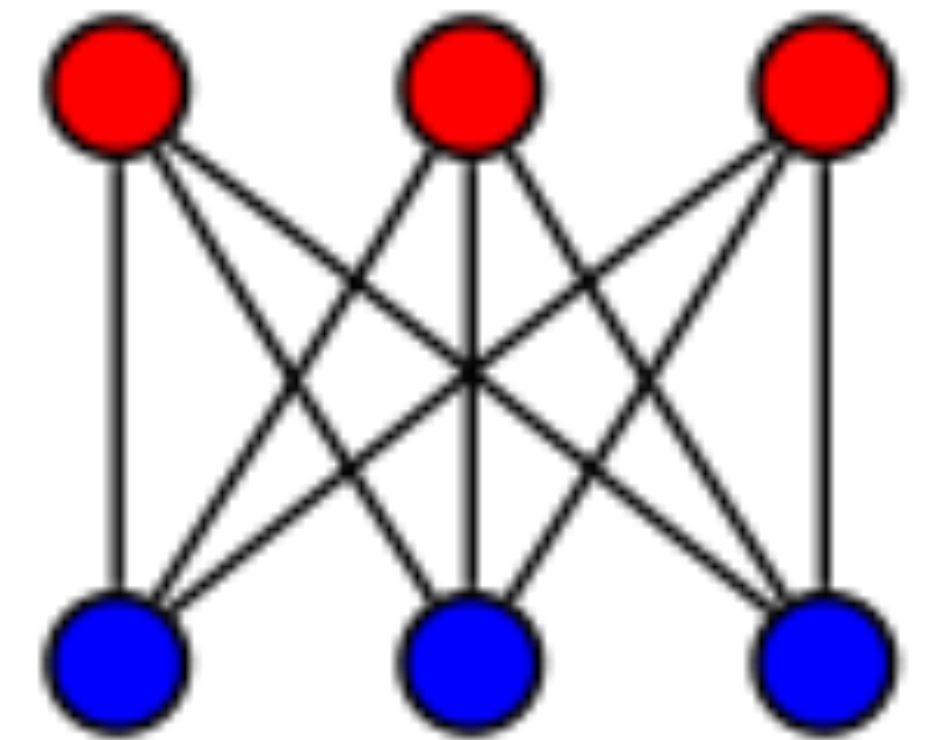
# Kuratowski's theorem gives an easy way to check planarity

## Kuratowski's theorem

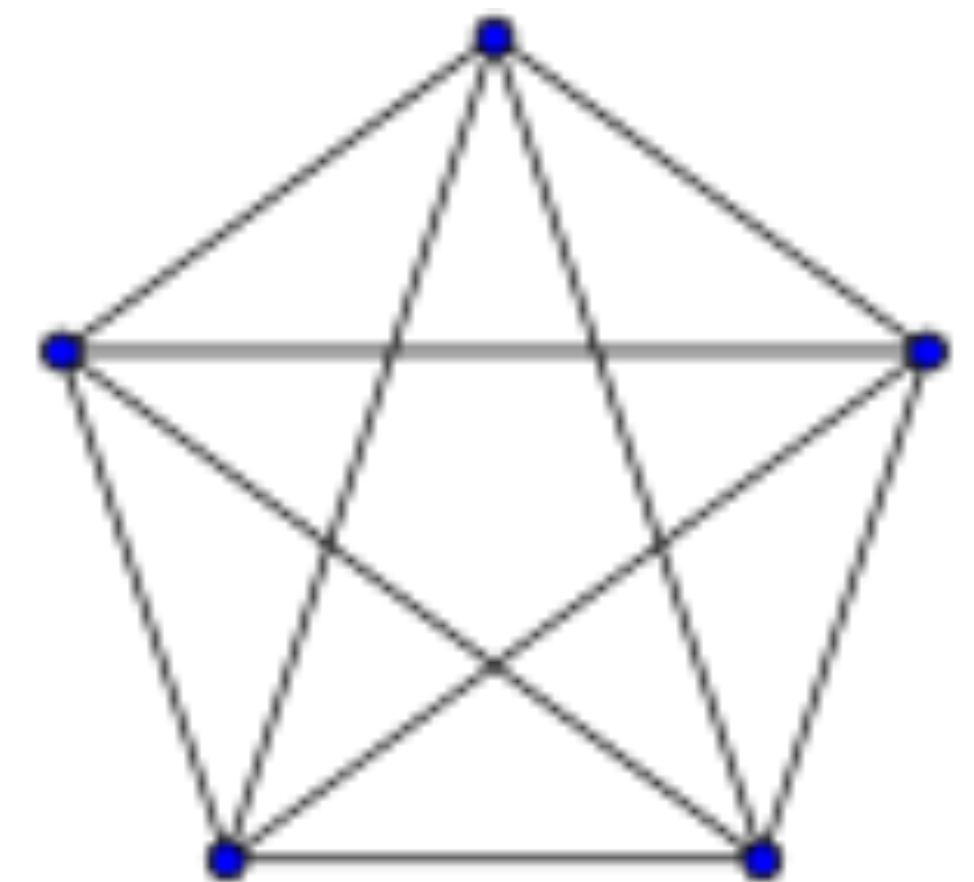
A graph is planar if and only if it does not contain (a subgraph that is a subdivision of)  $K_5$  or  $K_{3,3}$

Allows testing planarity in linear time.

$K_{3,3}$



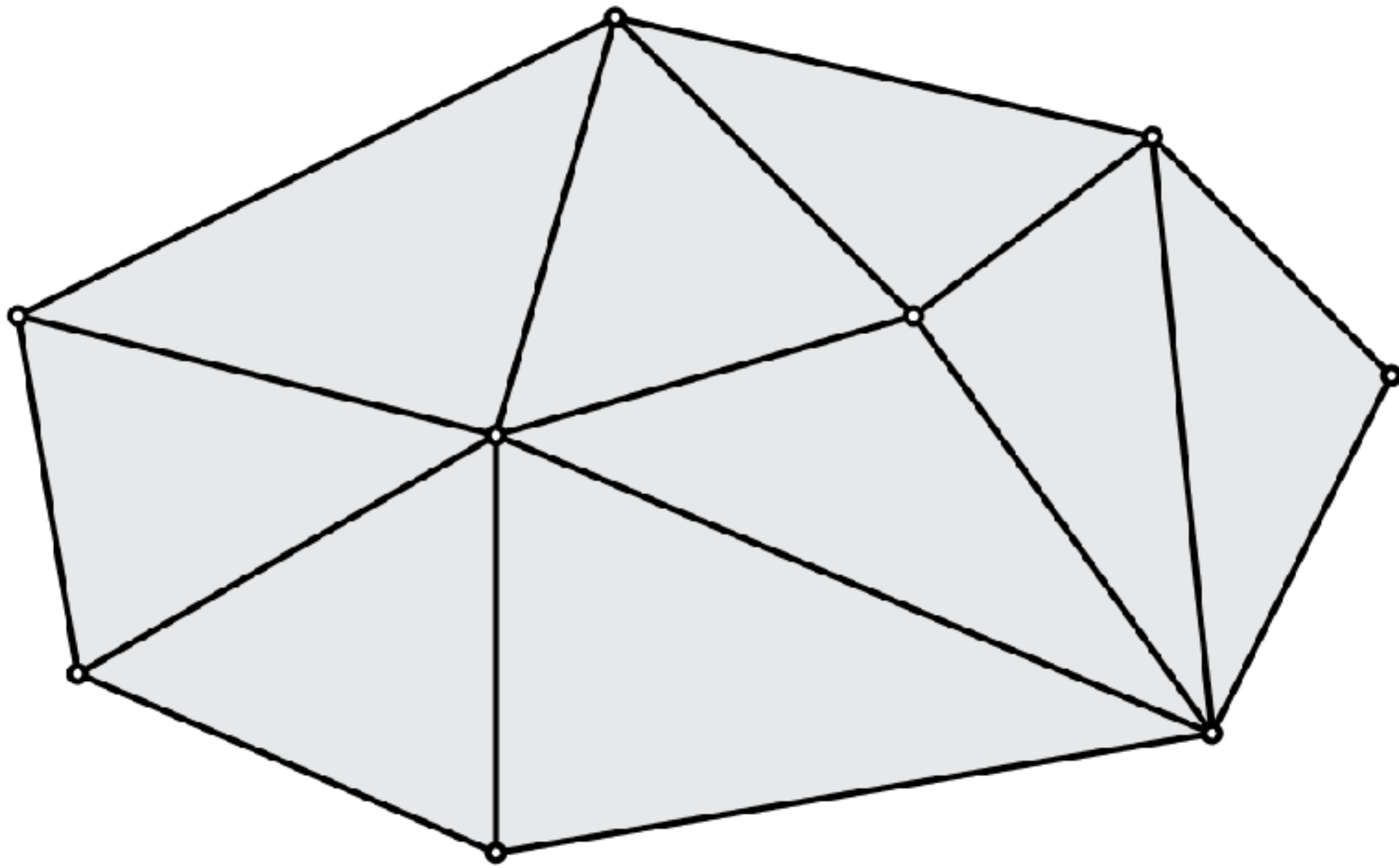
$K_5$





# Optimal networks

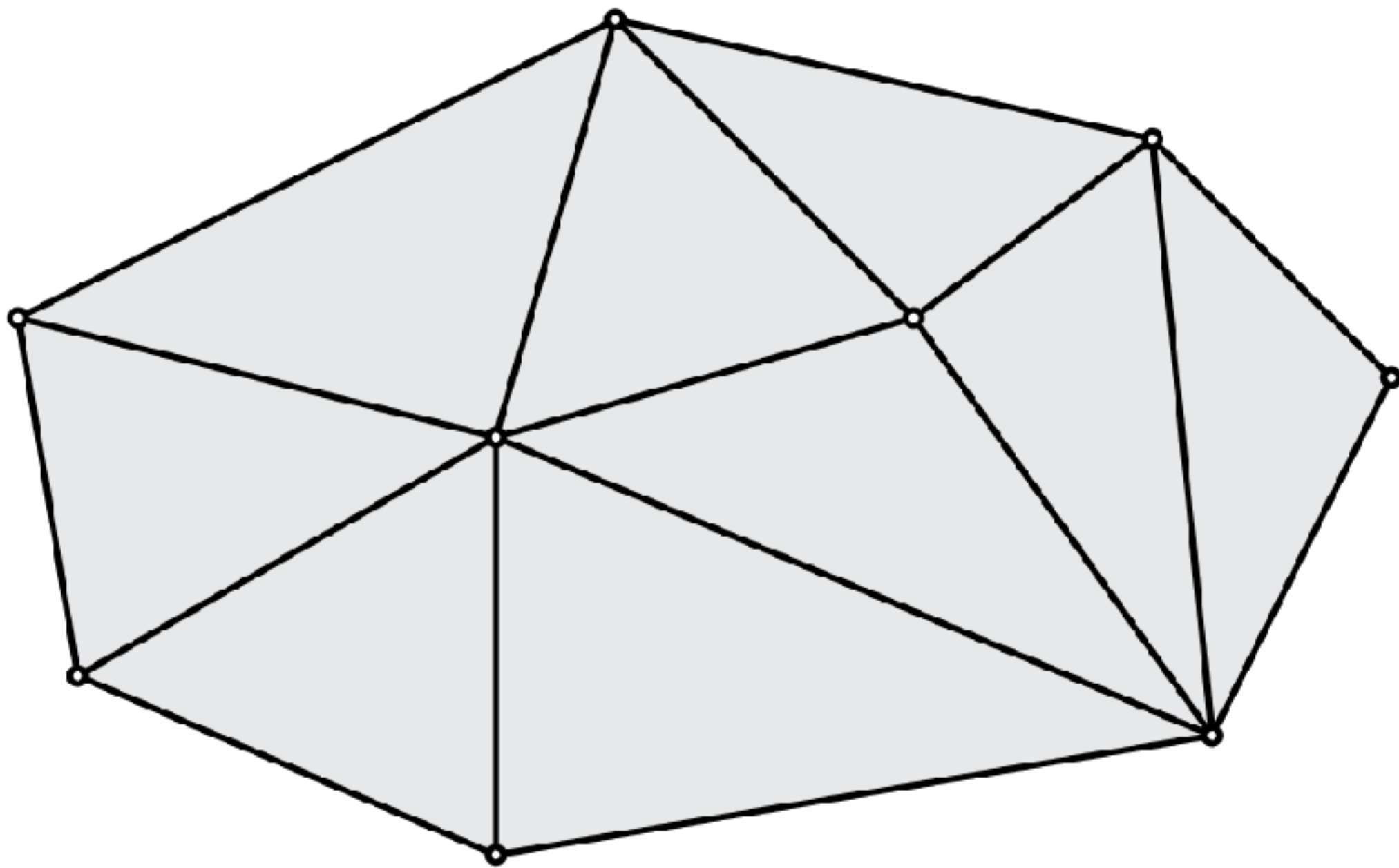
A **triangulation** of 2D points is a maximal set of non-crossing edges



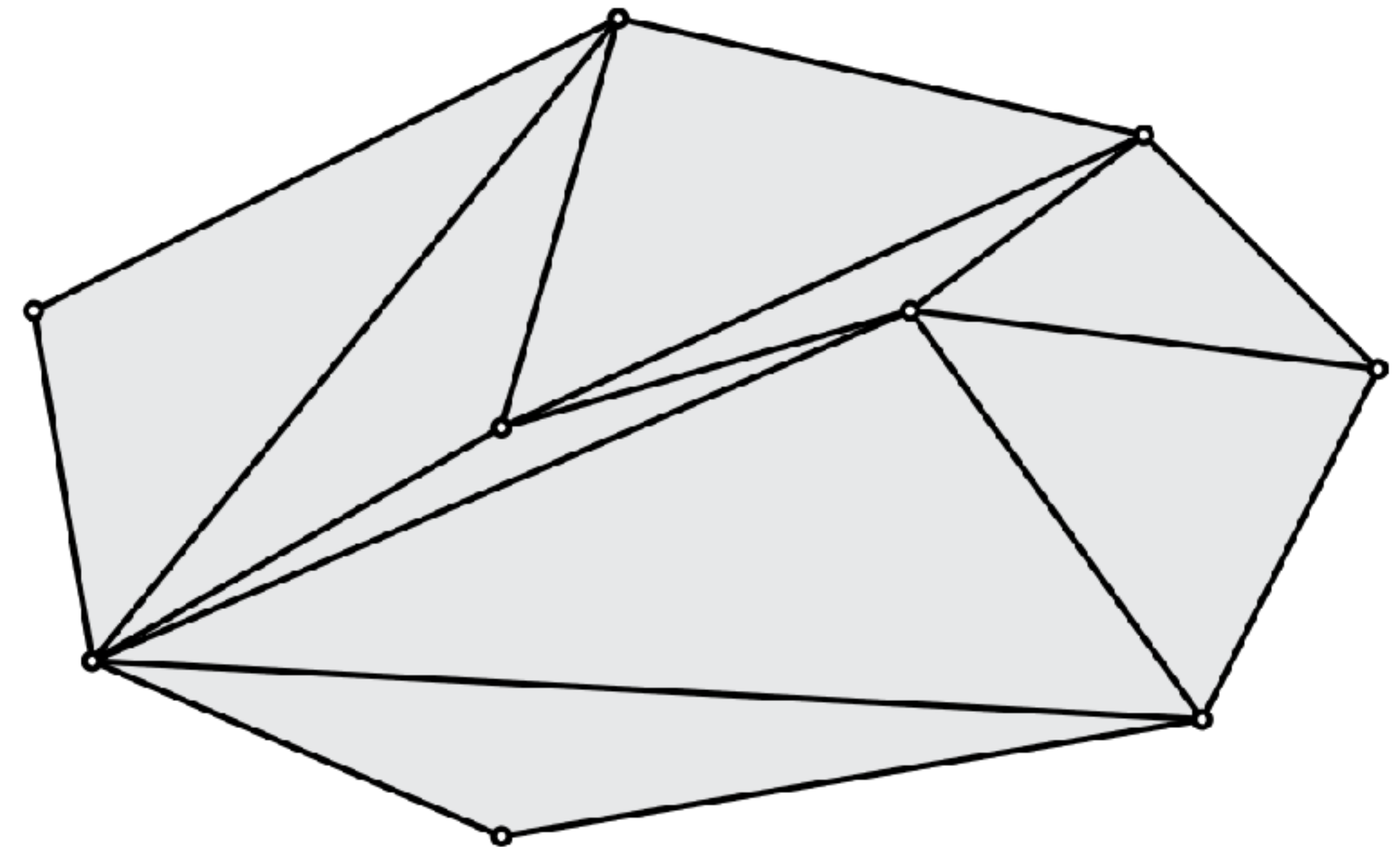
It is planar



A **triangulation** of 2D points is a maximal set of non-crossing edges



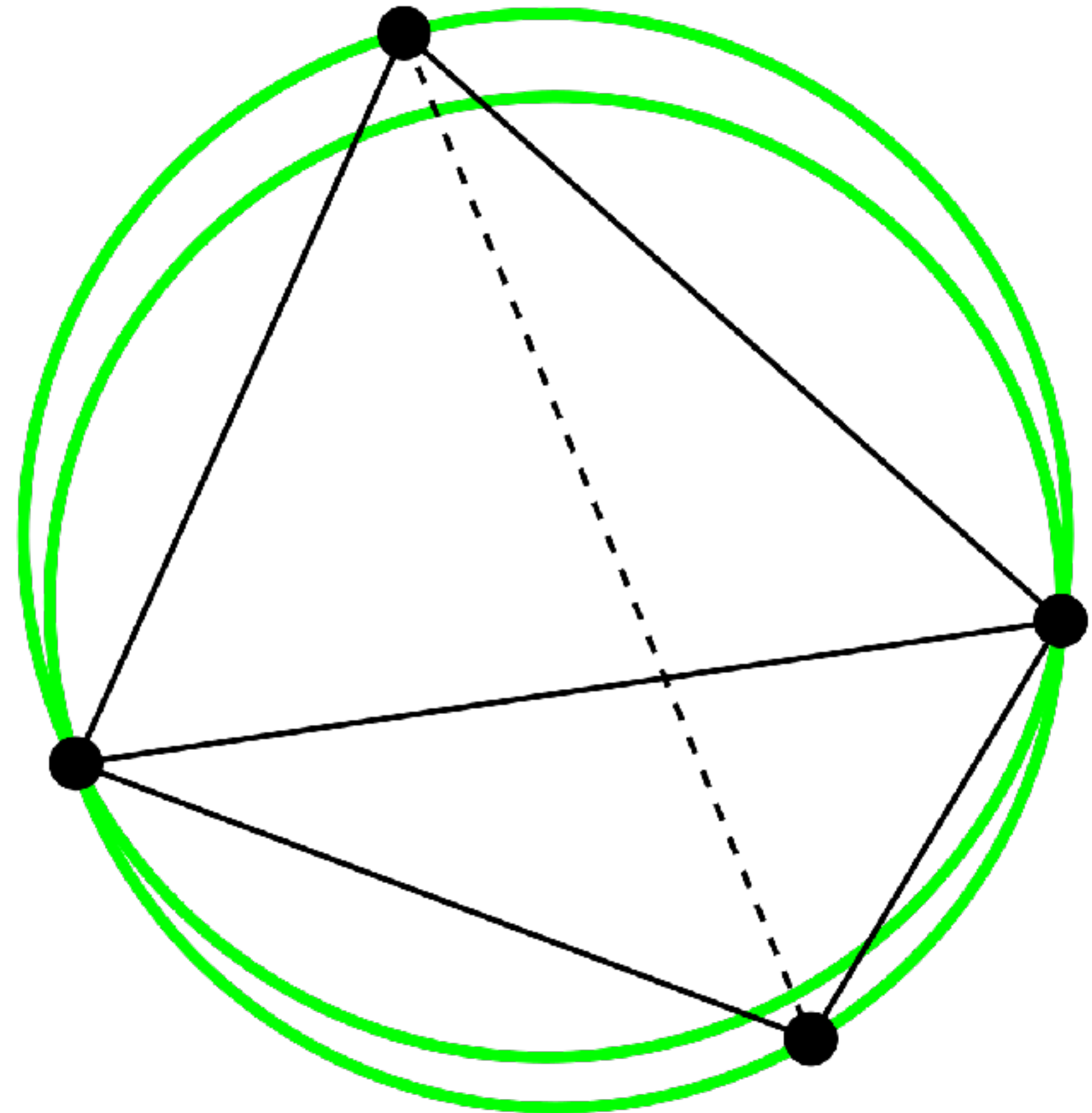
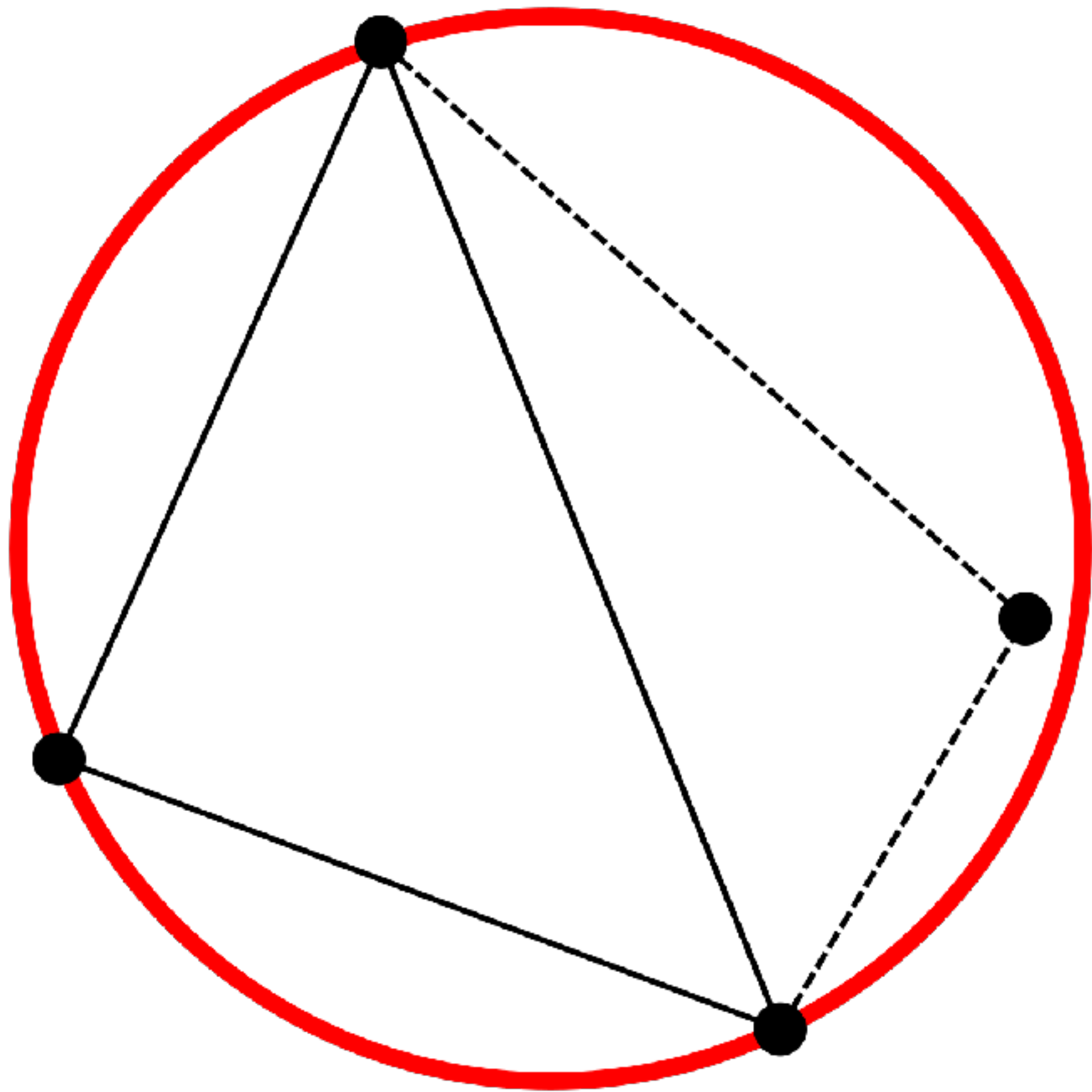
It is planar



It is not unique

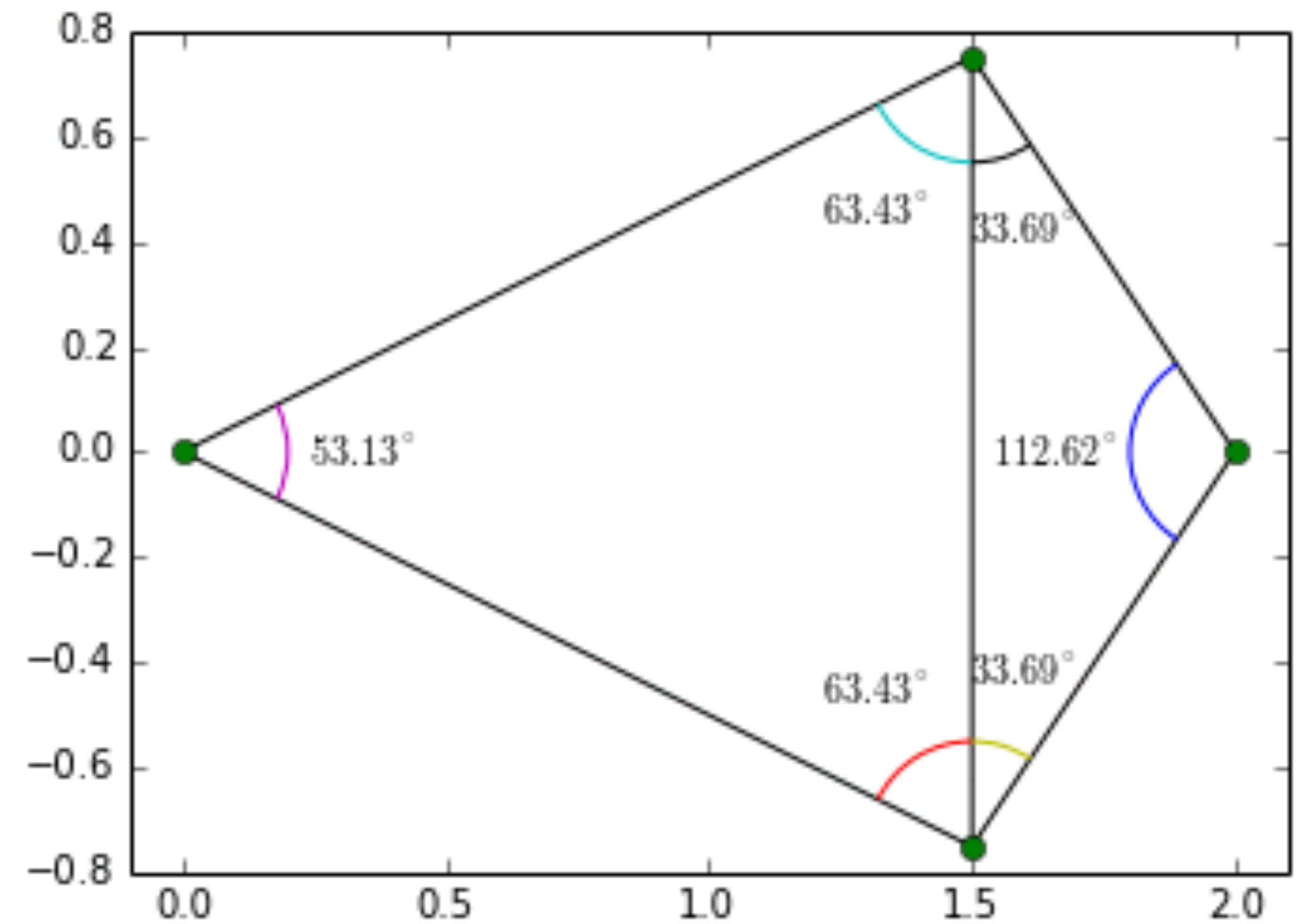
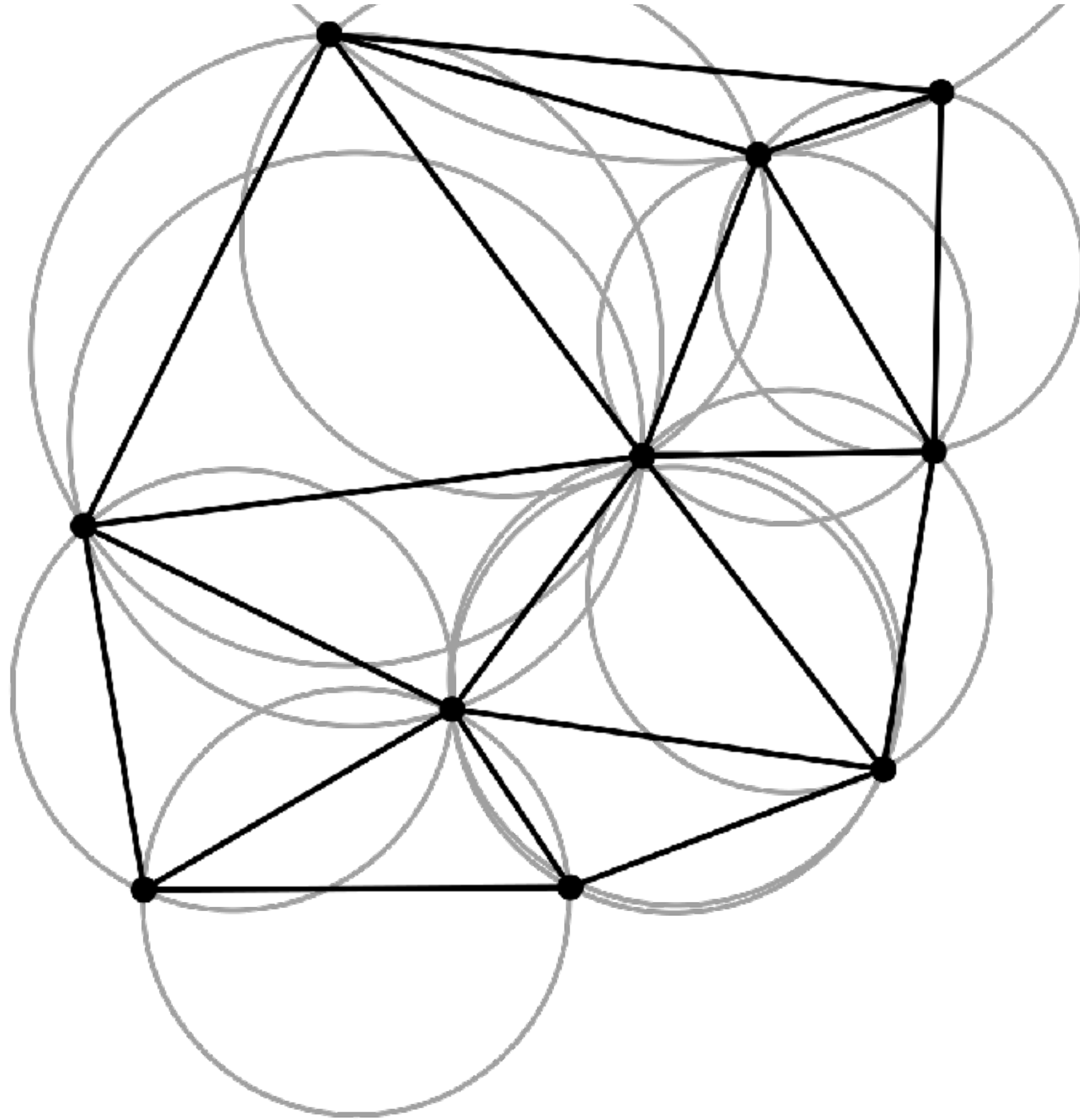
A **Delaunay triangulation** maximizes the minimum of all angles

No point can be inside the circumcircle of any triangle



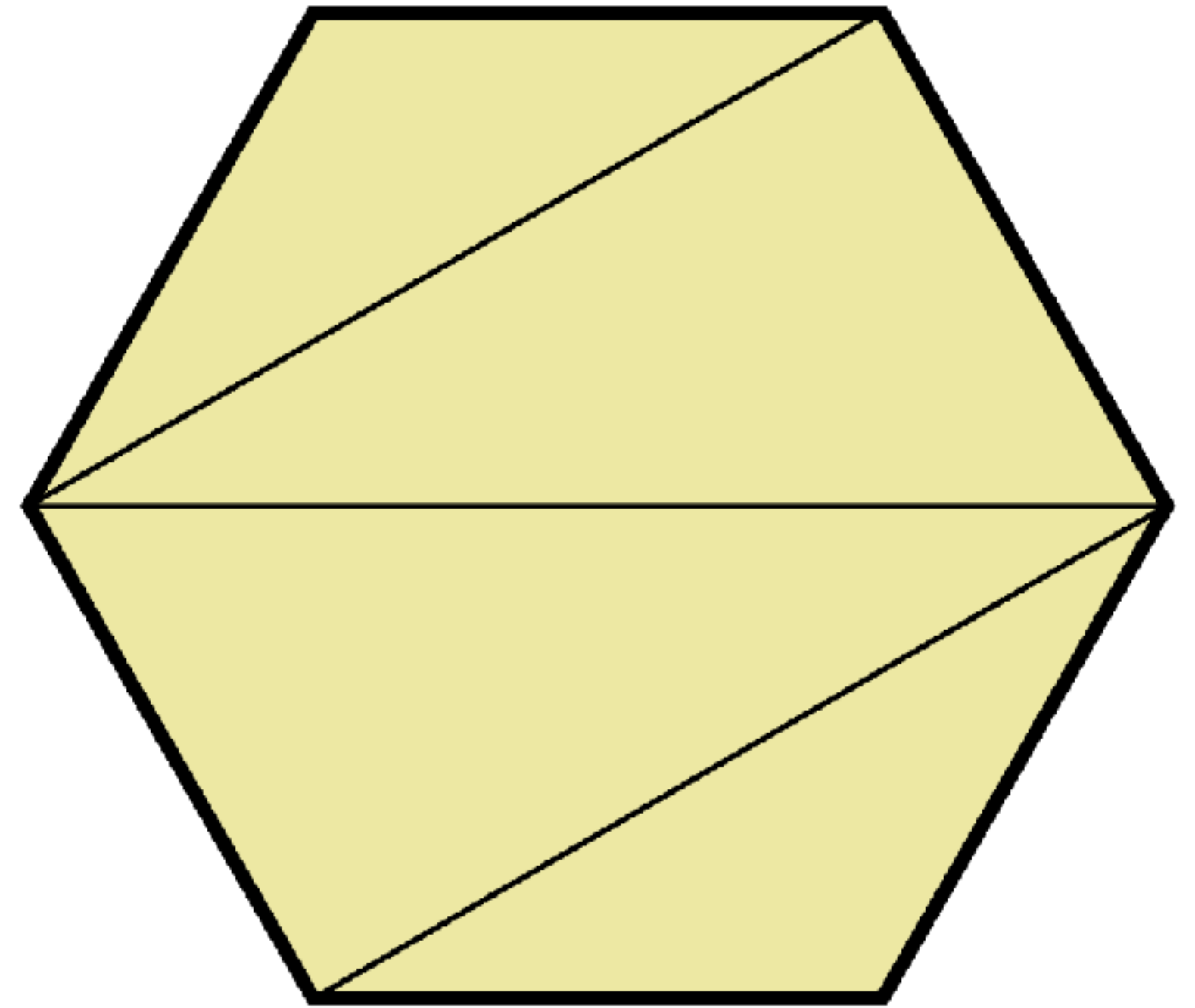
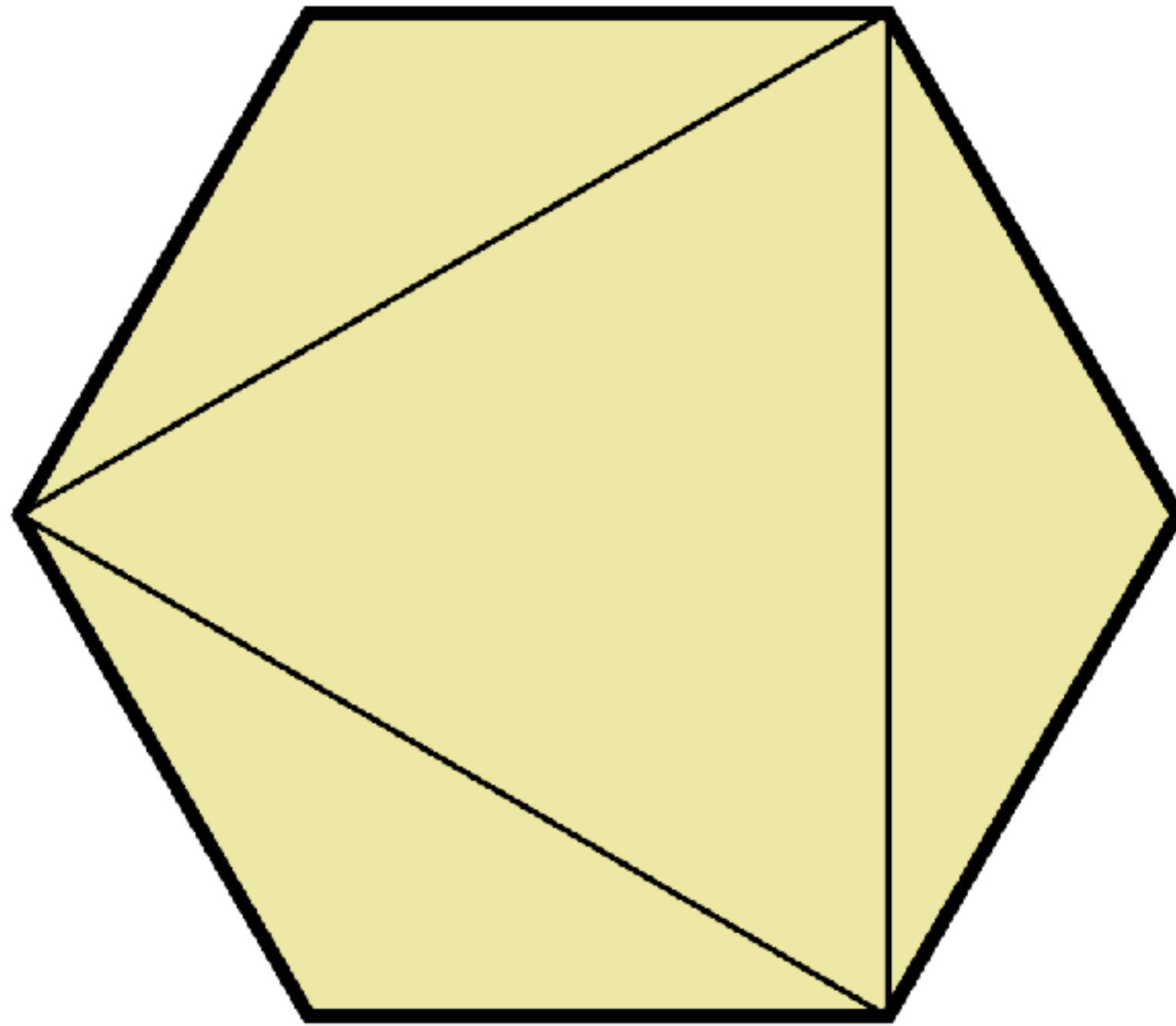
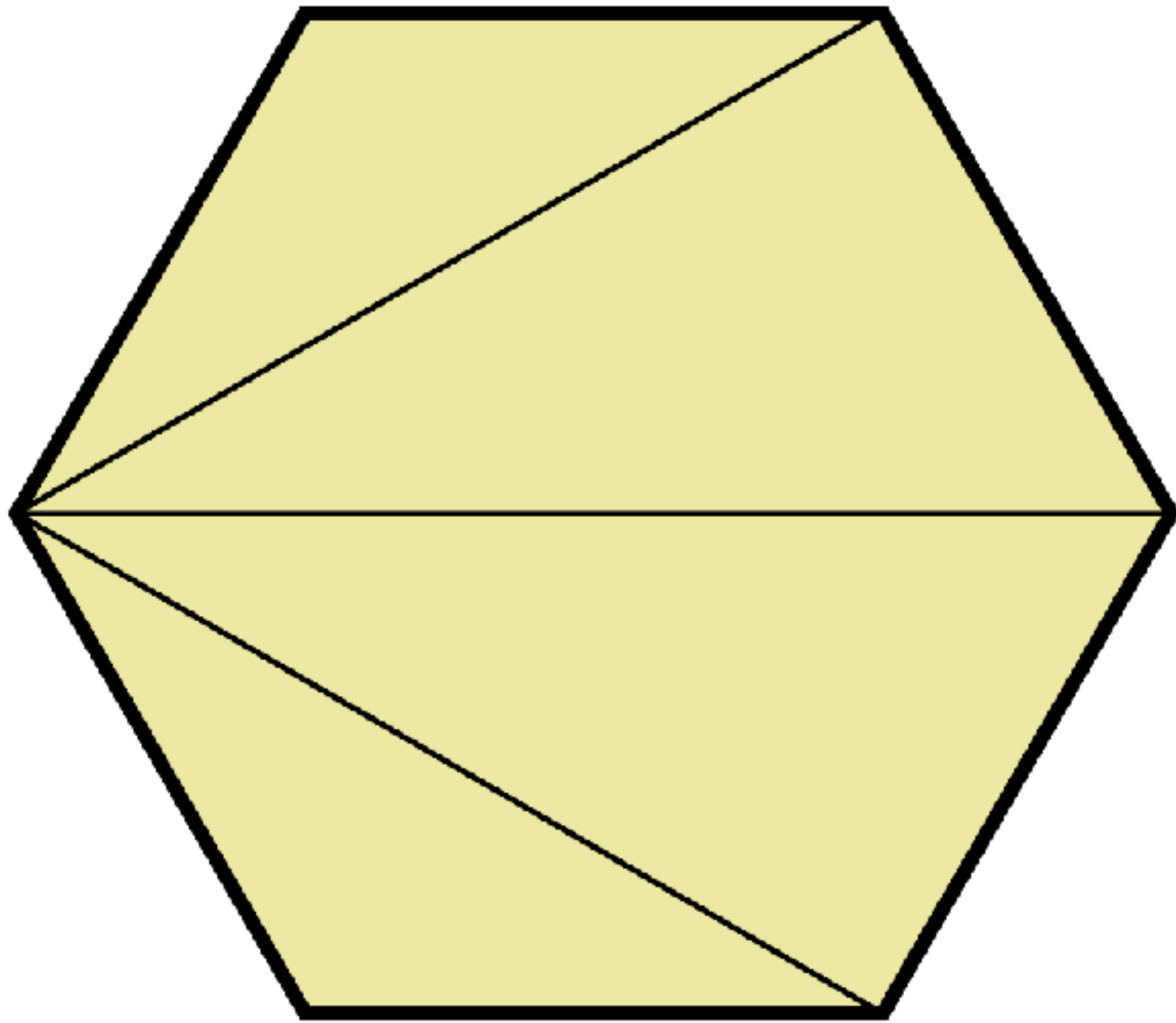
A **Delaunay triangulation** maximizes the minimum of all angles

No point can be inside the circumcircle of any triangle





A **minimum weight triangulation** minimizes sum of edge lengths

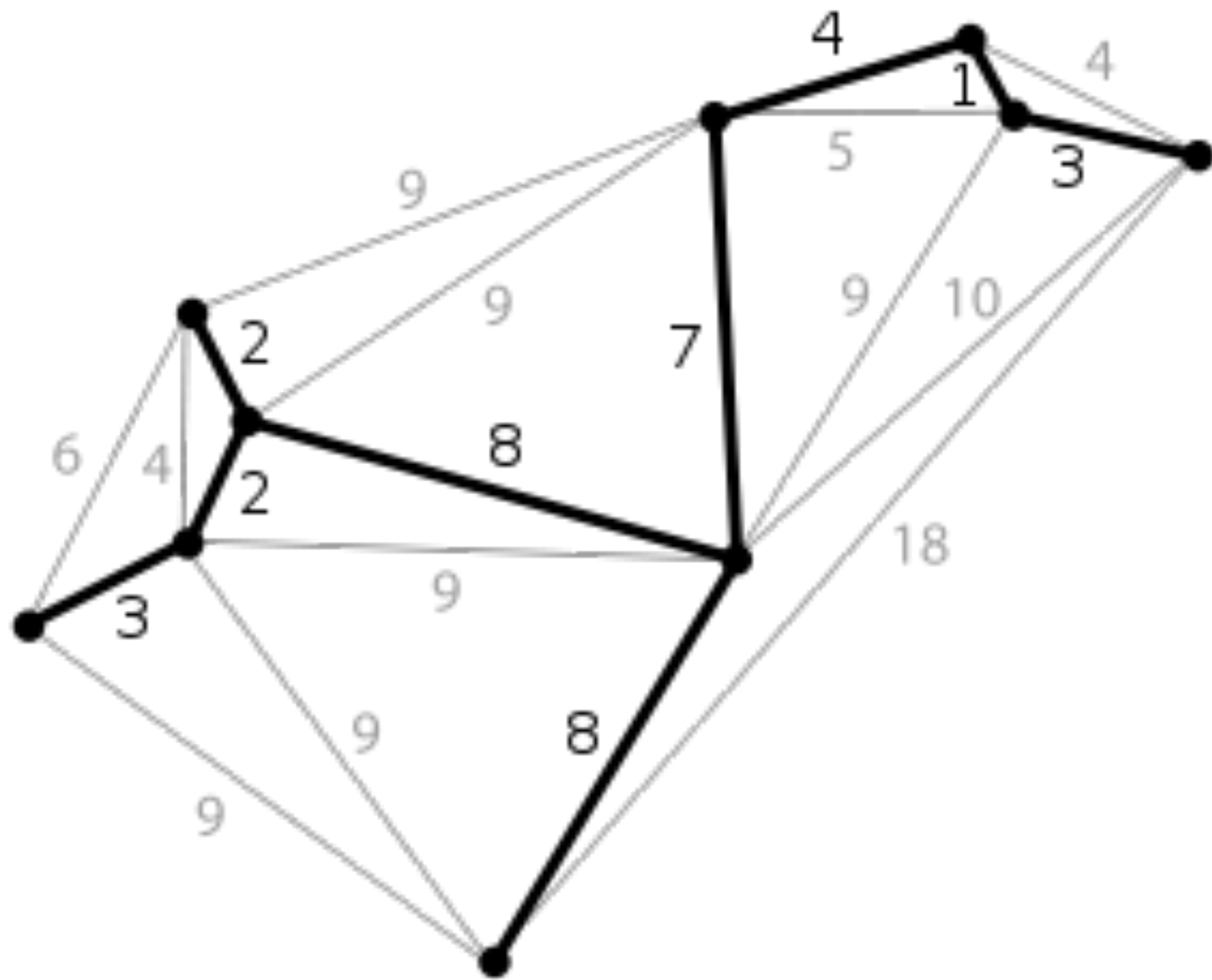




# Many other triangulations exist

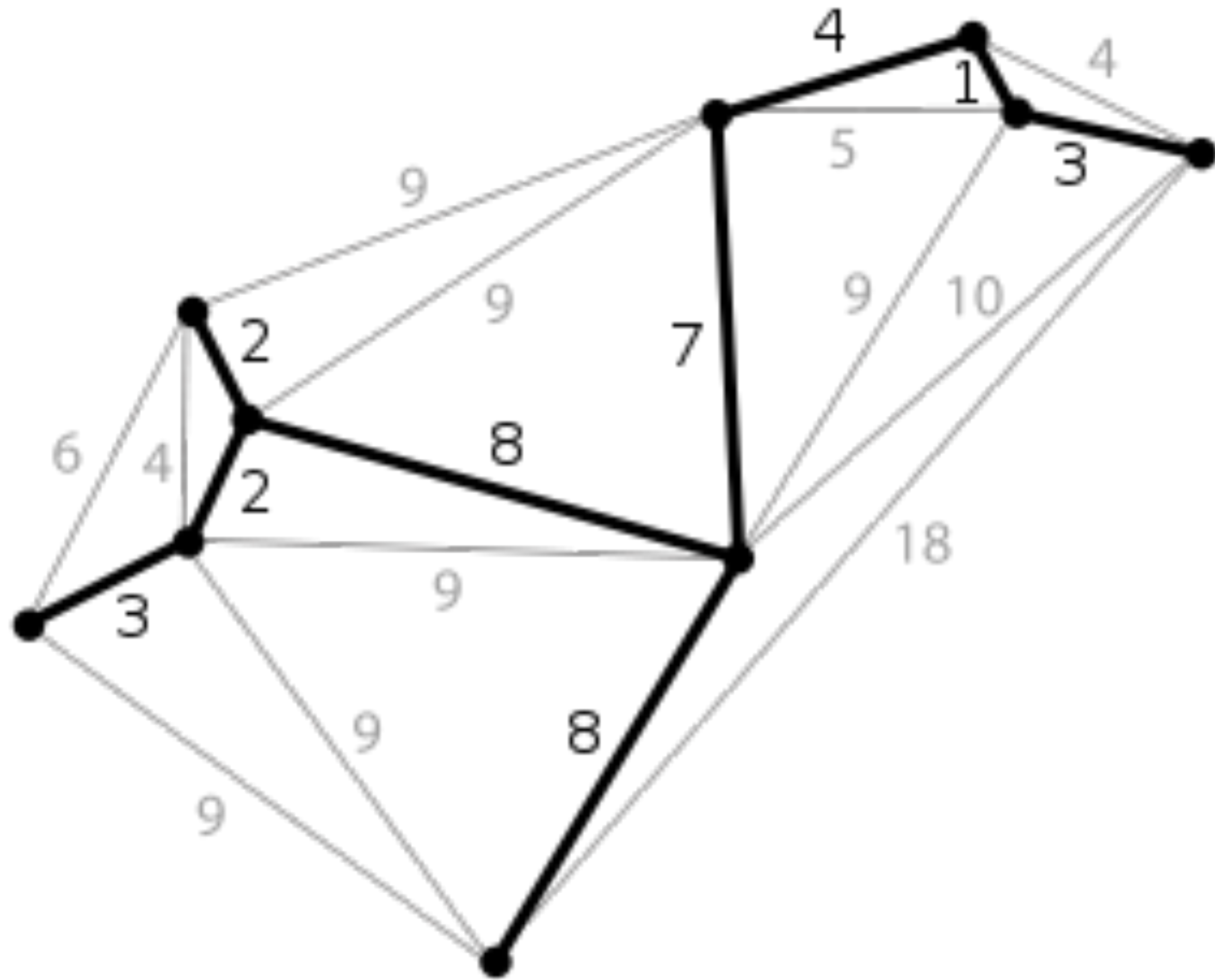
|         |              | minimize                                          | maximize                                  |
|---------|--------------|---------------------------------------------------|-------------------------------------------|
| minimum | angle        |                                                   | $O(n \log n)$<br>(Delaunay triangulation) |
| maximum |              | $O(n^2 \log n)$ [8] [9]                           |                                           |
| minimum | area         | $O(n^2)$ [10]                                     | $O(n^2 \log n)$ [11]                      |
| maximum |              | $O(n^2 \log n)$ [11]                              |                                           |
| maximum | degree       | NP-complete<br>for degree of 7 [12]               |                                           |
| maximum | eccentricity | $O(n^3)$ [9]                                      |                                           |
| minimum | edge length  | $O(n \log n)$<br>(Closest pair of points problem) | NP-complete [13]                          |
| maximum |              | $O(n^2)$ [14]                                     | $O(n \log n)$<br>(using the Convex hull)  |
| sum of  |              | NP-hard<br>(Minimum-weight triangulation)         |                                           |
| minimum | height       |                                                   | $O(n^2 \log n)$ [9]                       |
| maximum | slope        | $O(n^3)$ [9]                                      |                                           |





Given a connected, undirected, weighted graph, a **spanning tree** of the graph is a subgraph that is a tree and connects all the nodes.

The MST is a subgraph that connects all nodes optimally

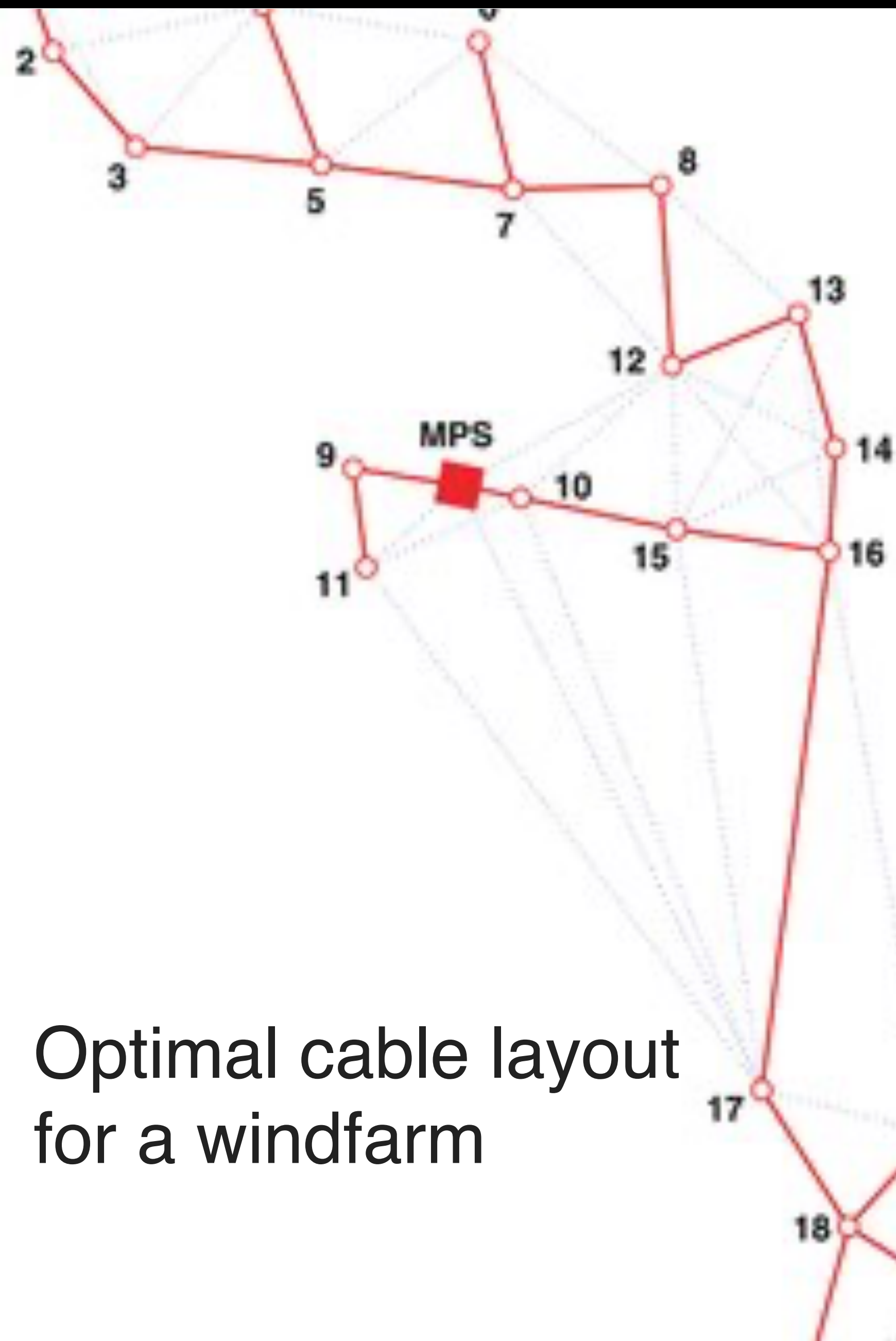


Given a connected, undirected, weighted graph, a **spanning tree** of the graph is a subgraph that is a tree and connects all the nodes.

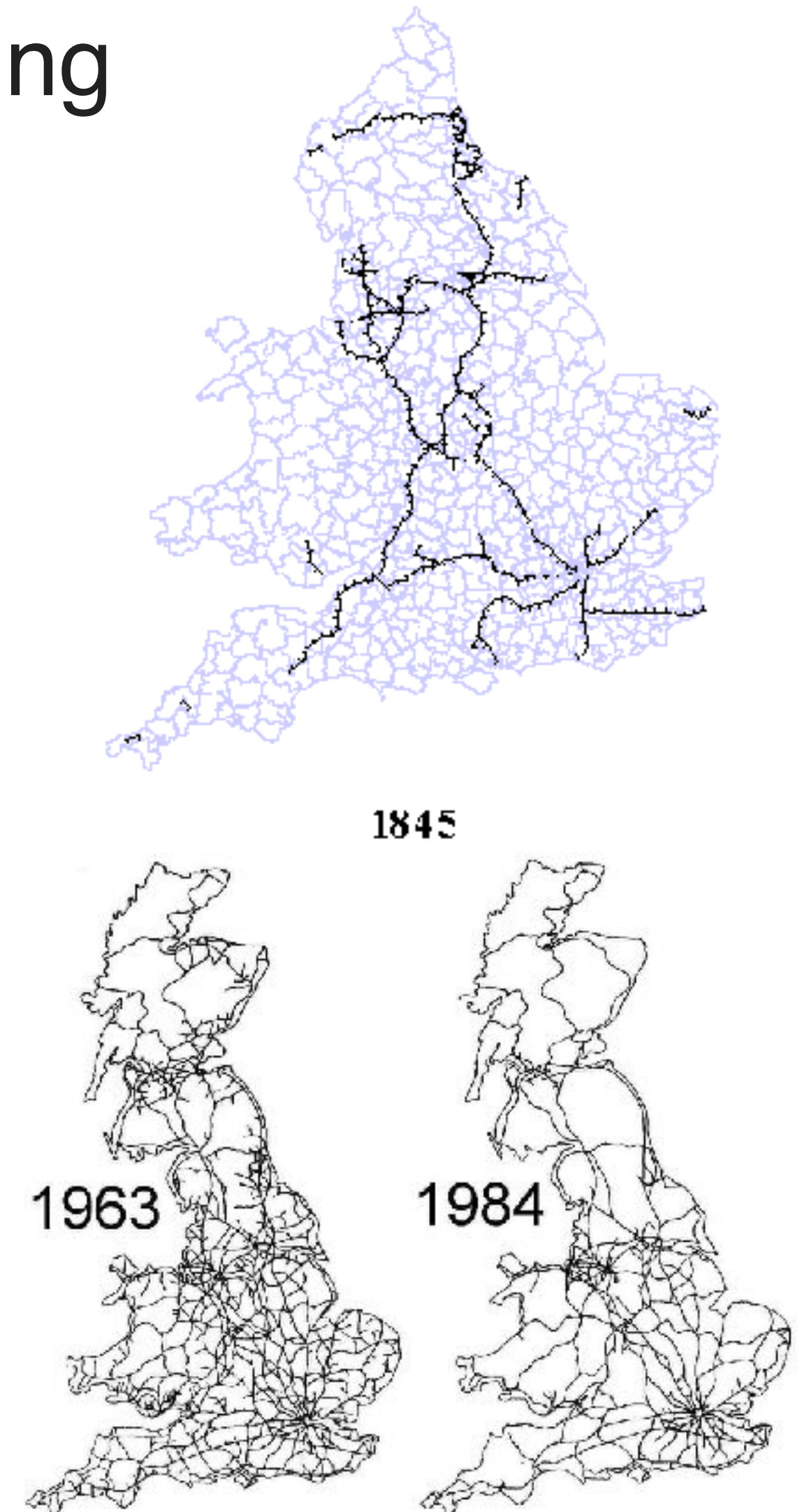
A **minimum spanning tree (MST)** is a spanning tree such that the sum of its link lengths is smaller than or equal to the sum of the link lengths of any other spanning tree.



# The MST has many applications for spatial networks



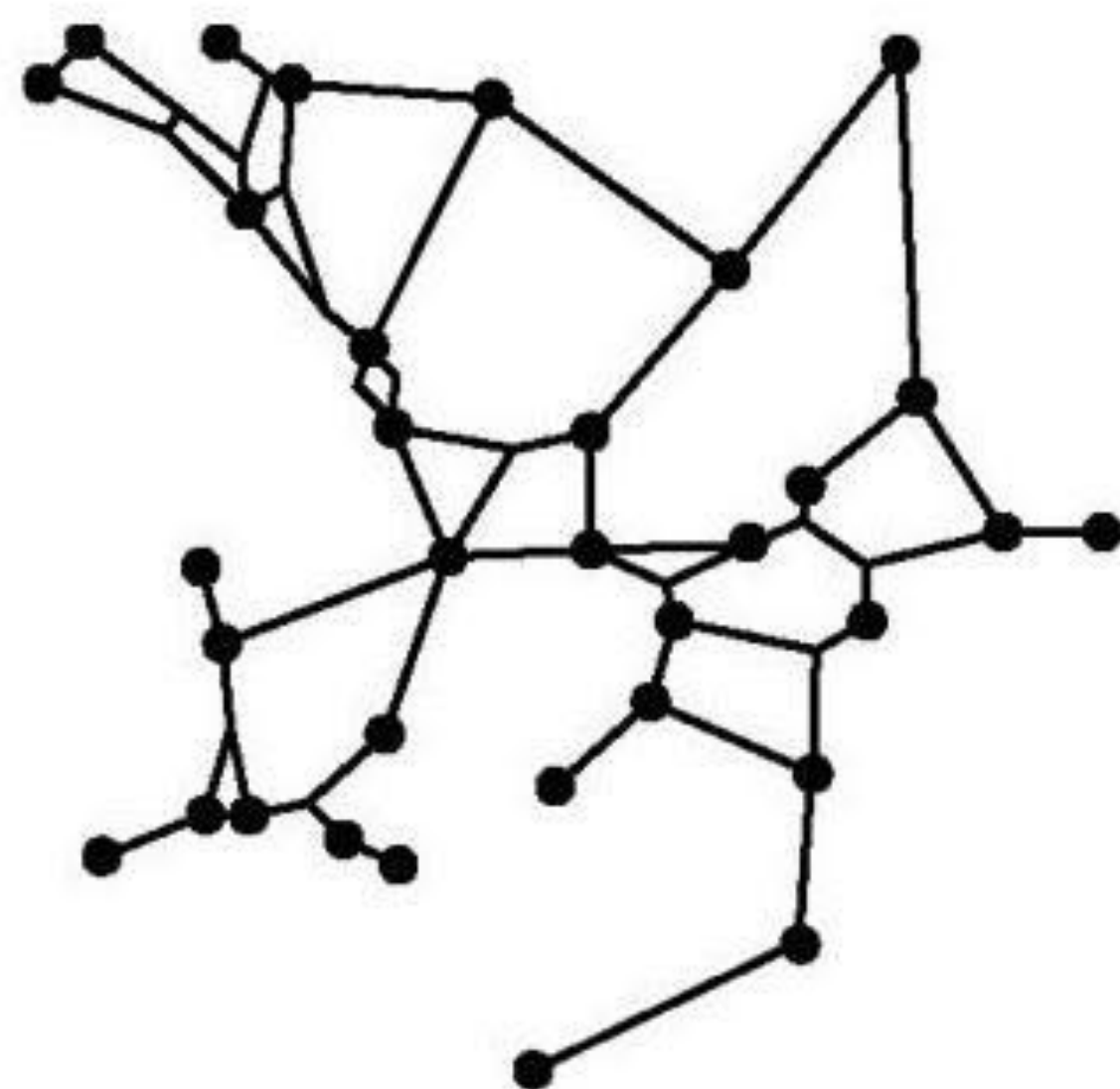
## Railroad network planning



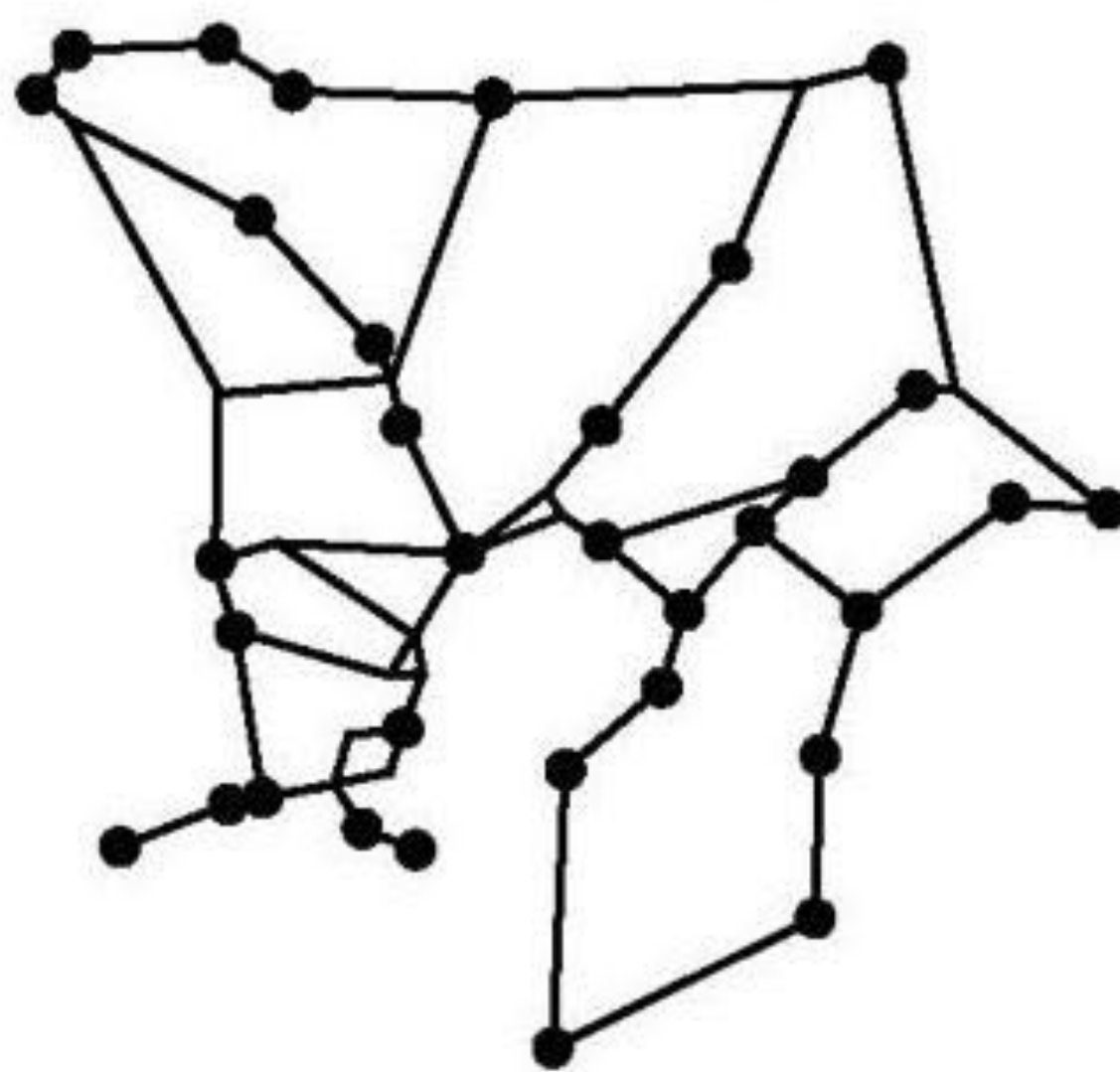
Optimal cable layout  
for a windfarm



Slime mold network



Actual Tokyo railway network

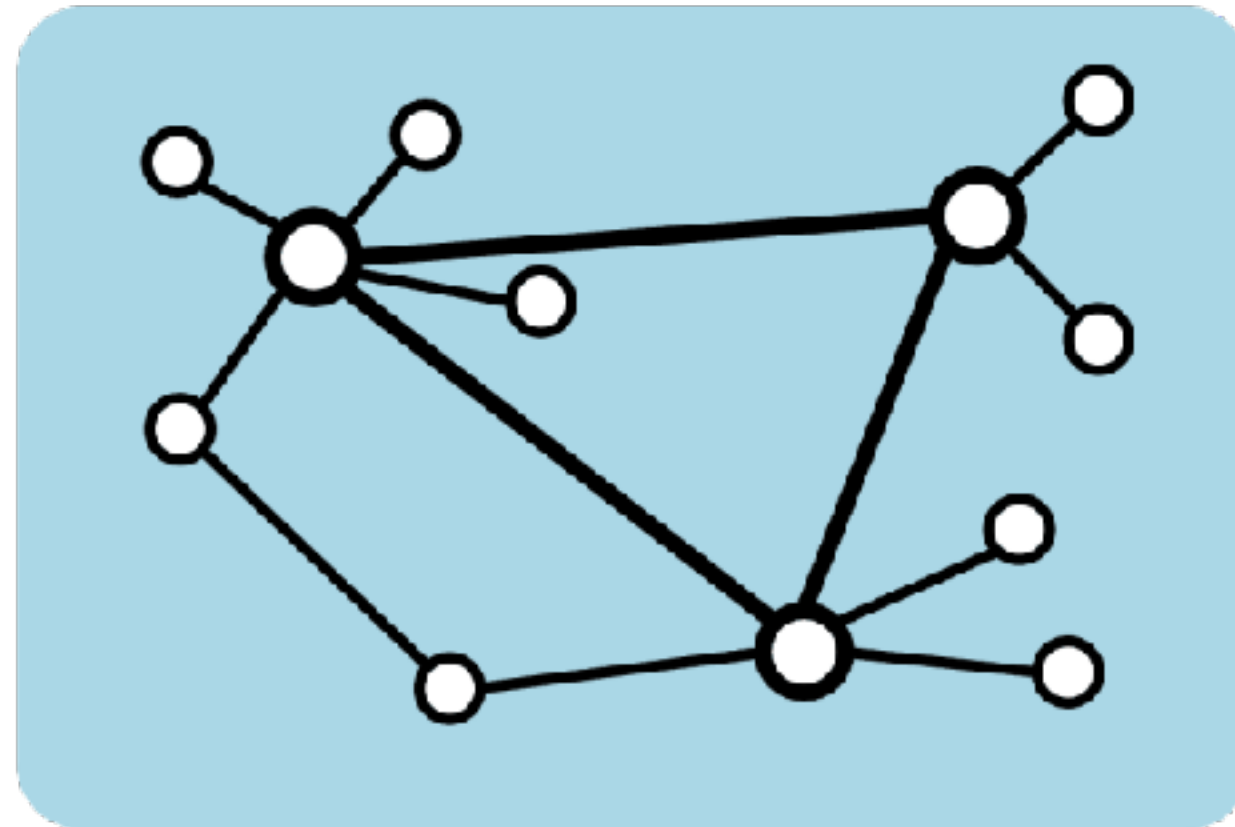


Tero et al. 2010



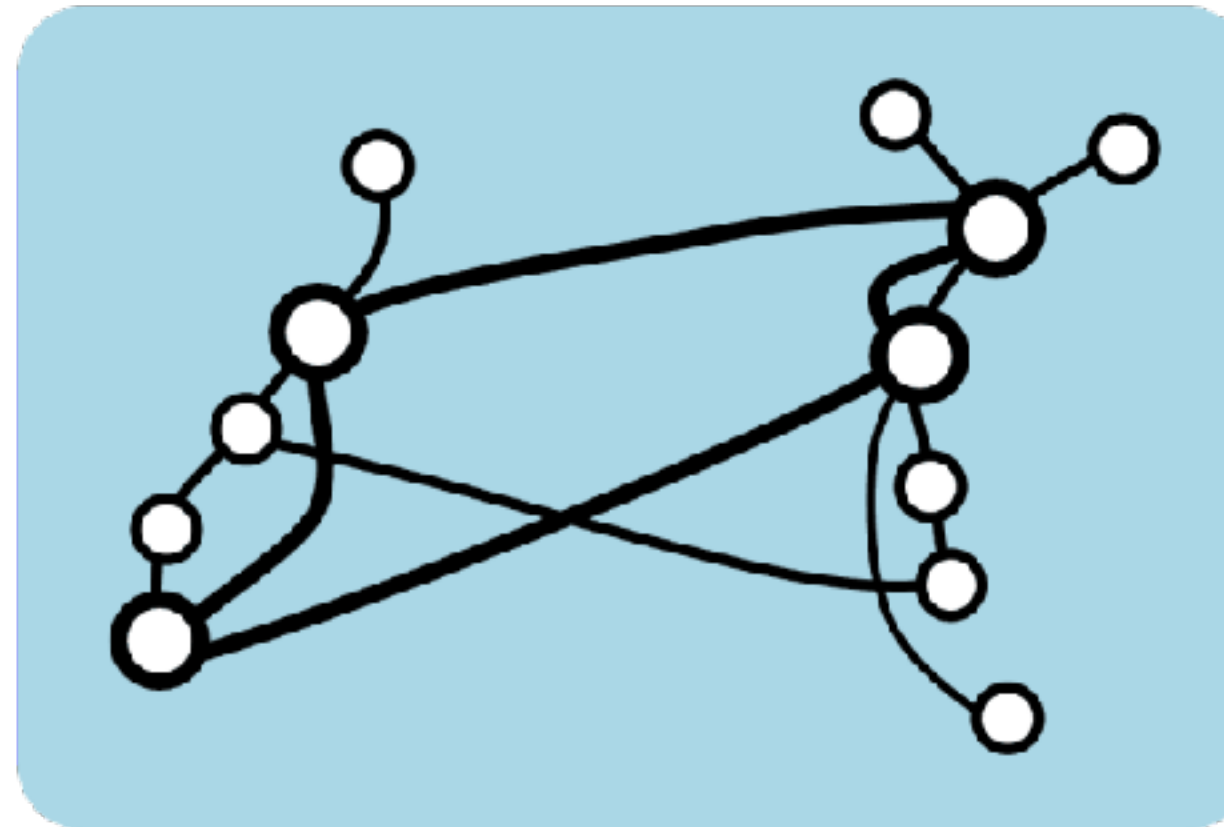
# Spatial networks have different shapes and vulnerabilities

**Air Networks**



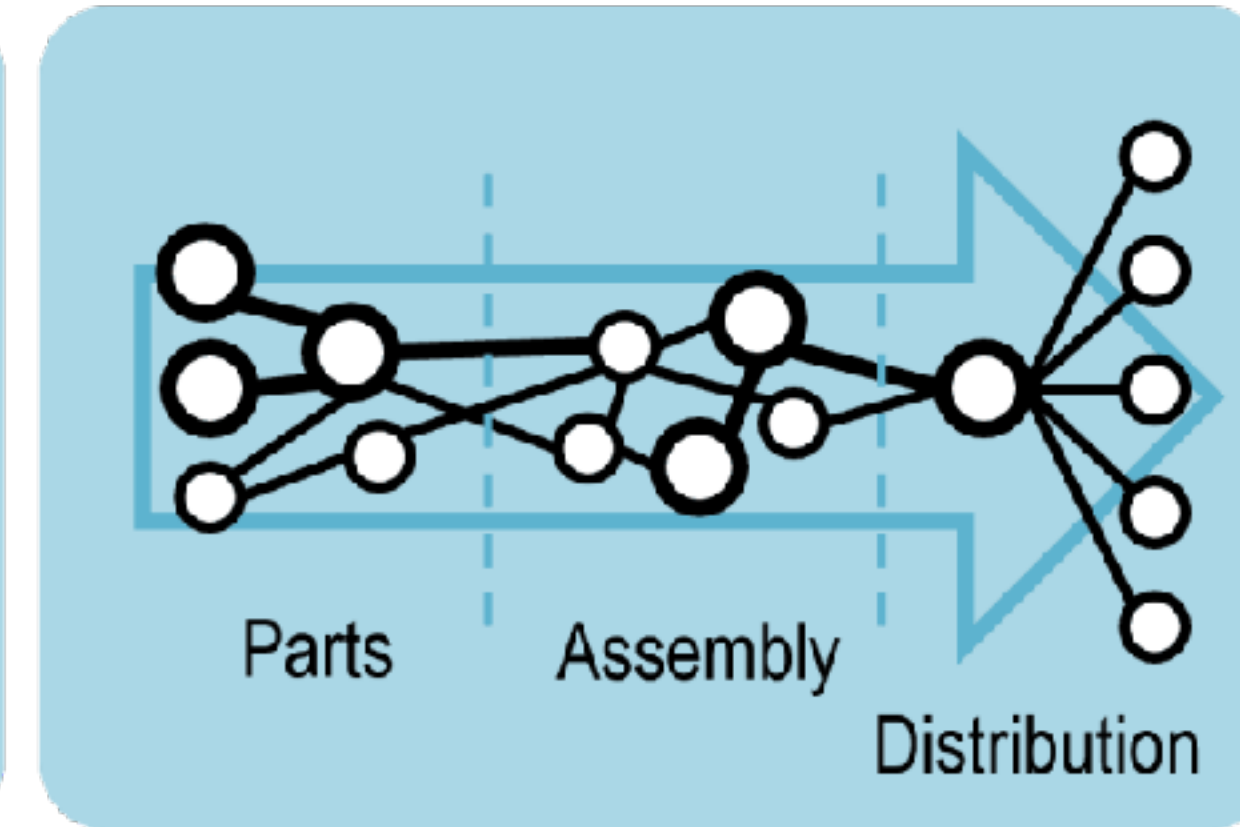
Nodal hierarchy (hub-and-spoke)

**Maritime Networks**



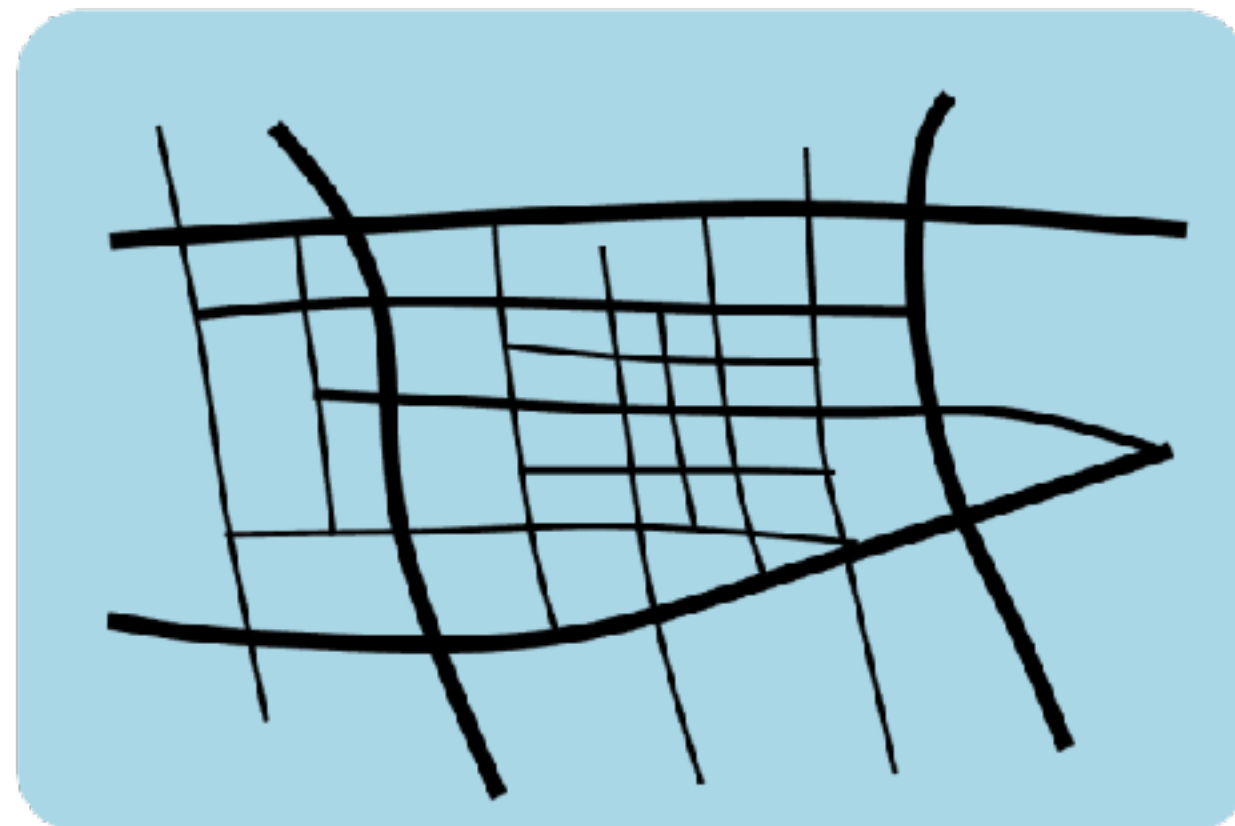
Circuitous nodal hierarchy

**Logistical Networks**



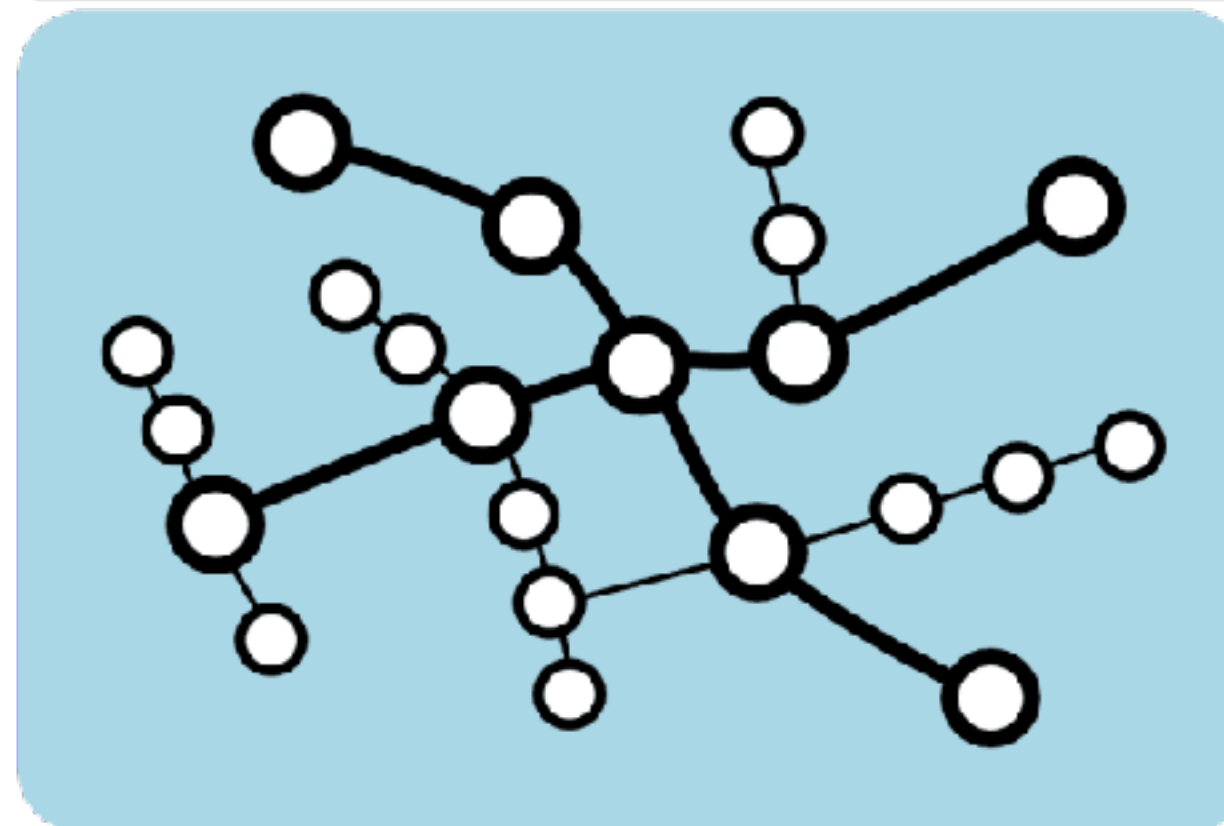
Sequential multi-nodal hierarchy

**Road Networks**



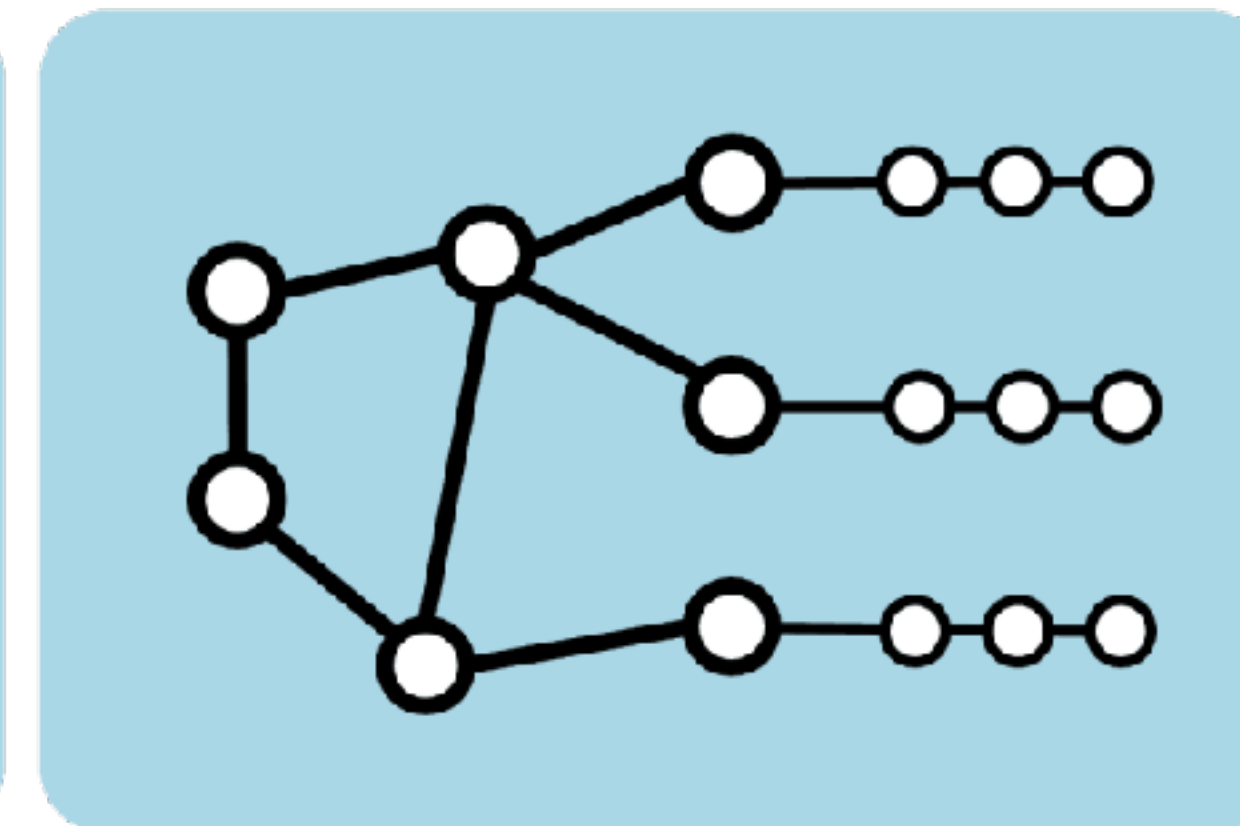
Hierarchical meshes

**Rail Networks**



Linear nodal hierarchy

**Power Grids**

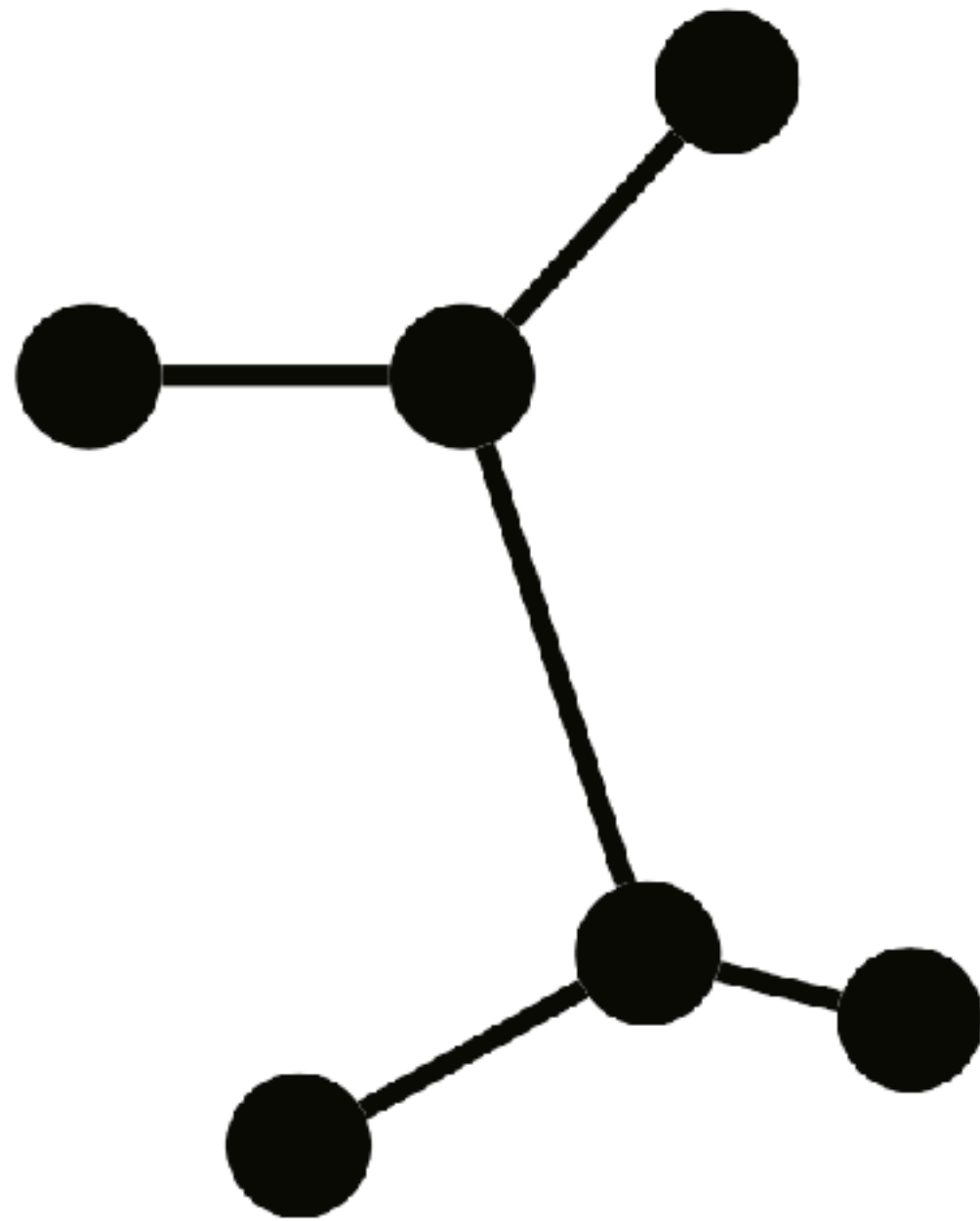


Sequential linear hierarchy



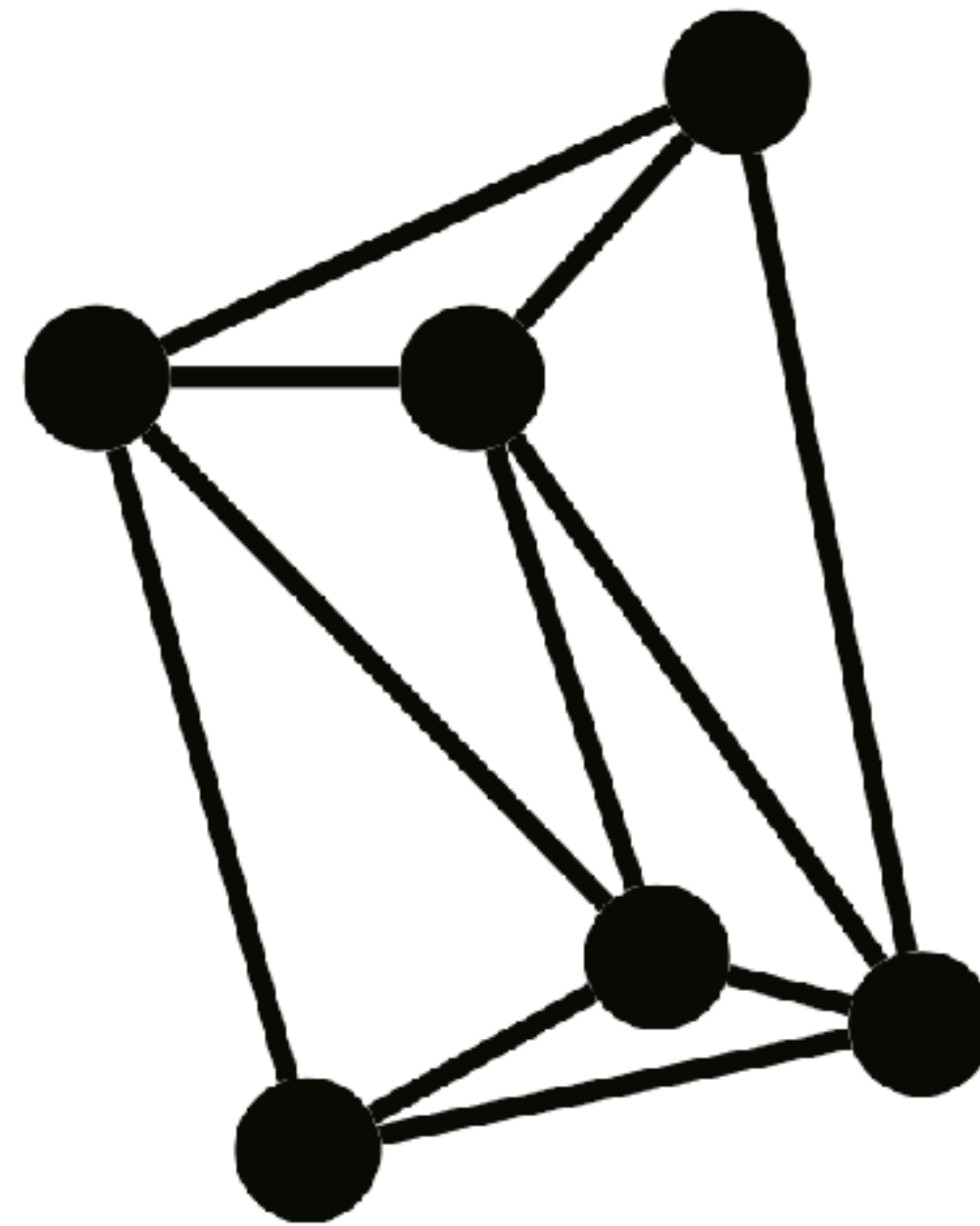
# There is a tradeoff between economic investment and resilience

Minimum spanning tree



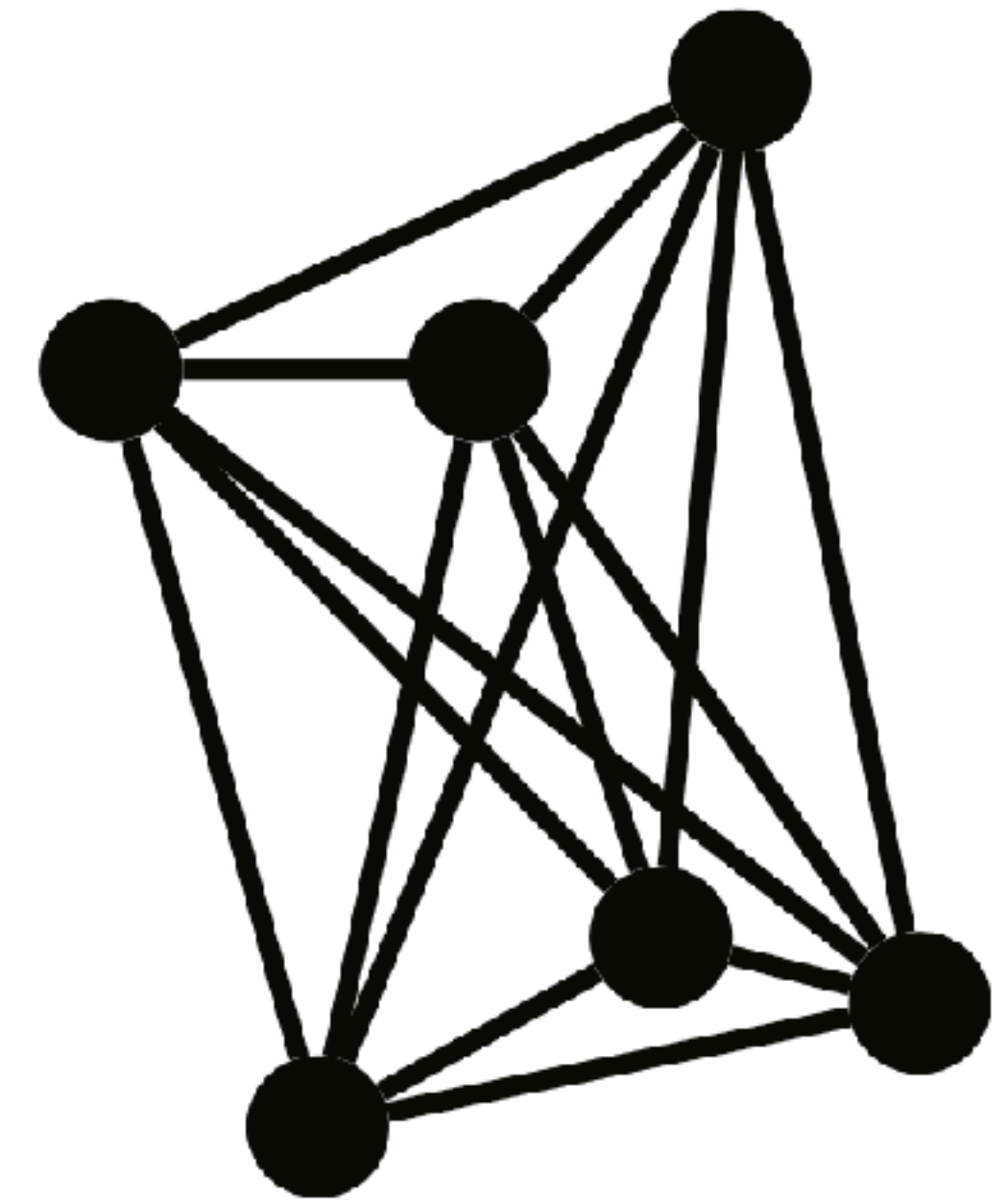
Investor's  
optimum

Triangulation



Cohesive  
planar network

Fully connected



Traveler's  
optimum

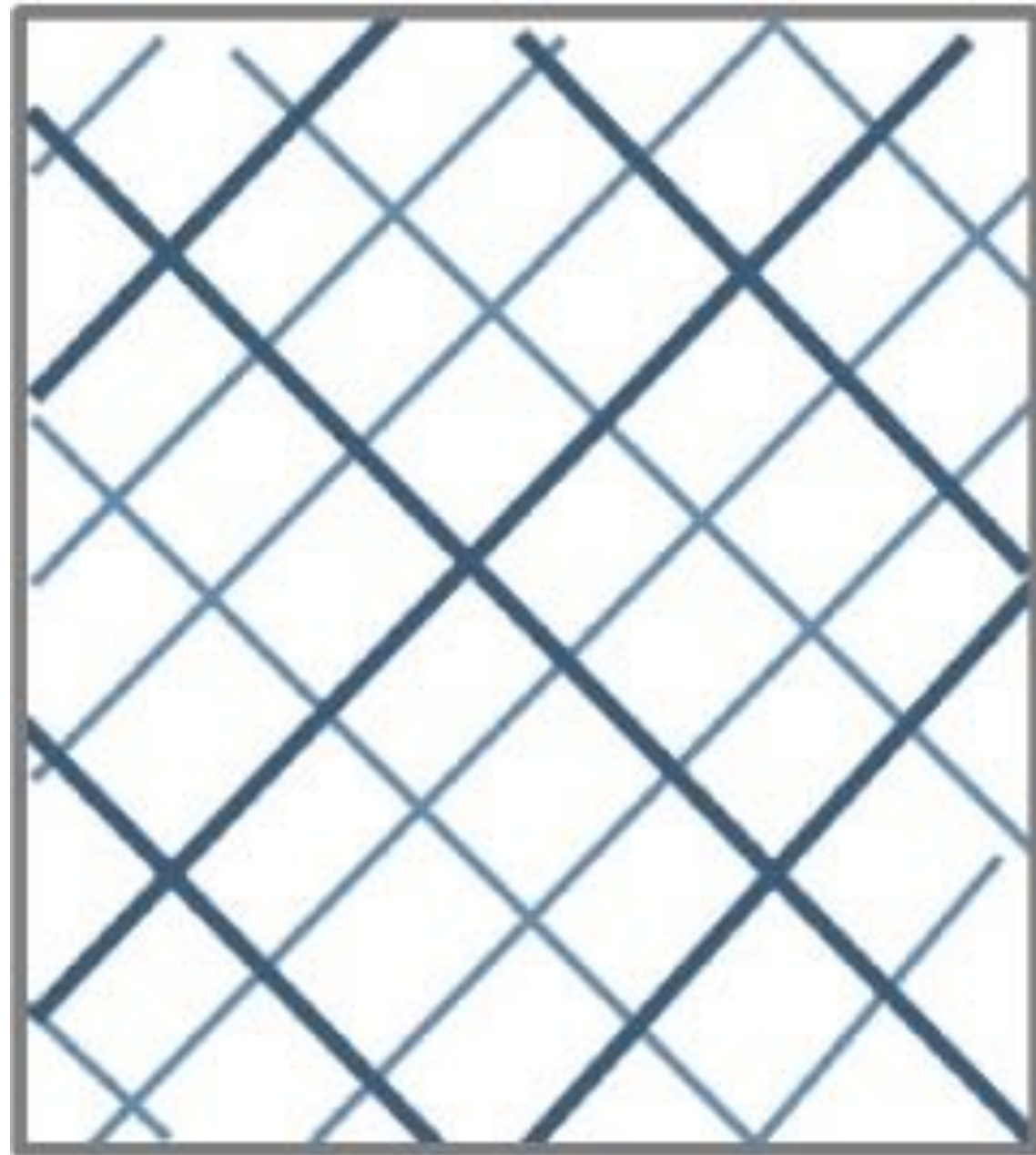
Economic

Resilient

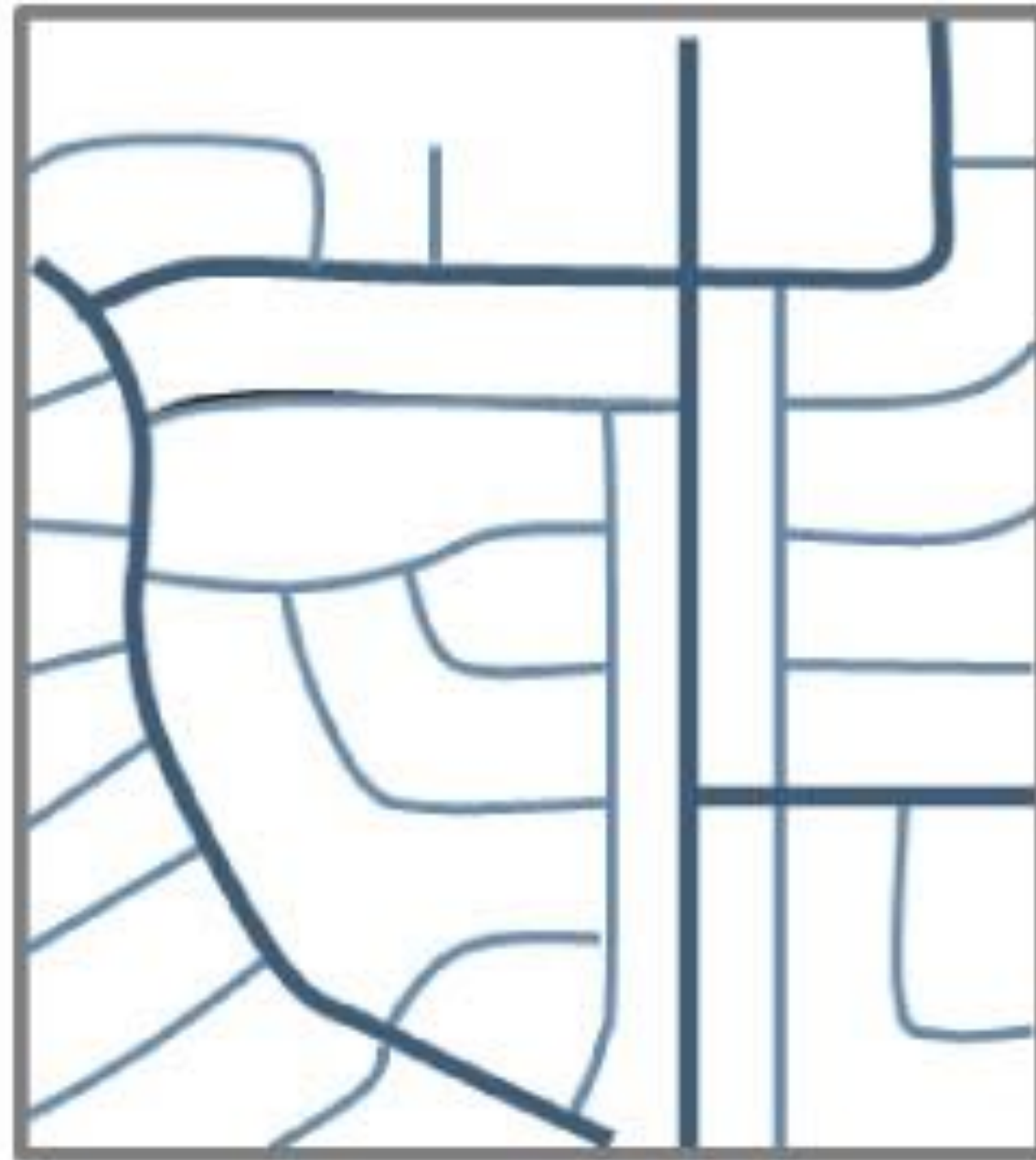
# Street networks



# Street networks can look very different



**Conventional Grid  
Pattern (c 1900)**



**Curvilinear Loop Pattern &  
Beginning of Cul-de-Sacs (1930-1950)**



**Conventional  
Cul-de-Sac Pattern  
(since 1950)**

— *Arterial road*

— *Local street*

# Spatial network measures can quantify these differences

**Detour index:** Straight distance versus network distance

$$DI = \frac{D(S)}{D(T)}$$

Also: circuitry, directness, stretch



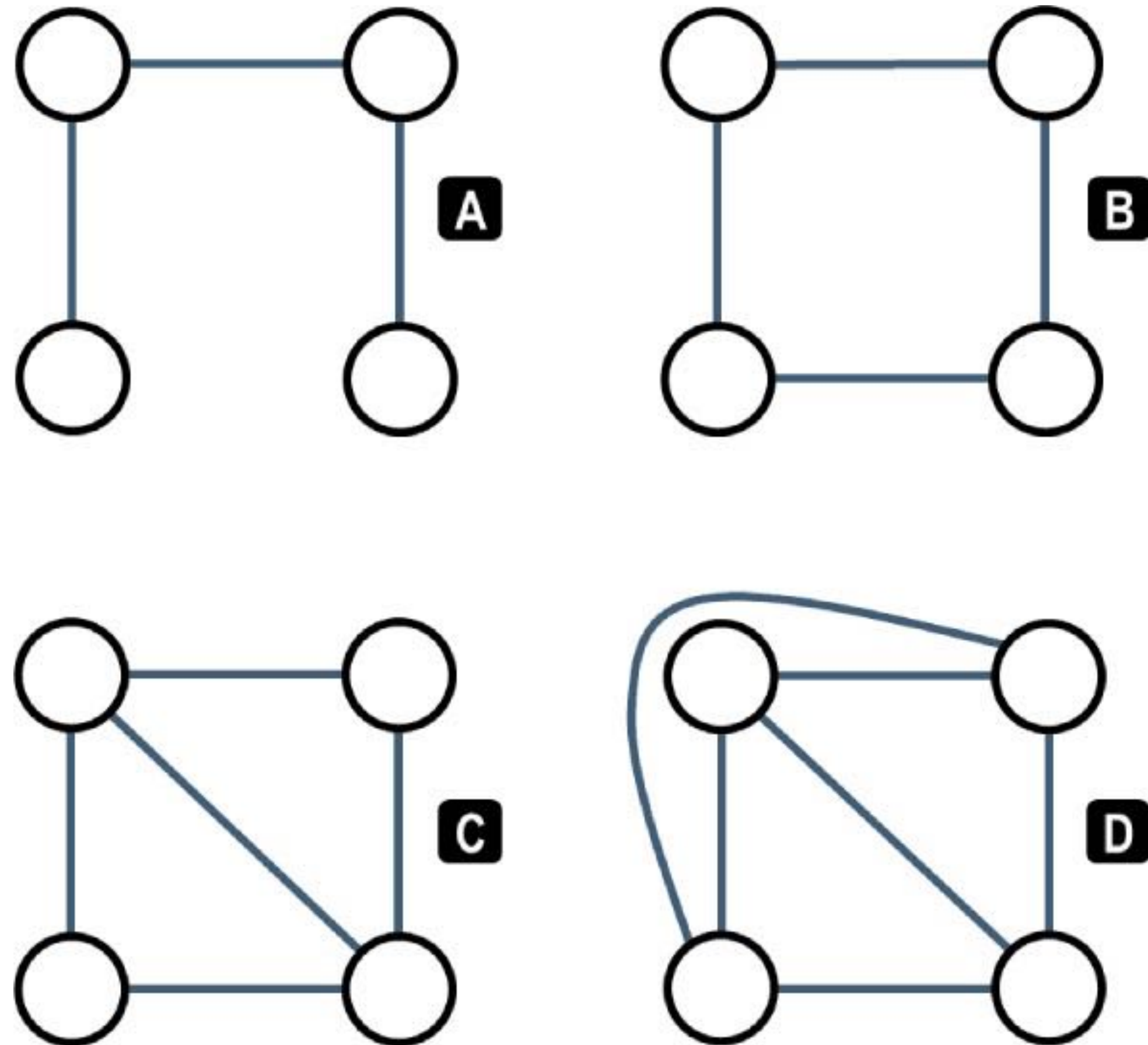
# Spatial network measures can quantify these differences

**Network density:** Links per surface area

$$ND = \frac{L}{S}$$

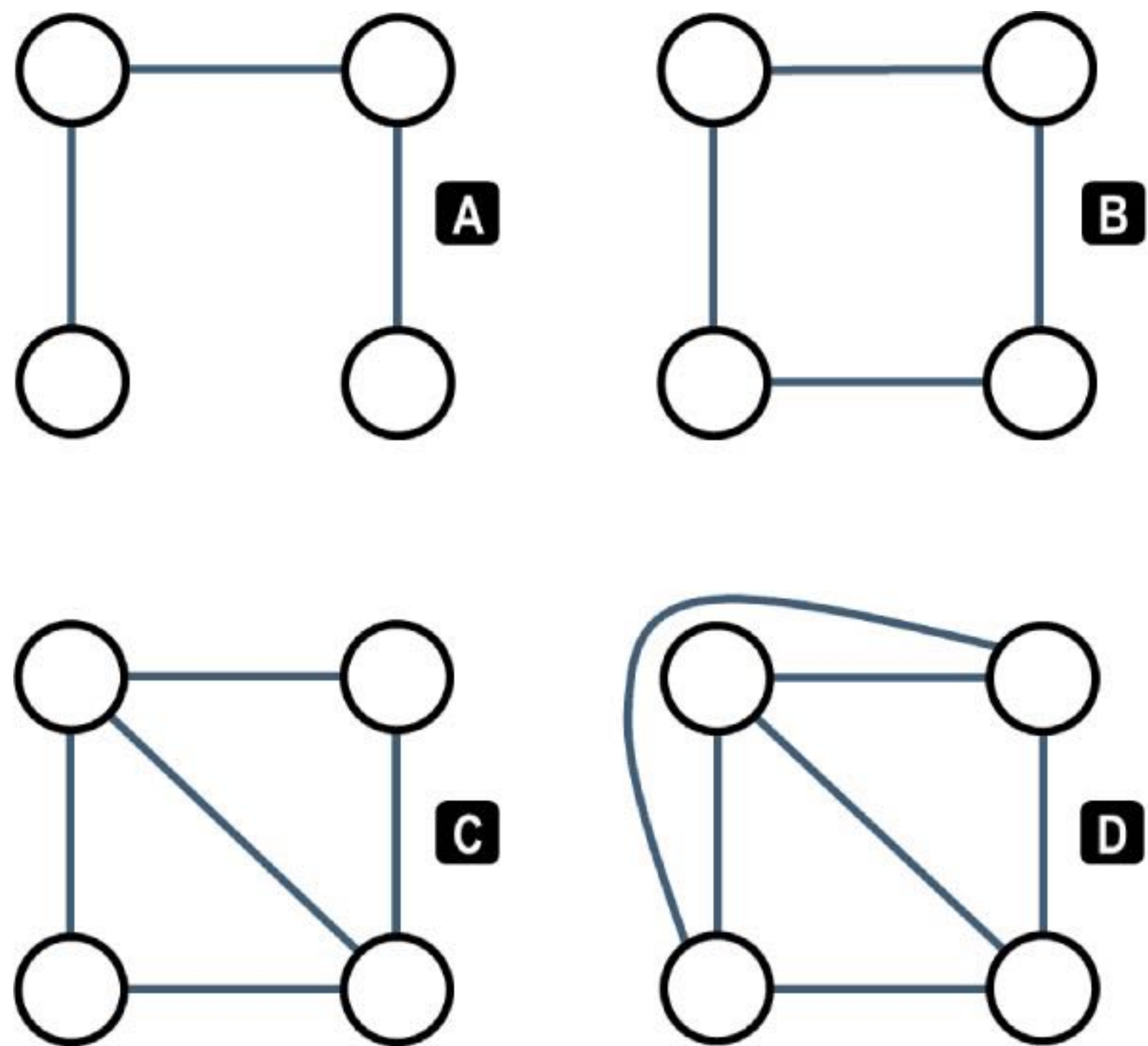
# Spatial network measures can quantify these differences

**Alpha index:** Existing versus possible elementary cycles



# Spatial network measures can quantify these differences

Alpha index: Existing versus possible elementary cycles



$$\alpha = \frac{u}{2v - 5}$$

|   | $u (e-v+p)$ | $2v-5$ | Alpha |
|---|-------------|--------|-------|
| A | 0           | 3      | 0.0   |
| B | 1           | 3      | 0.33  |
| C | 2           | 3      | 0.66  |
| D | 3           | 3      | 1.0   |

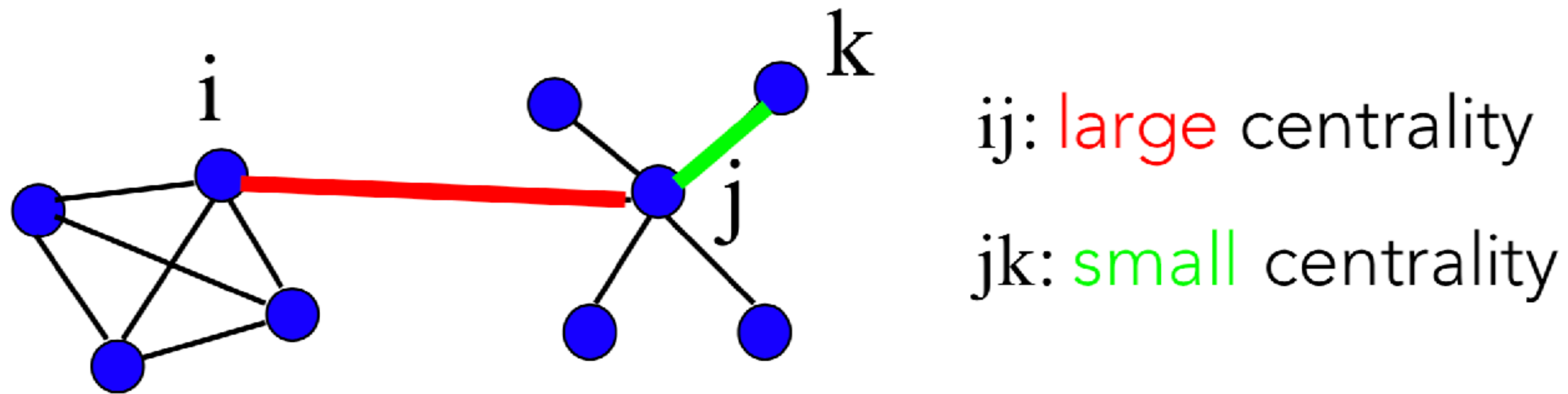
Tree

Max. planar



# Betweenness centrality measures flow

Freeman (1977): **Betweenness centrality** measures how many times a link is part of all shortest paths in the network.

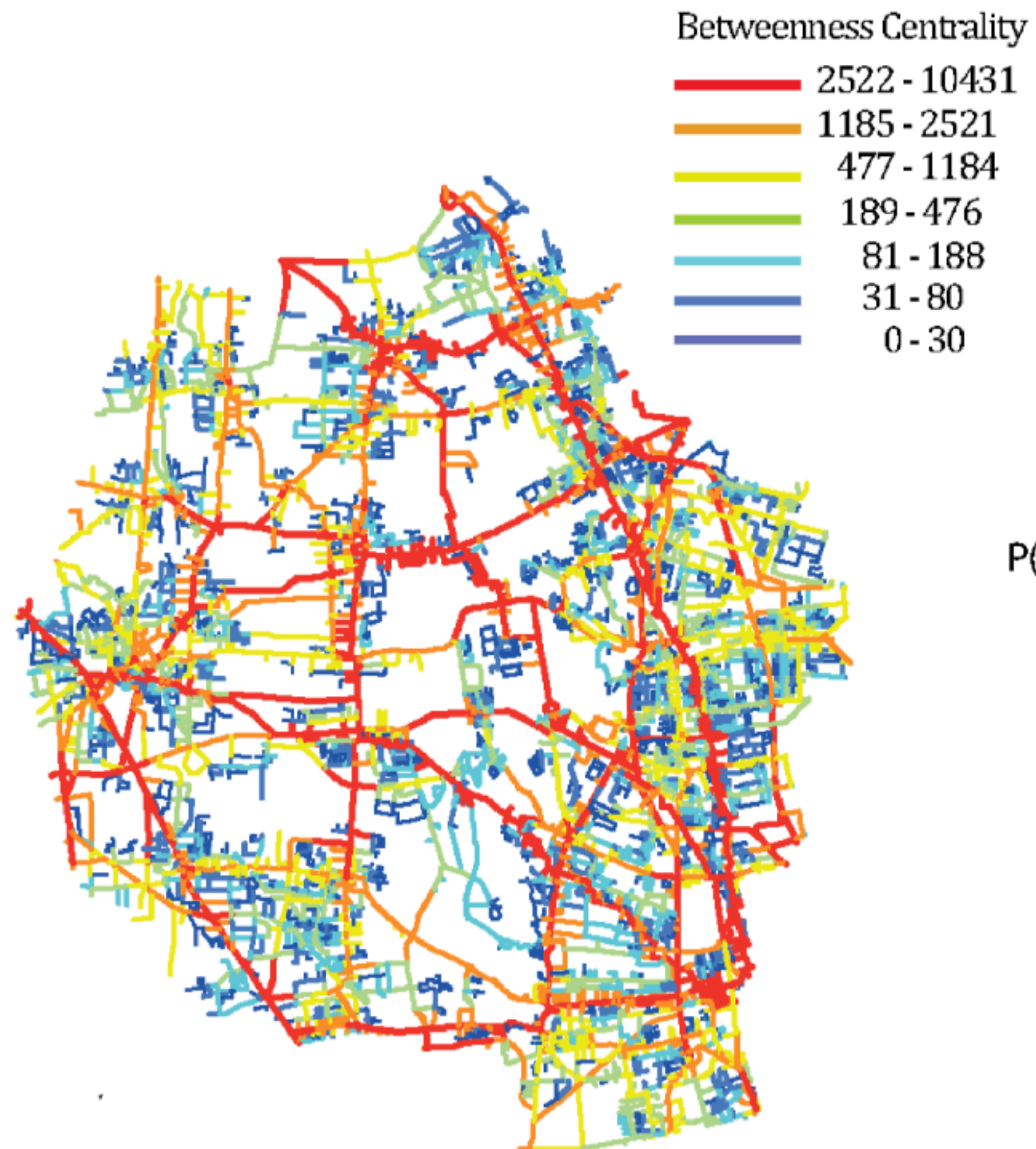


$$g(ij) = \sum_{s,t} \frac{\sigma_{st}(ij)}{\sigma_{st}}$$



# Betweenness centrality measures flow

Freeman (1977): **Betweenness centrality** measures how many times a link is part of all shortest paths in the network.



Measures the importance of a segment in the shortest paths flow

Gives the backbone of stable central roads

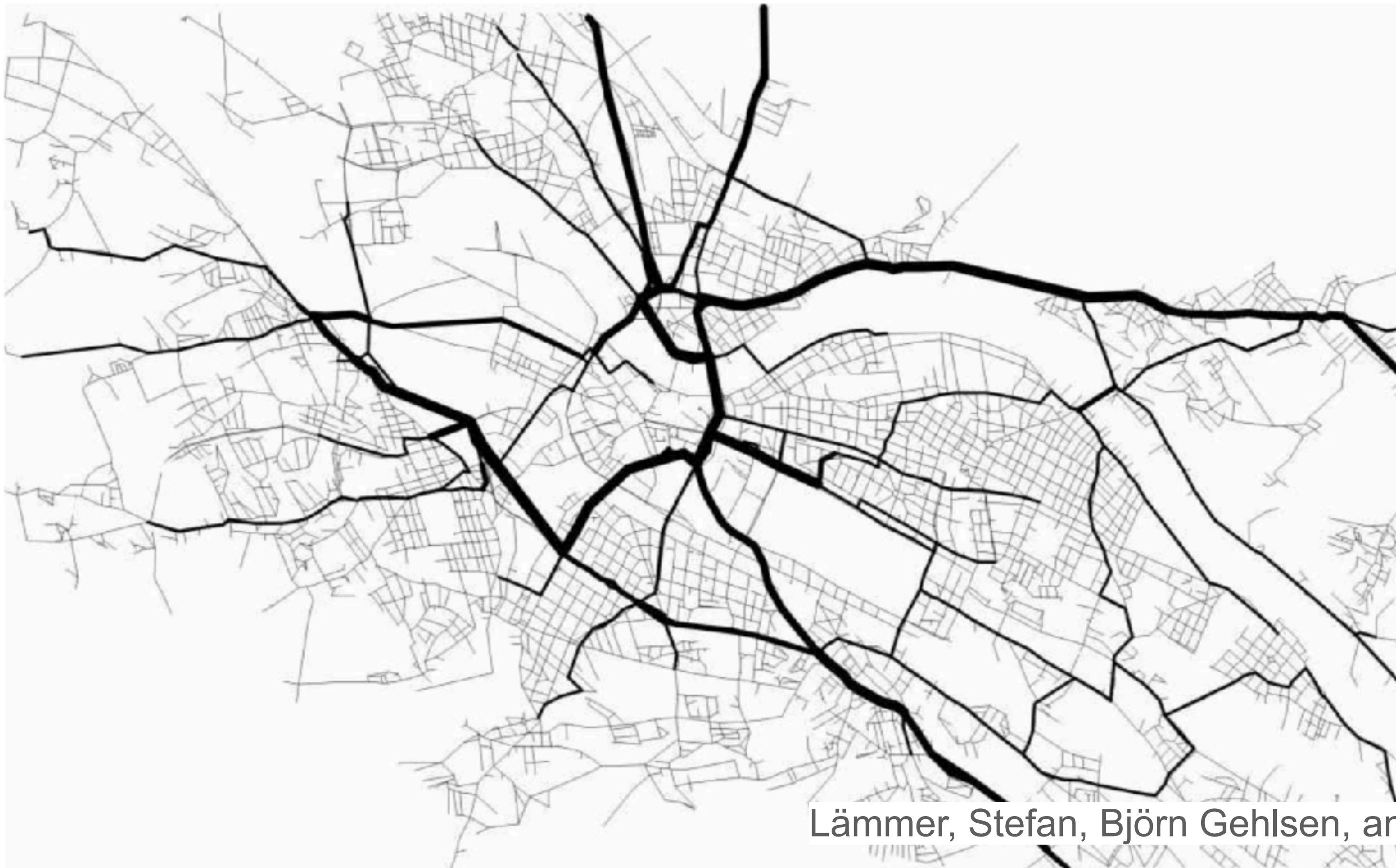
Can be broadly distributed



# Betweenness centrality measures flow

Freeman (1977): **Betweenness centrality** measures how many times a link is part of all shortest paths in the network.

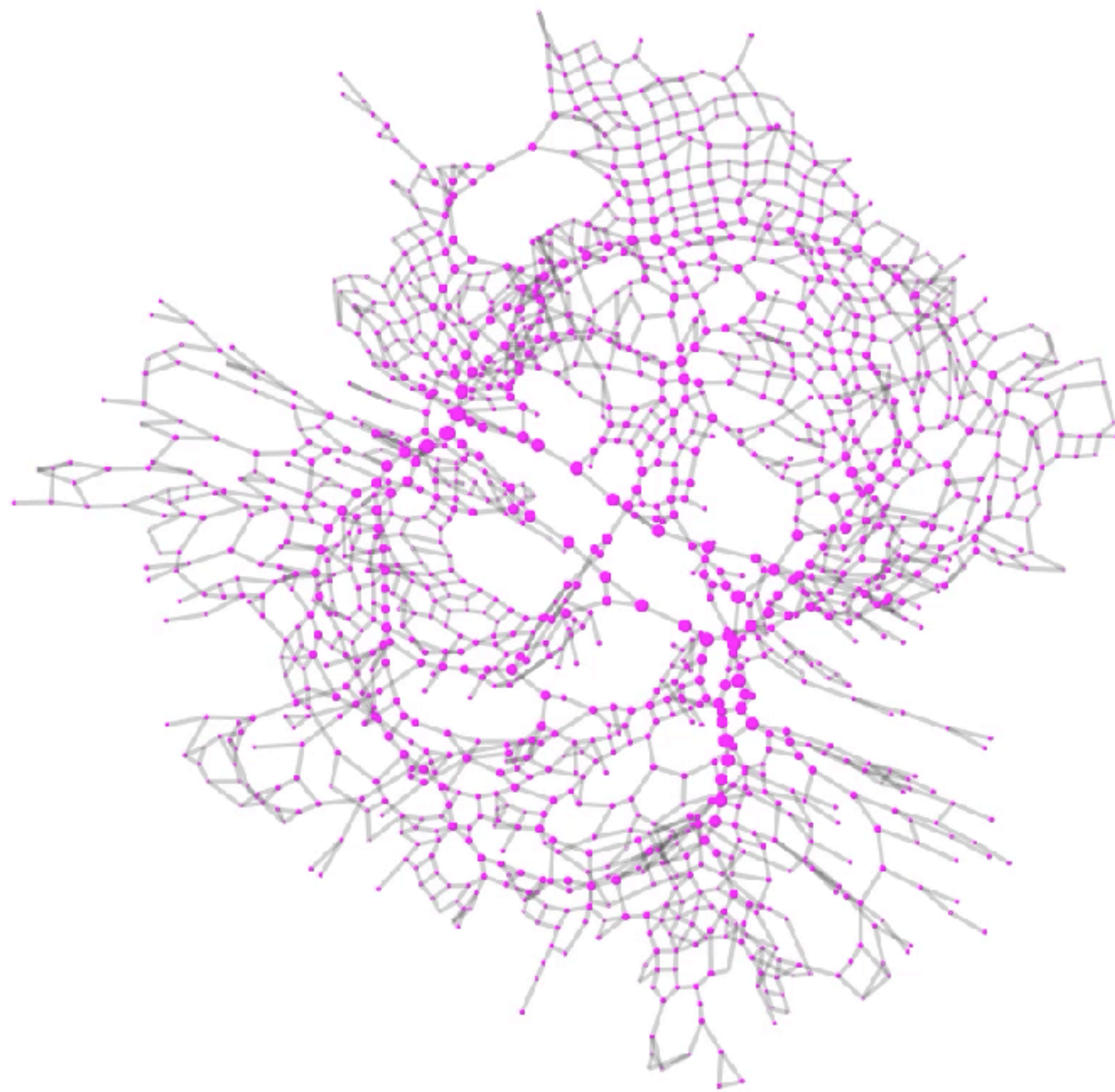
Can point to problems with congestion:





# Betweenness centrality identifies bottlenecks in transport networks

Budapest attack tolerance



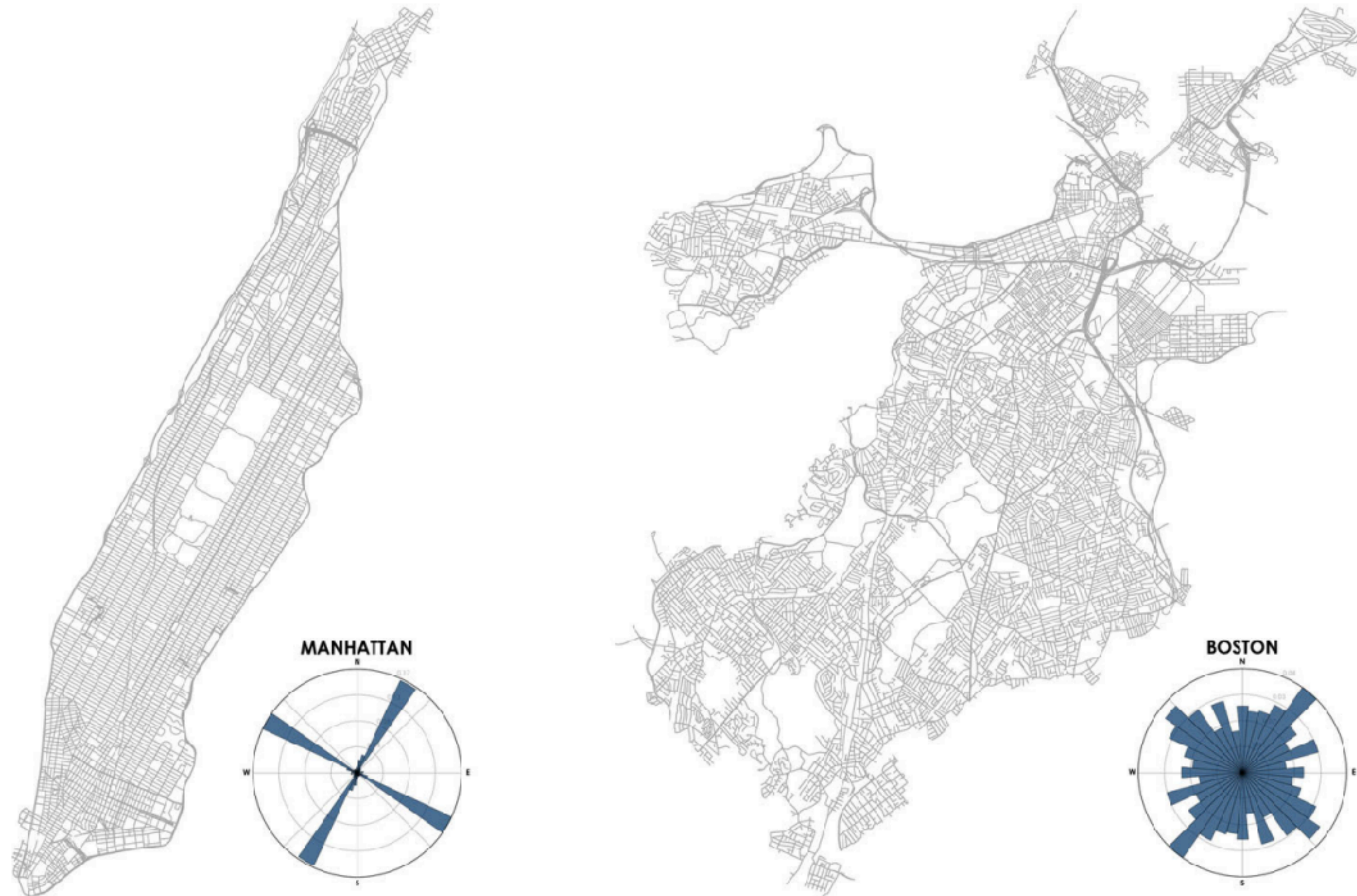
Fraction of intersections removed: 0.0%  
Connected Components: 1



Video by Luis Guillermo Natera Orozco



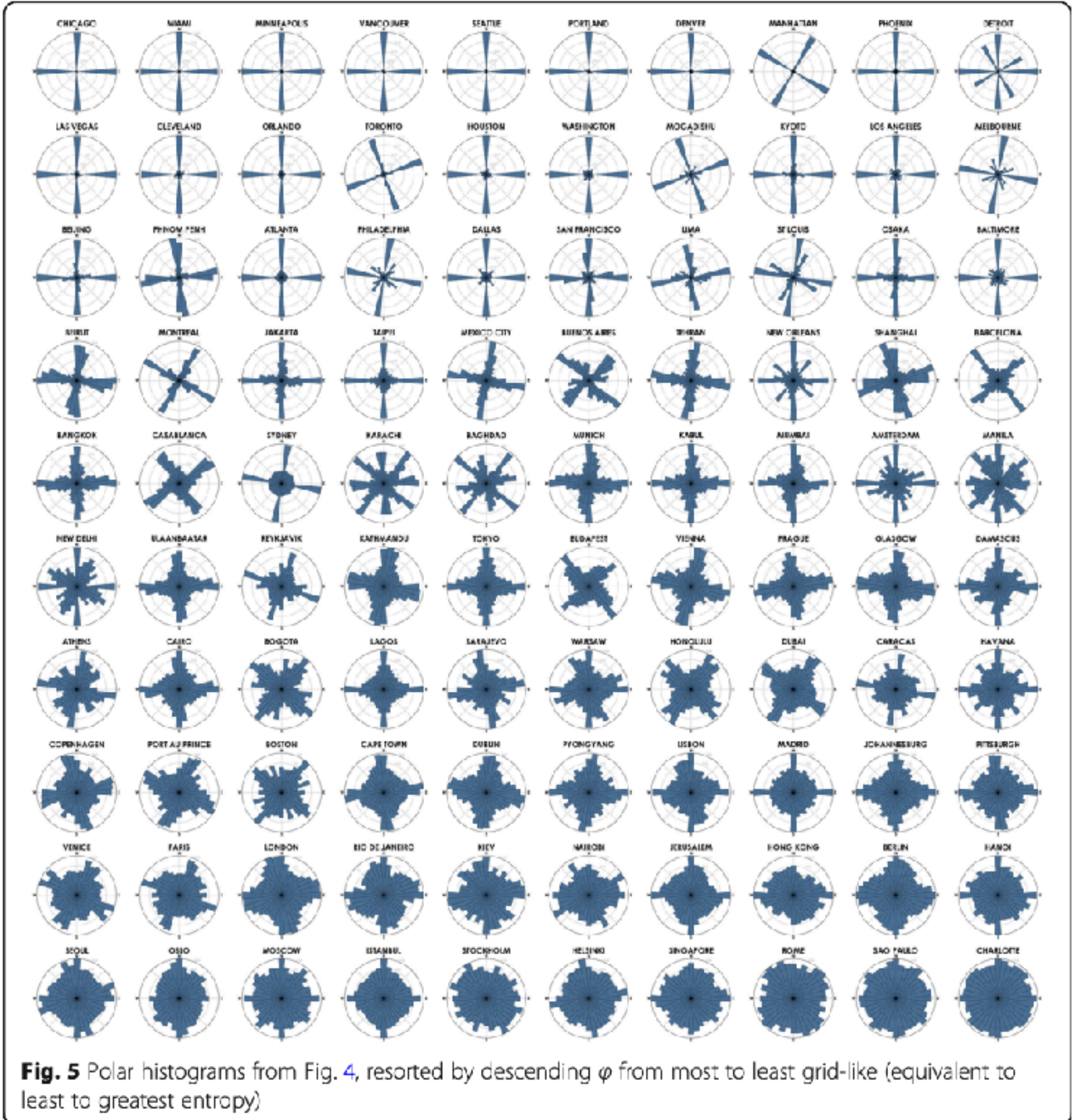
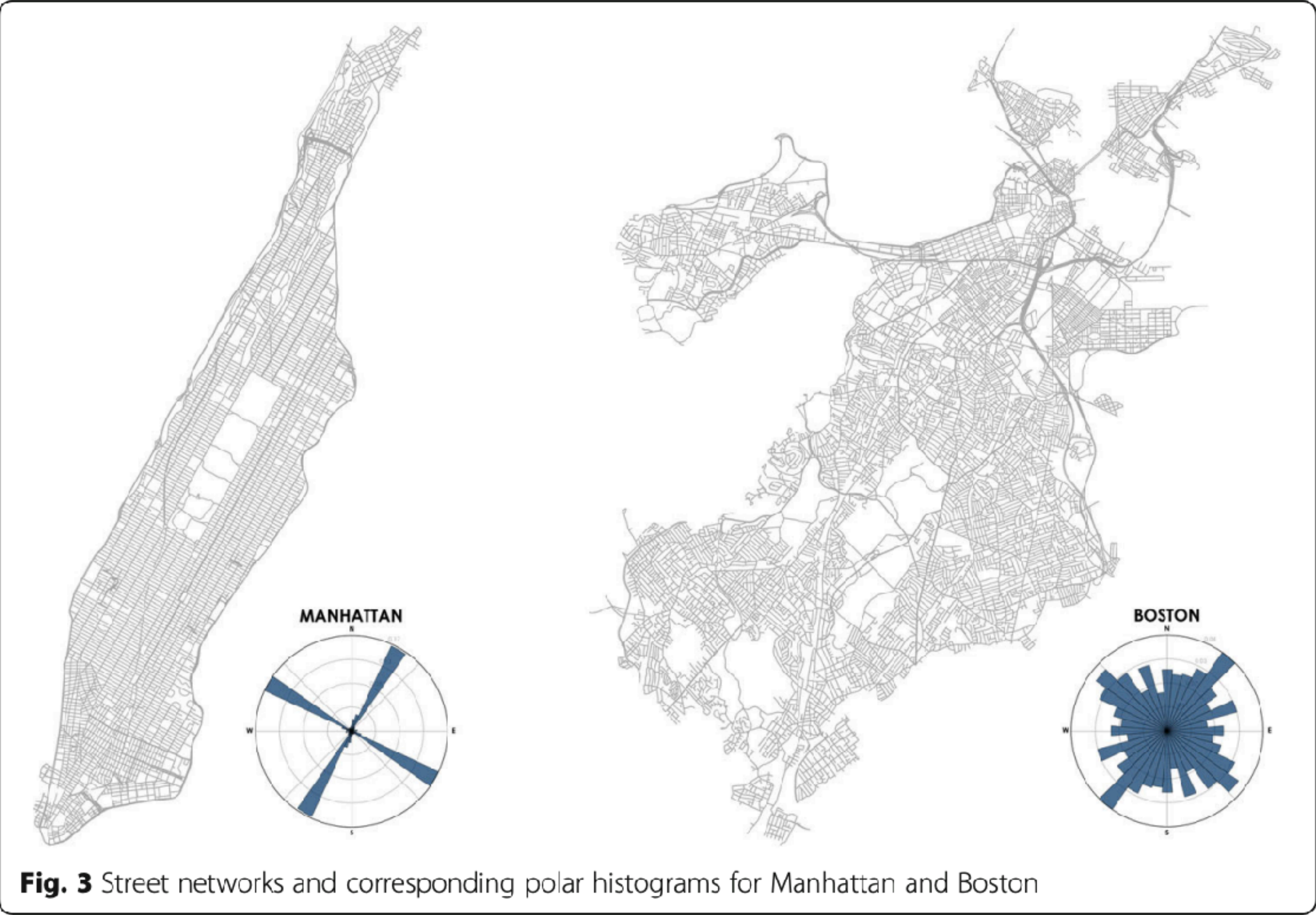
# Street network orientation



**Fig. 3** Street networks and corresponding polar histograms for Manhattan and Boston



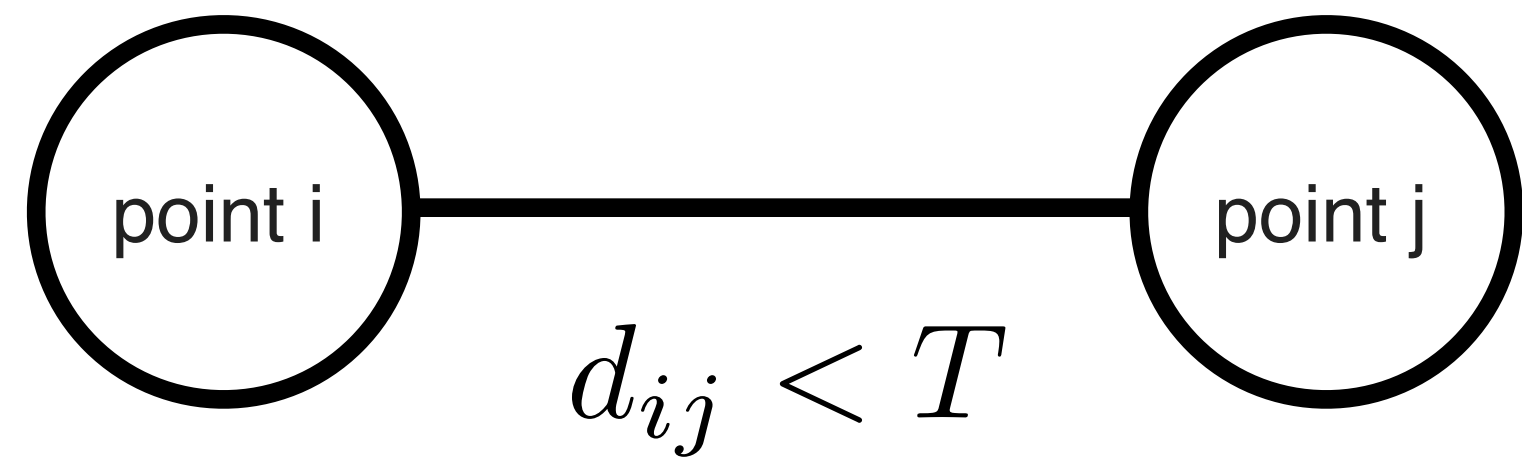
# Street network orientation



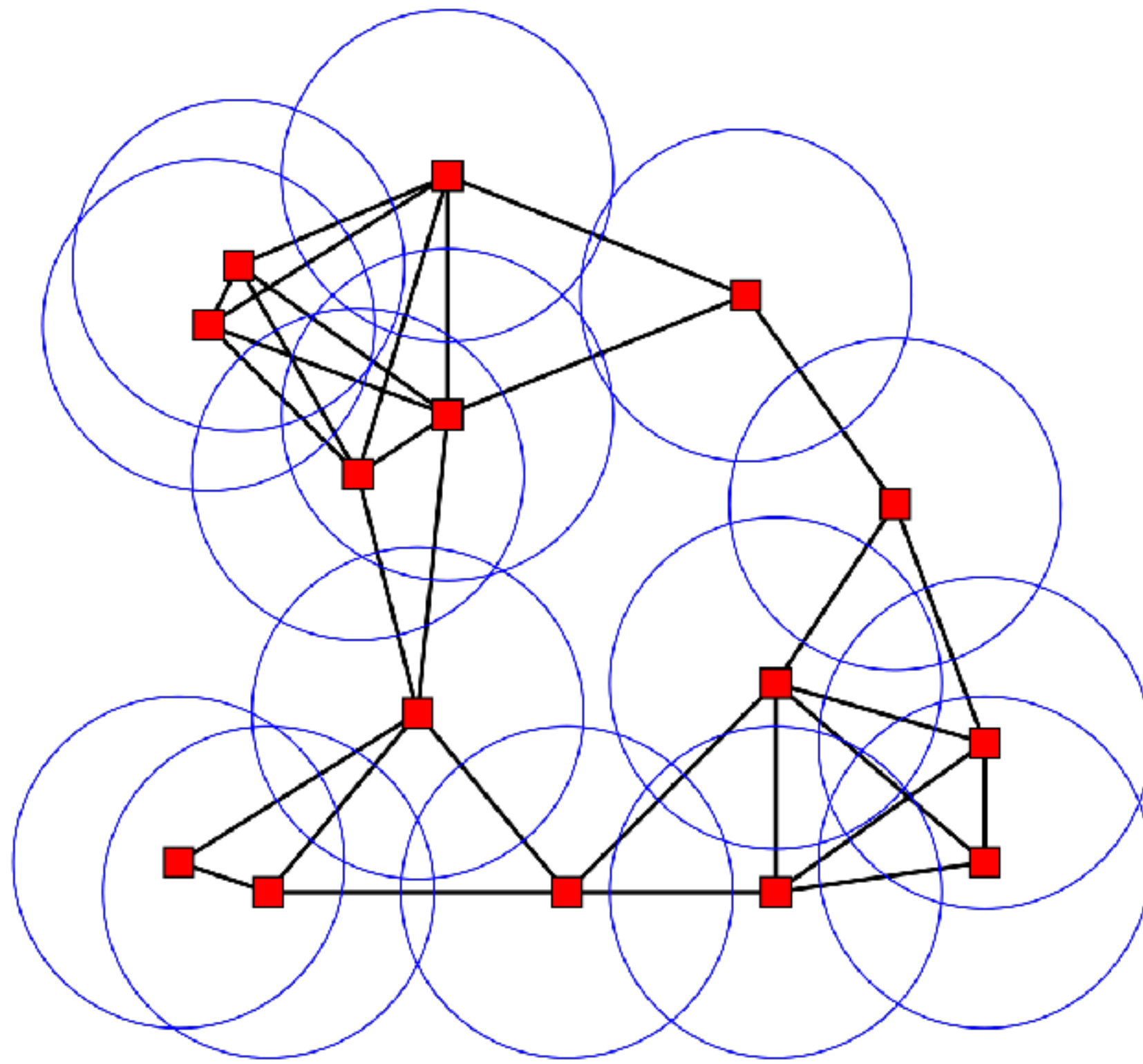


# Spatial network models

# Model: Random geometric graph



distance-based



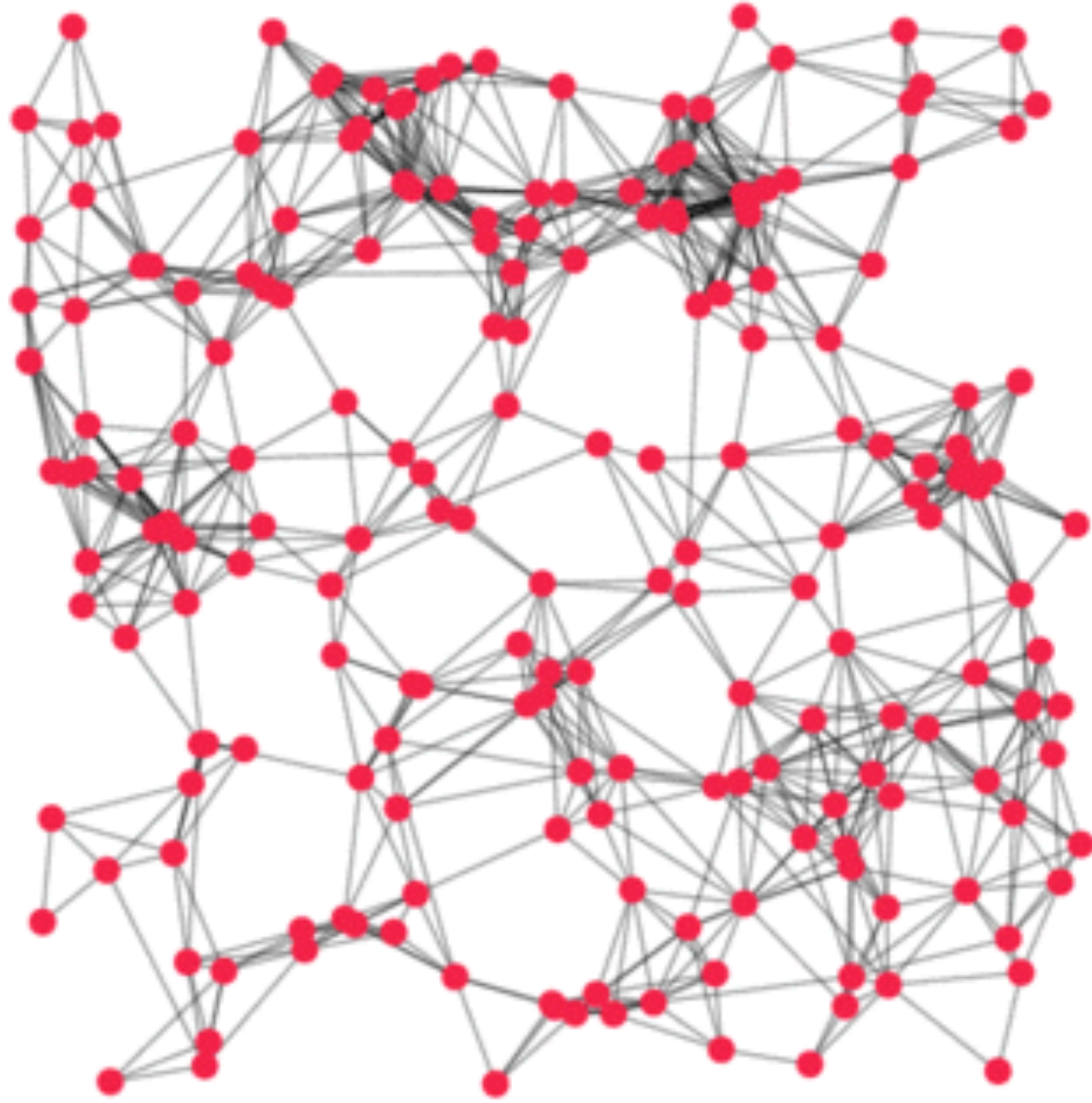
1) Generate random nodes

2) Place links between nodes with distance  $< T$

Model for wireless ad-hoc mobile/vehicle networks



# Model: Waxman



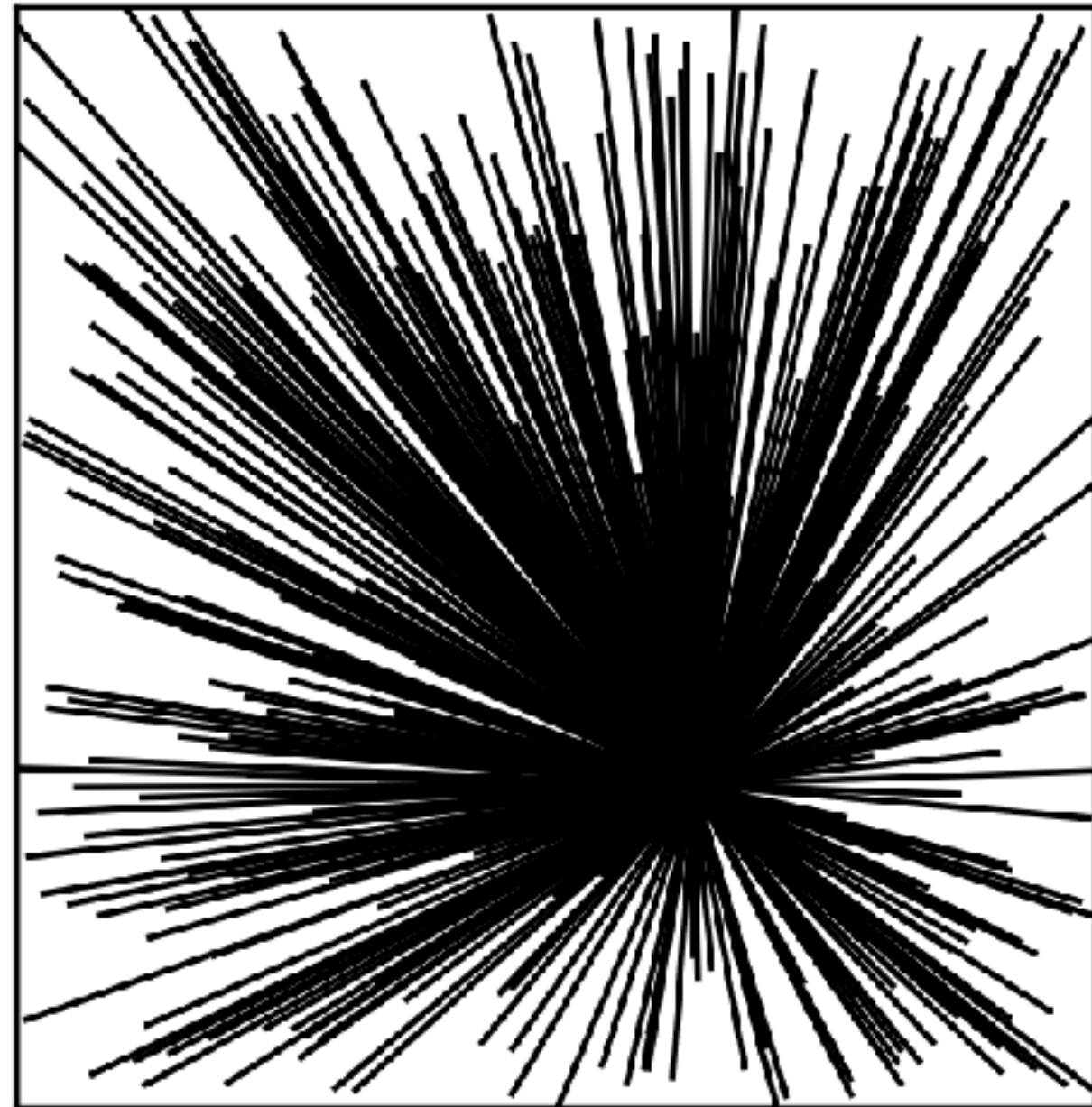
1) Generate random nodes

2) Add links depending on the Euclidian distance

$$P(i,j) = \beta e^{d(i,j)/d_0}$$

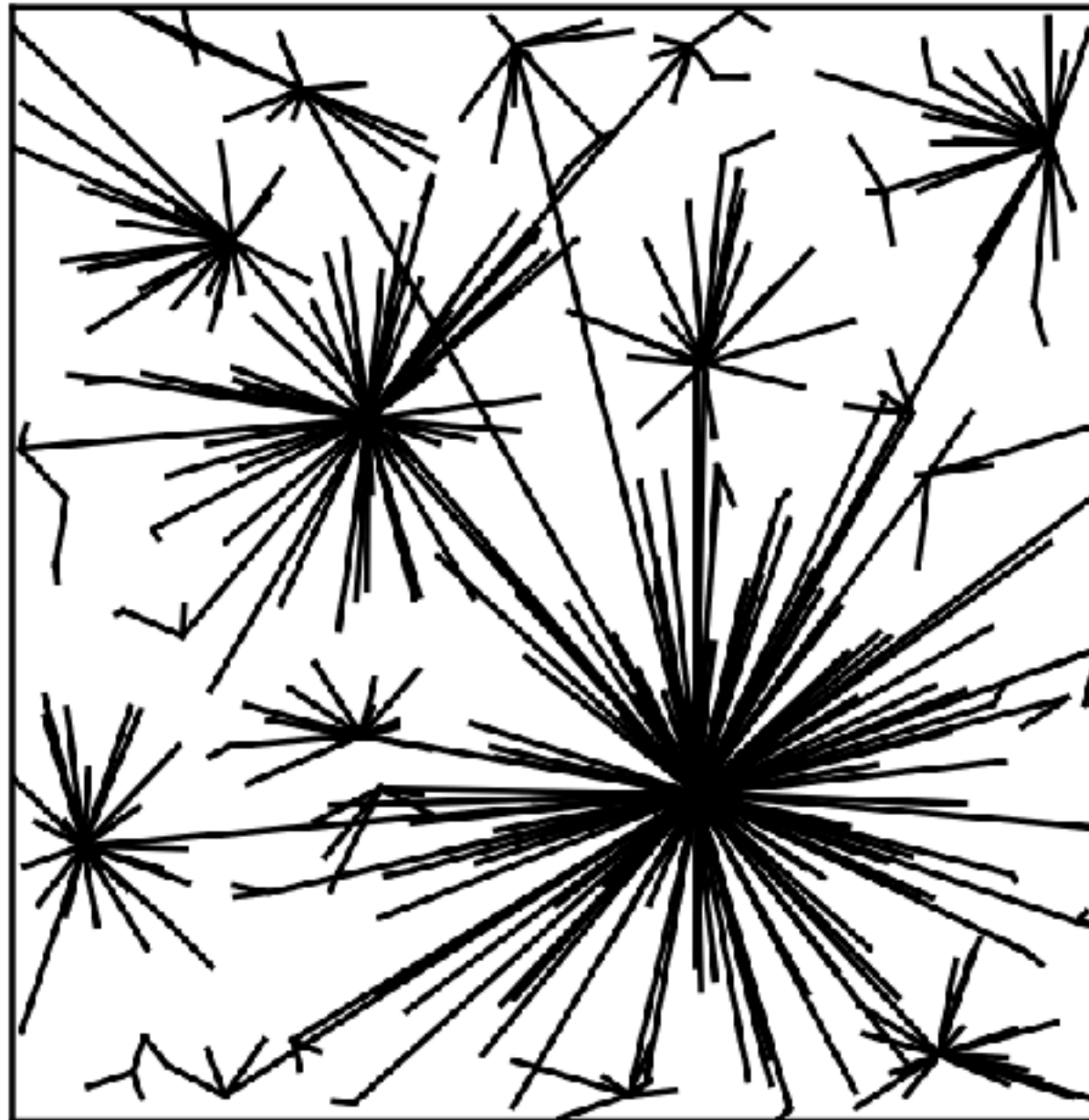
# Model: Hierarchical

$$\beta/\beta^* = 0.0$$



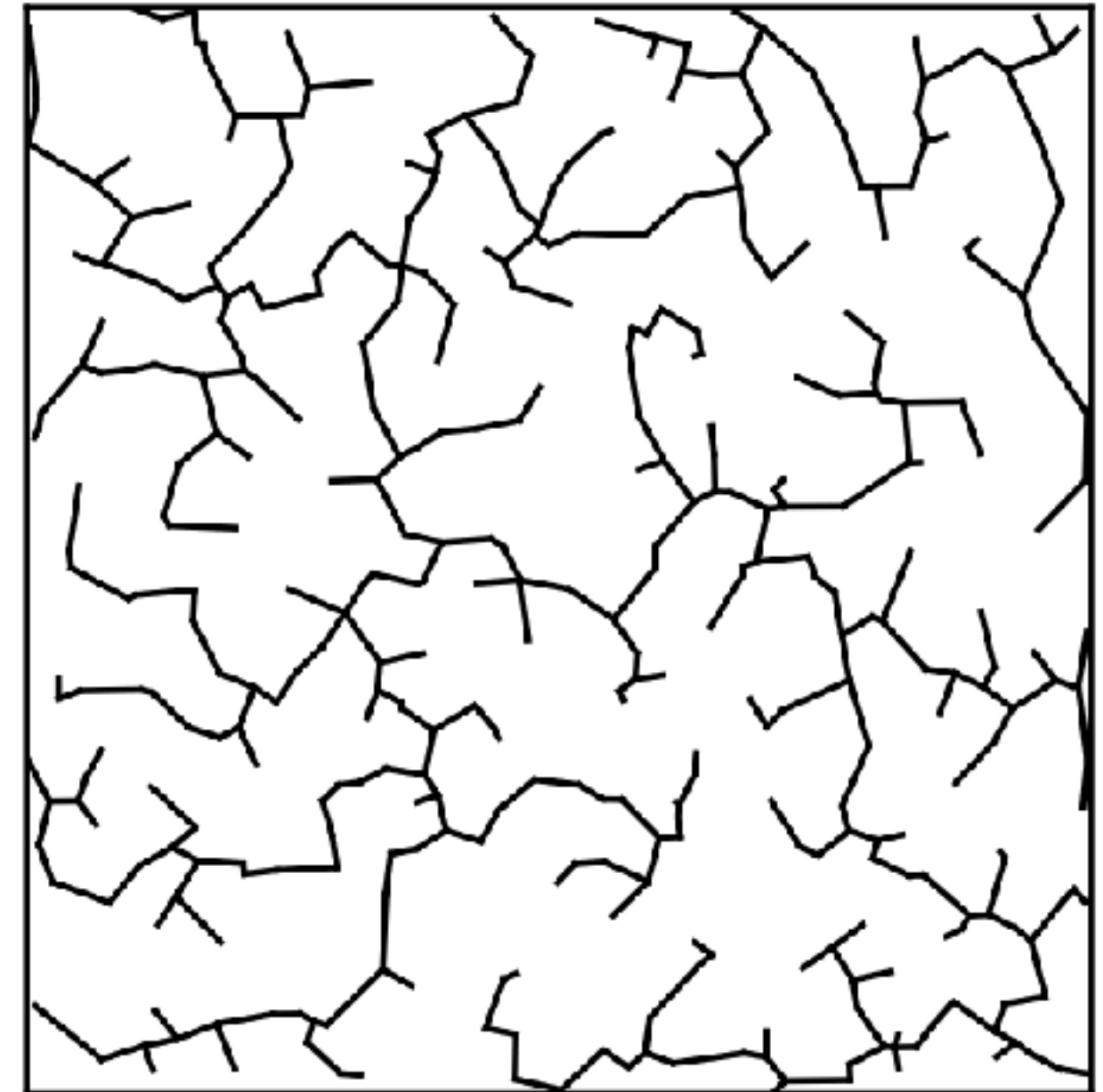
Star graph

$$\beta/\beta^* = 1.0$$



Spatial hierarchy

$$\beta/\beta^* = 1000.0$$



Minimum spanning tree

The average detour  
is minimal here

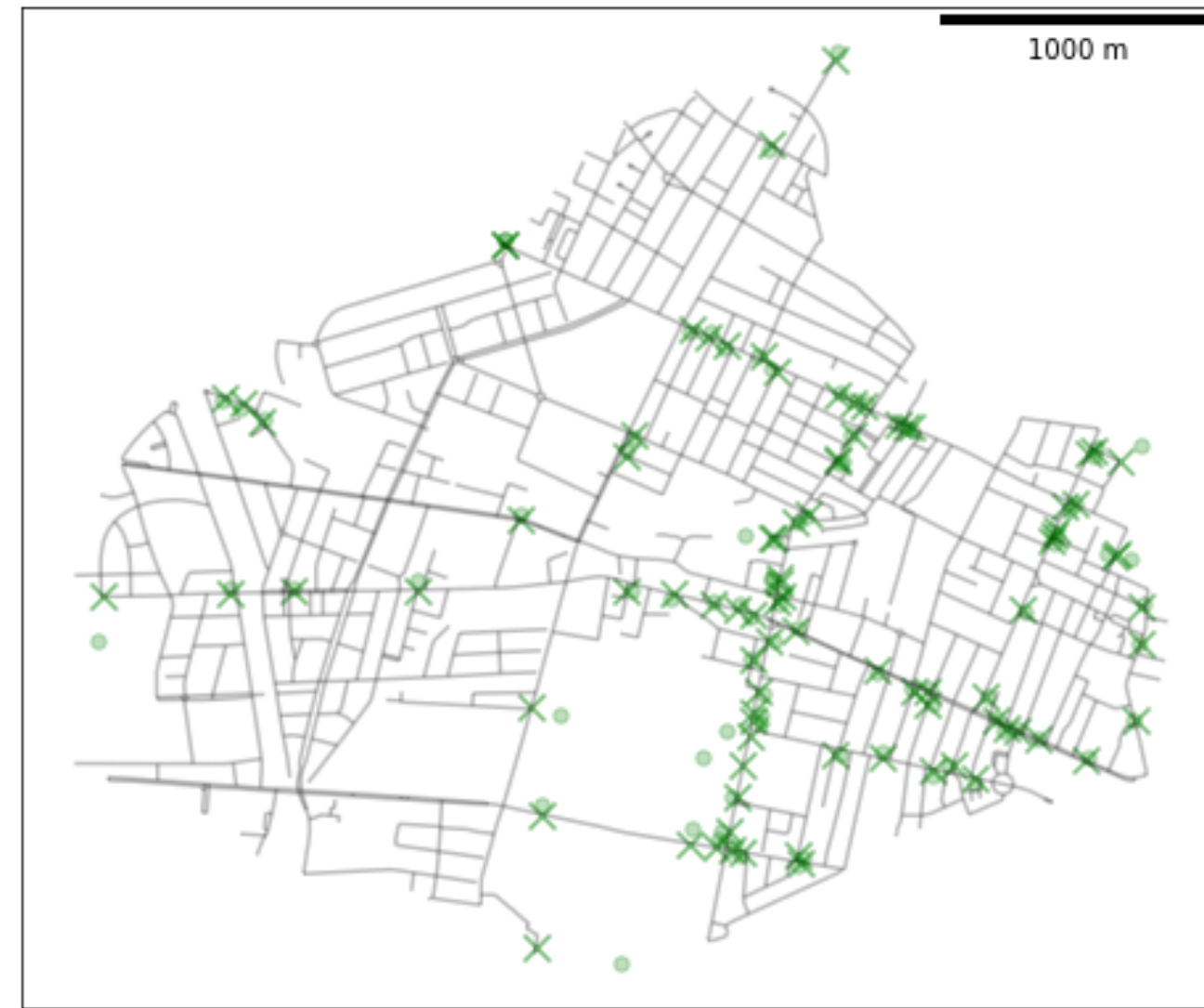
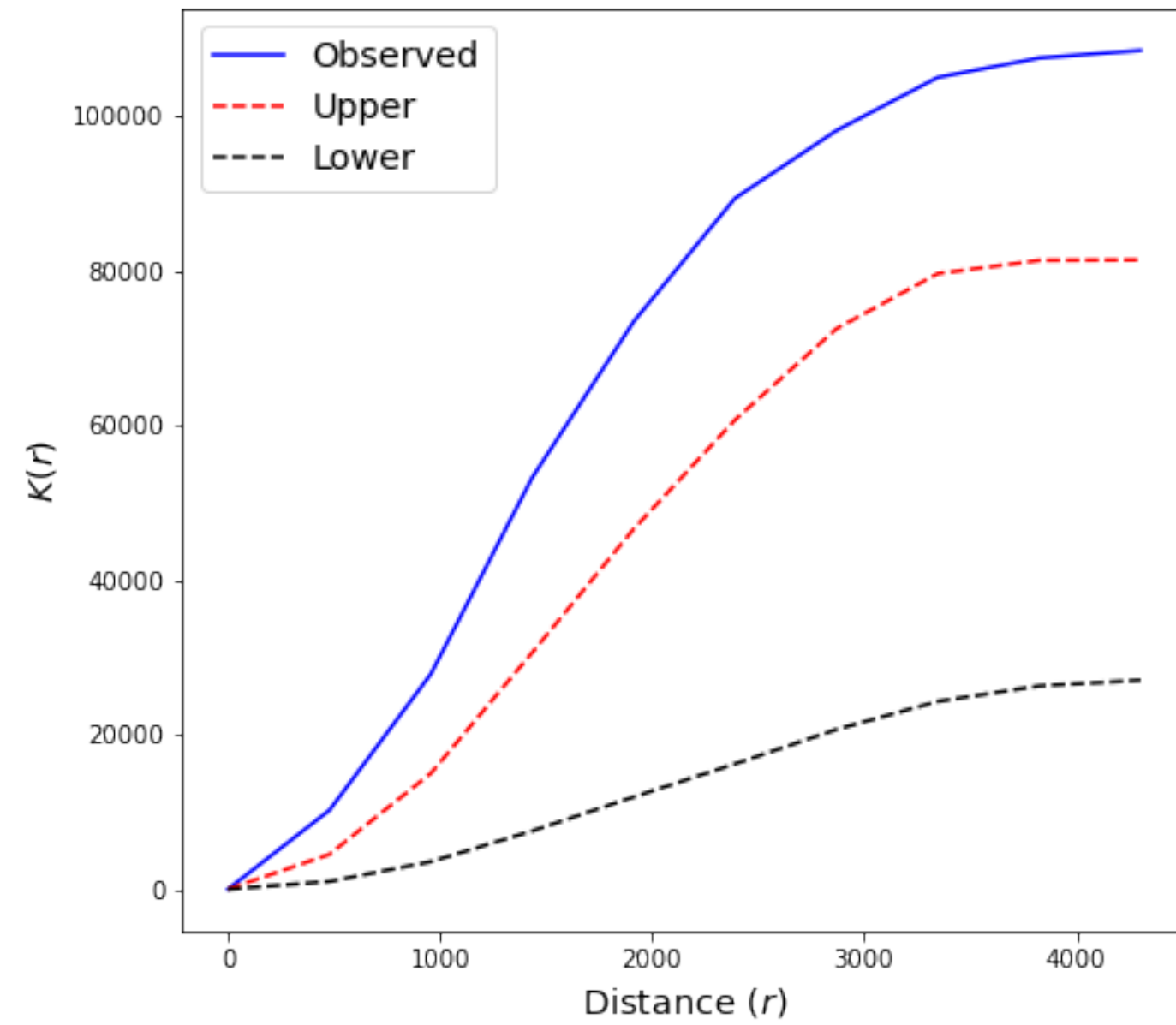
Model railway networks  
between cities

Ripley's  $K(d)$  function measures clustering for a distance  $d$

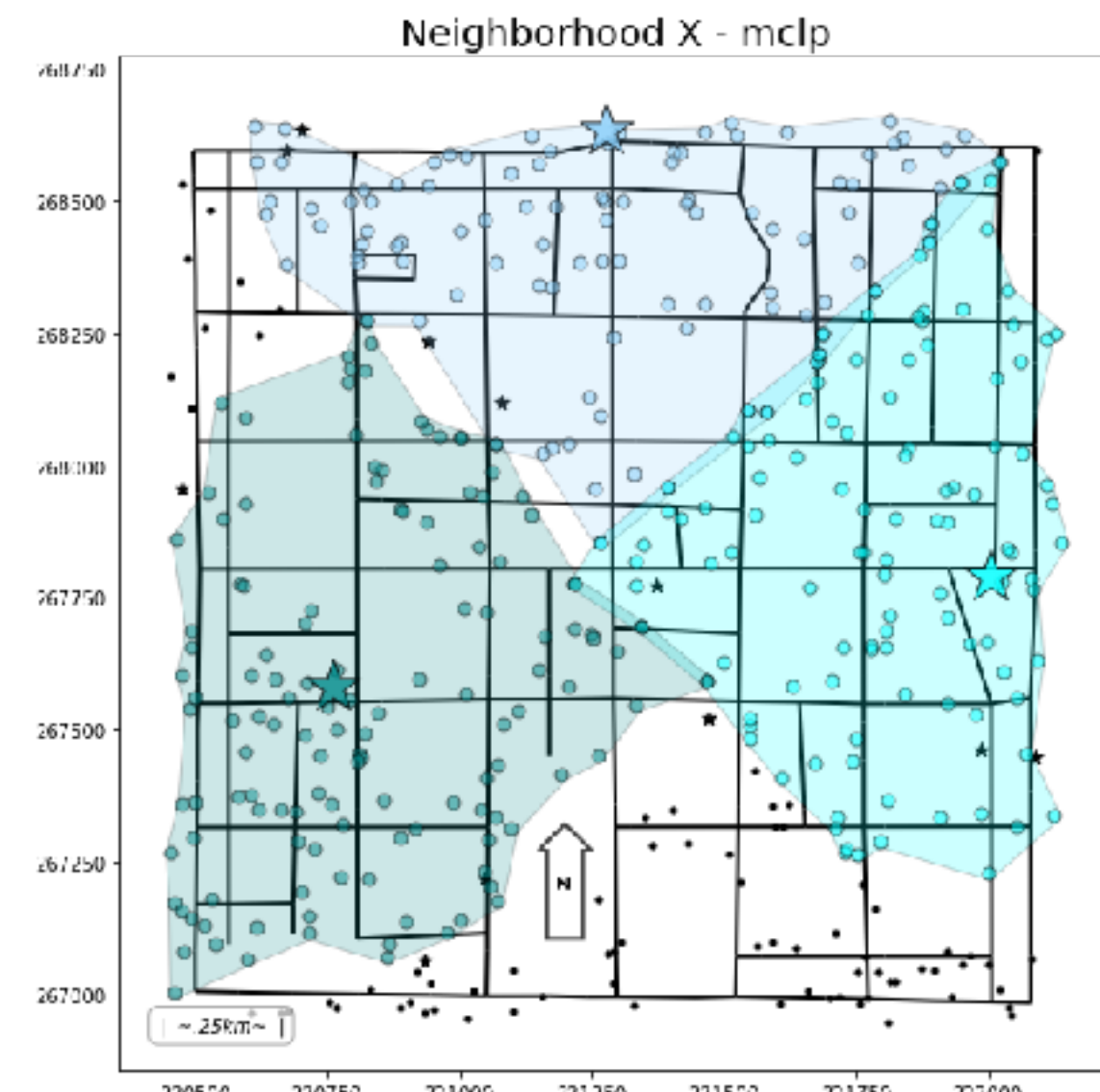
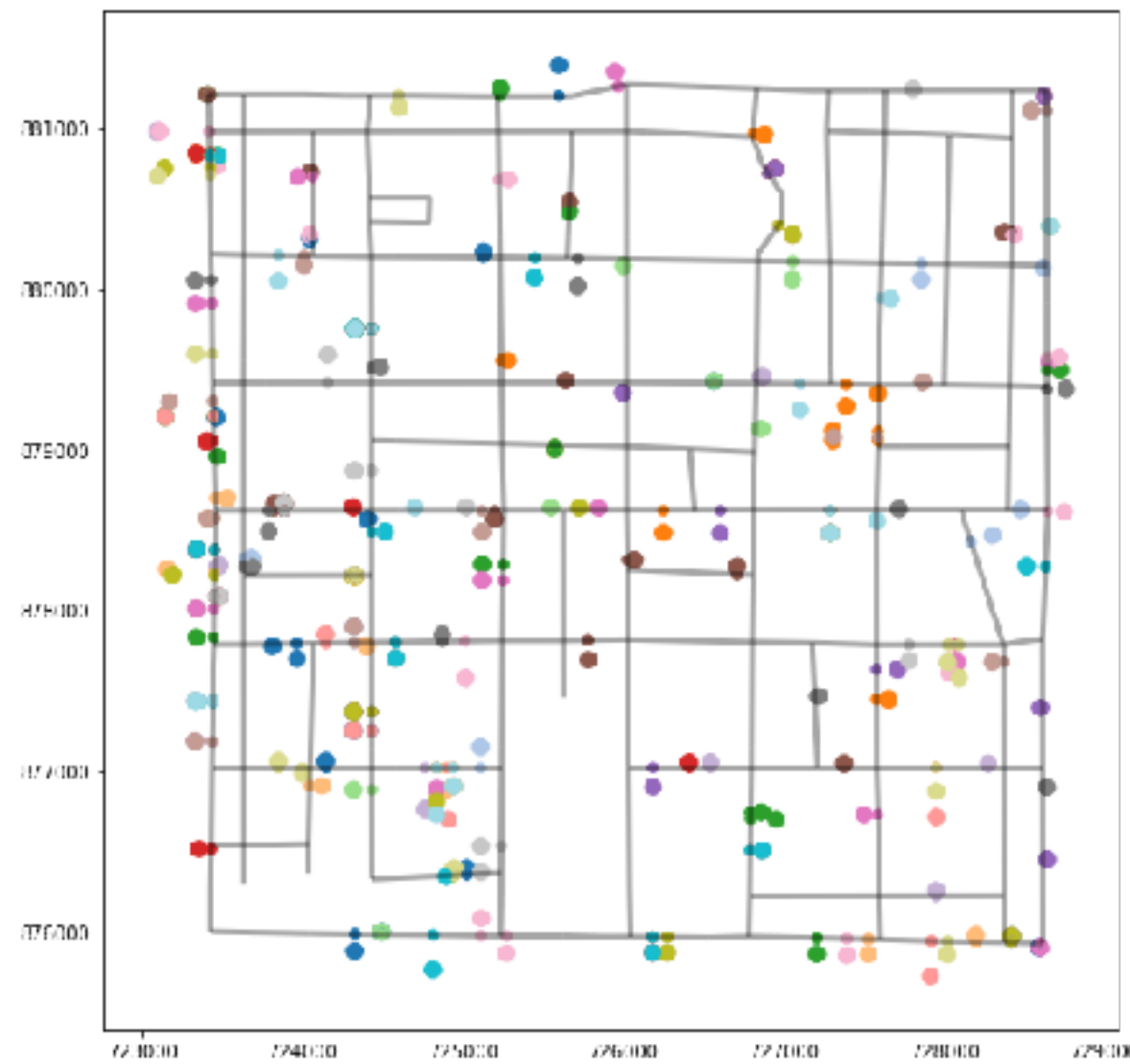
Ripley's  $K(d)$  function is the expected number of points within distance  $d$  from any randomly sampled point, corrected for density.



# Useful Python libraries: spaghetti and snkit

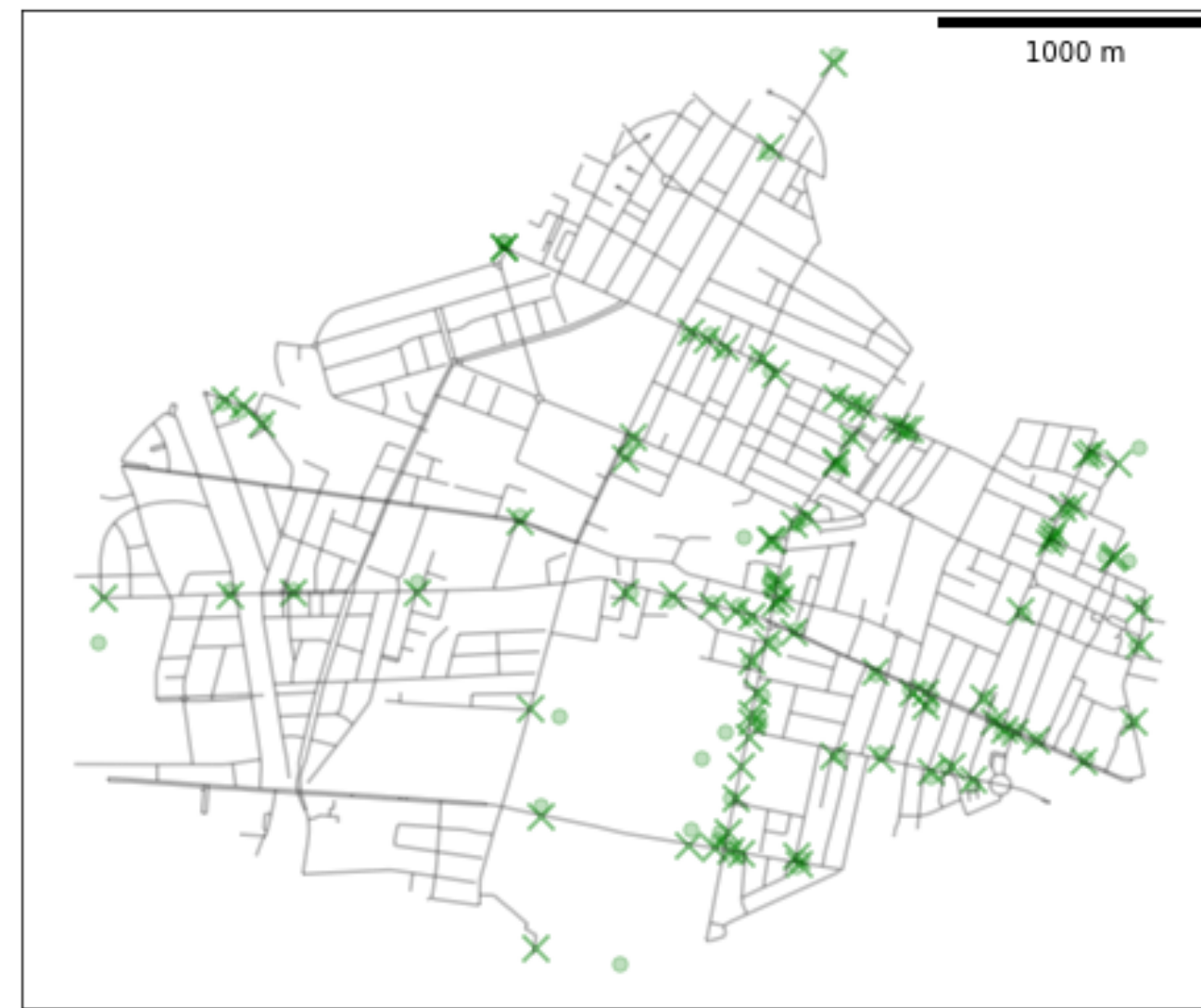
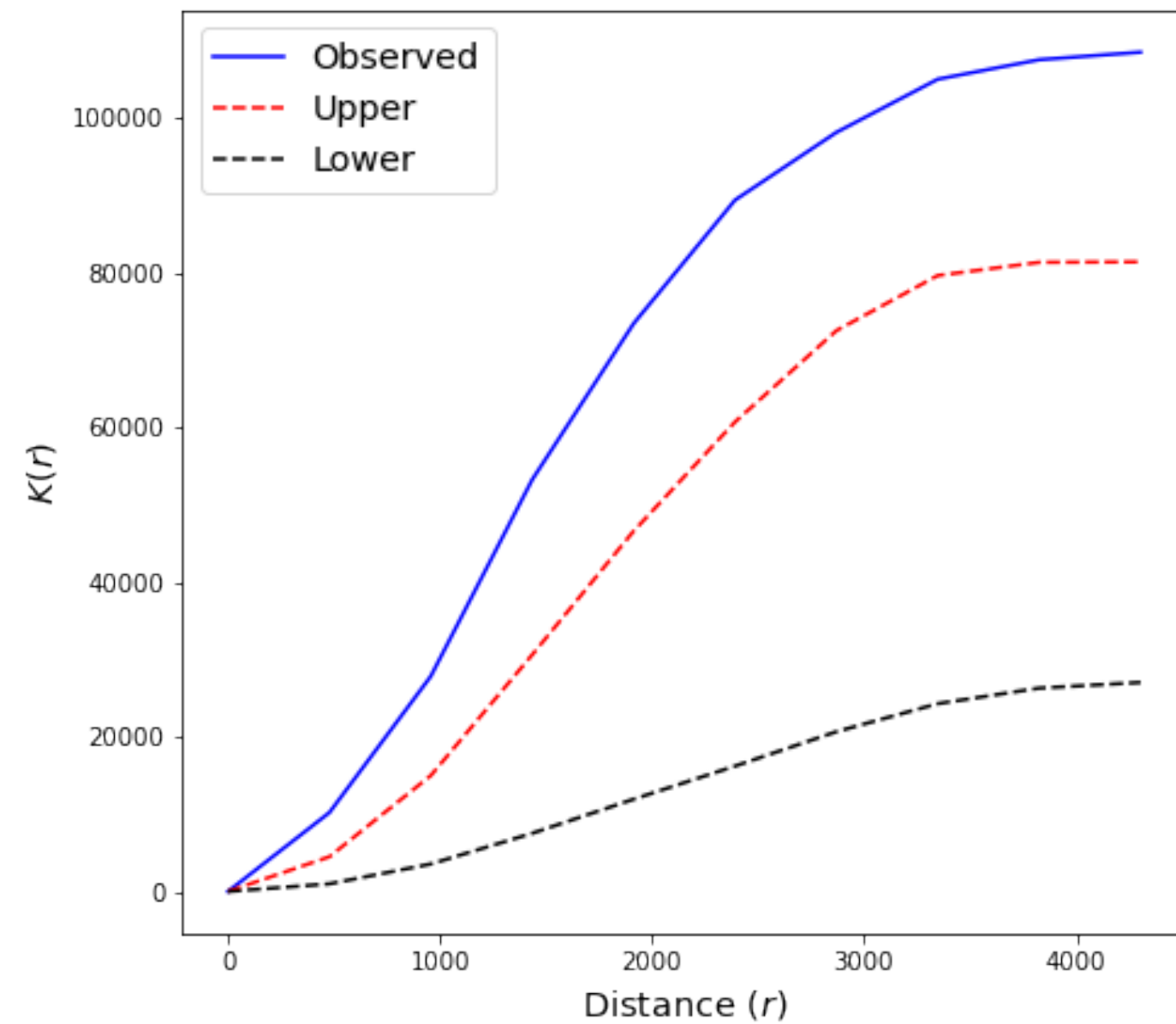


spaghetti allows spatial analysis on networks



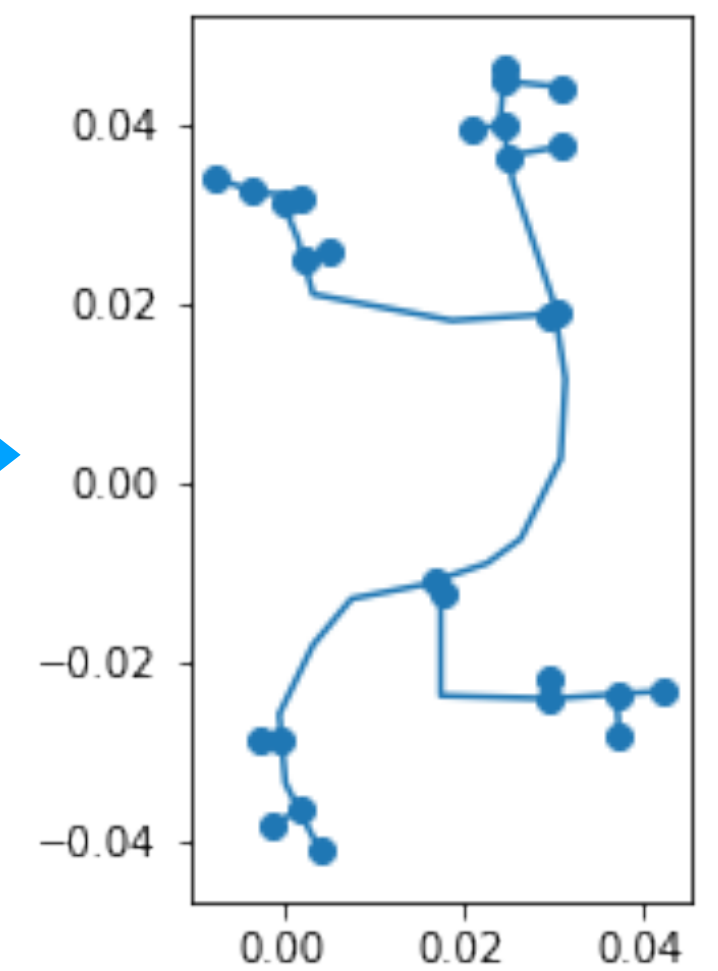
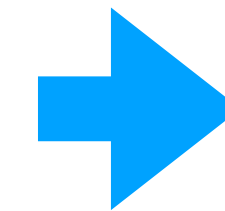
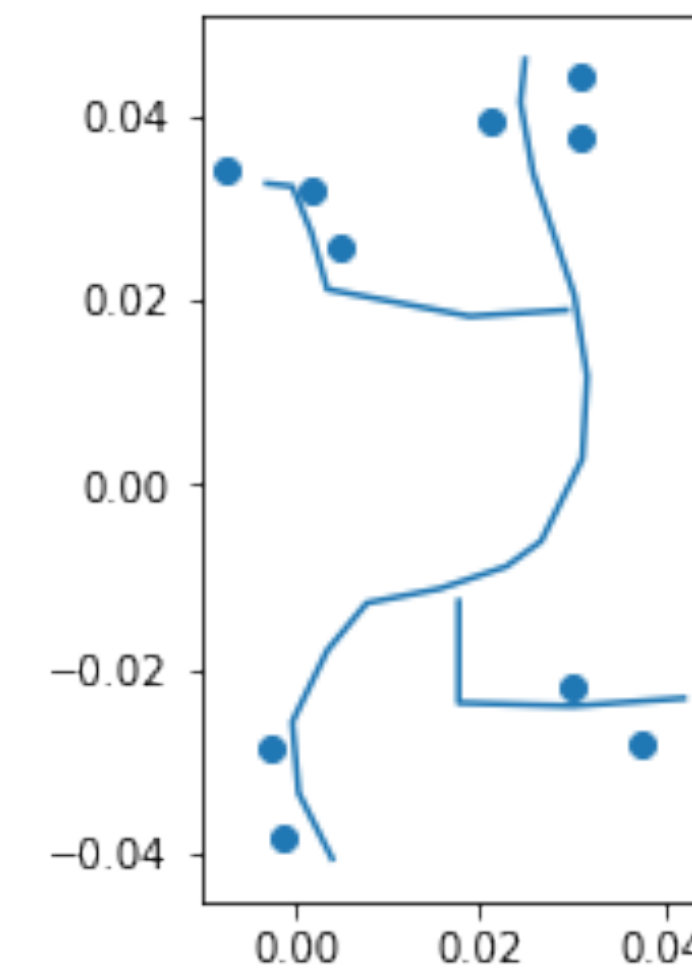
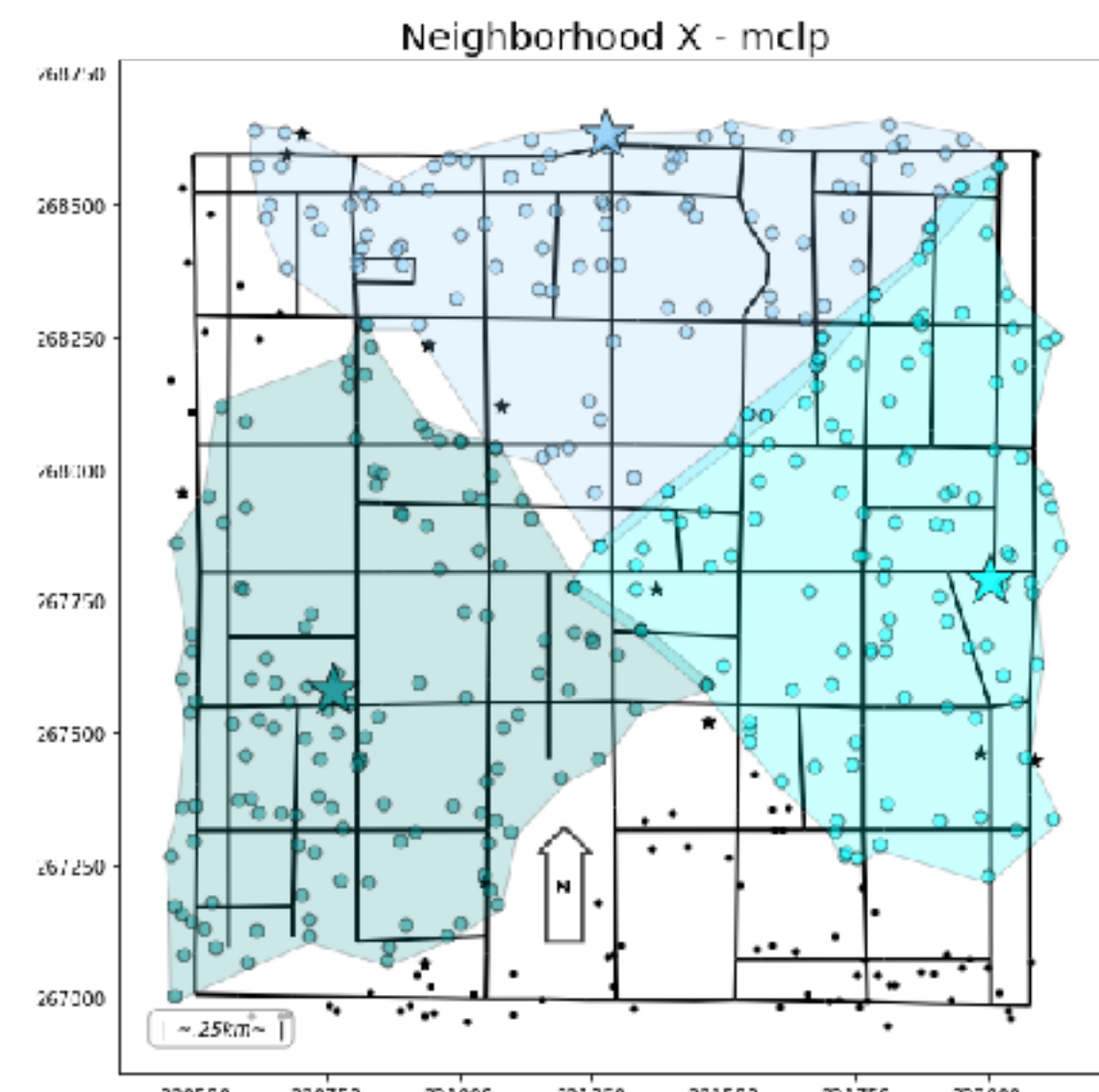
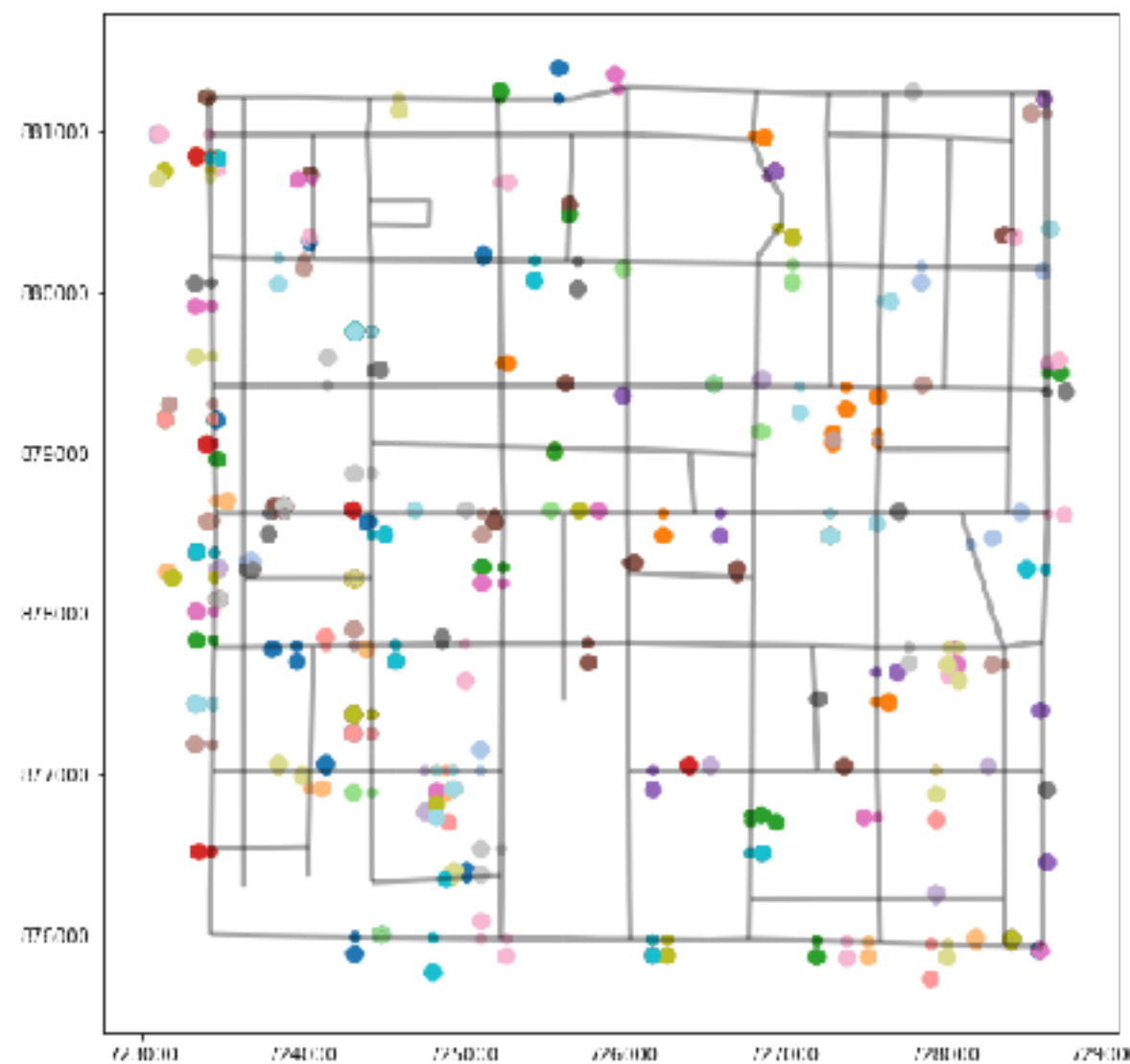
<https://pysal-spaghetti.readthedocs.io/en/latest/index.html>

# Useful Python libraries: spaghetti and snkit



spaghetti allows spatial analysis on networks

snkit helps tidy spatial network data



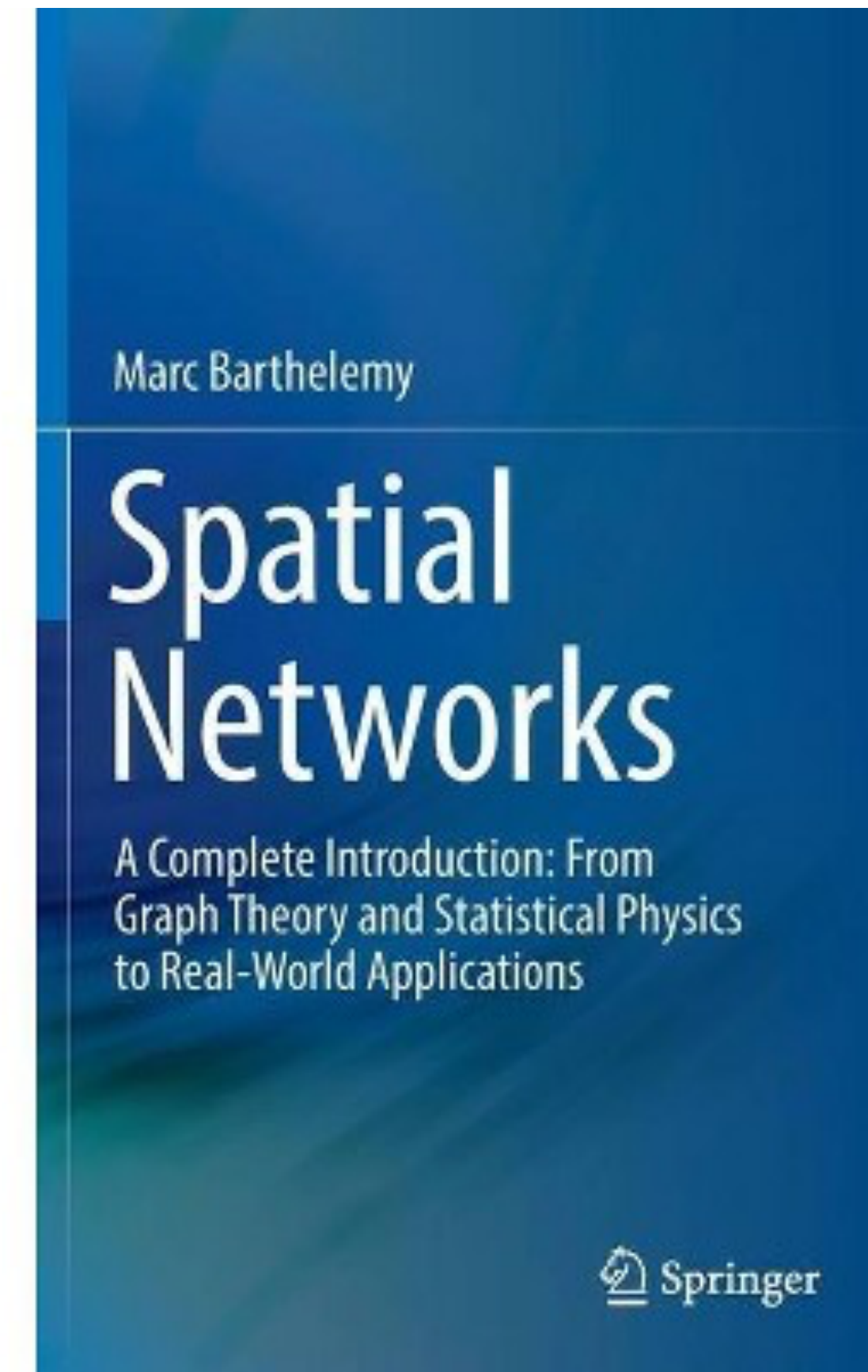
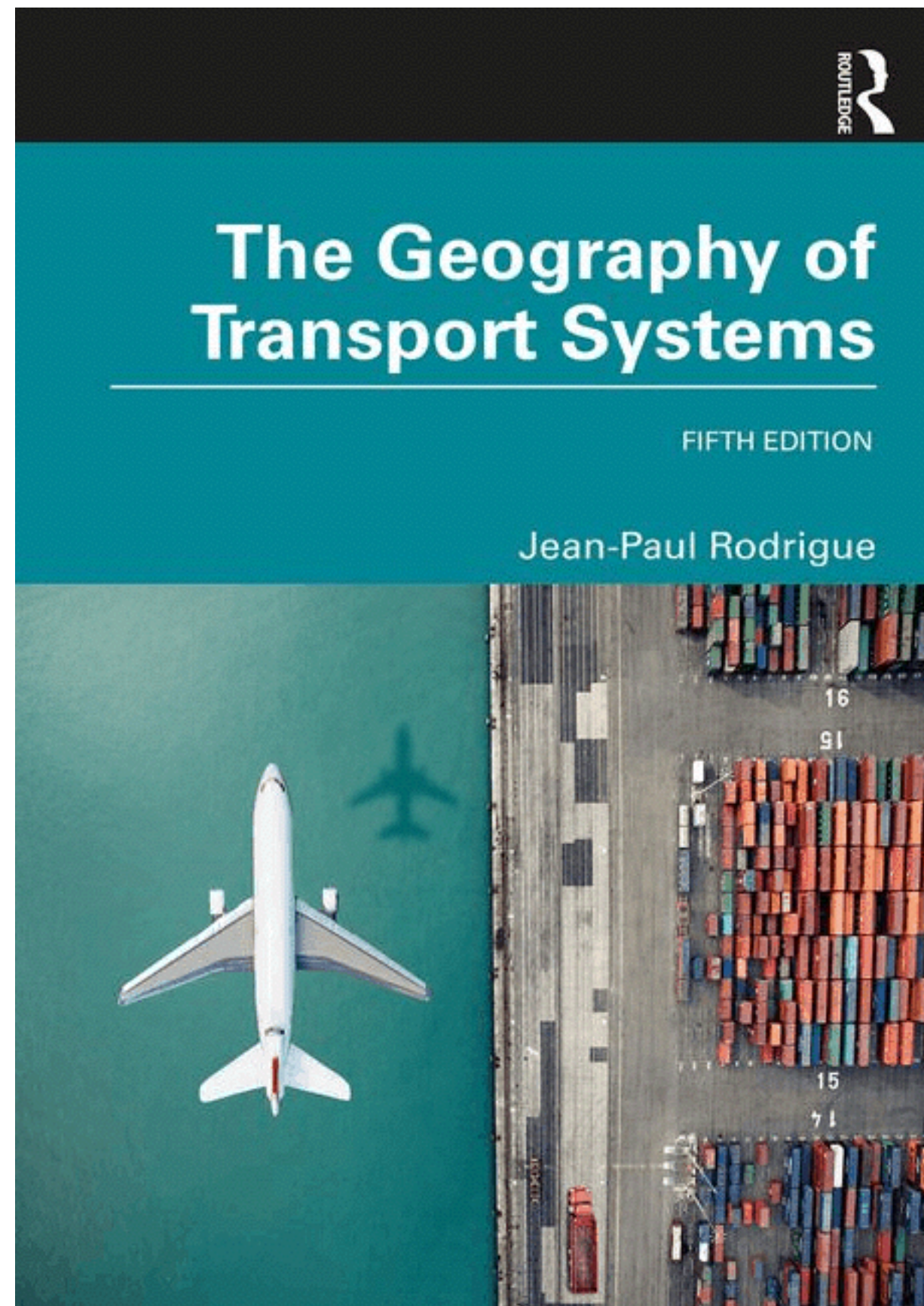
<https://github.com/tomalrussell/snkit>

<https://pysal-spaghetti.readthedocs.io/en/latest/index.html>

# Jupyter



# Sources and further materials for today's class



<https://transportgeography.org/>